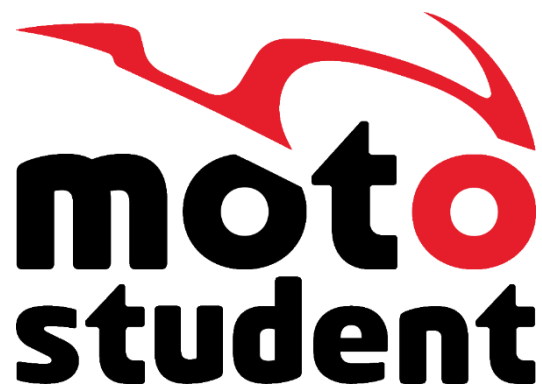


**IX** **EDITION**



**INTERNATIONAL COMPETITION**  
**2026 - 2027**

**Competition Regulations**

**Rev. 01/2026**



| MS2627 IX MotoStudent International Competition |                    |                   |
|-------------------------------------------------|--------------------|-------------------|
| Version                                         | Applicable as from | Modified articles |
|                                                 |                    |                   |
|                                                 |                    |                   |

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# **SECTION A**

## **ADMINISTRATIVE REGULATIONS**

## ARTICLE 1: INTRODUCTION TO MOTOSTUDENT COMPETITION

### A.1.1 MotoStudent objectives

A.1.1.1 | MotoStudent is an international academic challenge aimed at student teams from higher education institutions. For the purposes of this Competition, higher education institutions include universities, polytechnic institutes, technical colleges, and vocational training institutions providing post-secondary education. Teams must design and develop a competition motorbike project (Electric or eFuel), which will be evaluated throughout the different milestones of the Competition and tested in a Final Event to be held at the MotorLand Aragón facilities in Alcañiz (Teruel), Spain.

A.1.1.2 | MotoStudent is an advanced multidisciplinary educational program designed to develop technical, managerial and innovative talent among undergraduate and graduate students. Through a real motorsport challenge, participants conceive, design, develop and manufacture a competition motorbike project (Electric or eFuel), applying engineering knowledge, strategic thinking and professional standards within an international competitive environment.

Throughout the different phases of the competition, teams strengthen and demonstrate their capabilities in areas including:

- Project management, with emphasis on economic planning, budget control, marketing strategy, resource management, technical documentation and milestone execution.
- The design, development and manufacturing of a real competition prototype under defined regulations, with particular focus on innovation, technical rigor and performance.
- Competitive mindset, teamwork, leadership and international collaboration with student teams from higher education institutions worldwide.

### A.1.2 Event Promoter and Organization

A.1.2.1 | The MotoStudent International Competition is promoted and organized by the Moto Engineering Foundation (hereinafter “MEF”) together with TechnoPark MotorLand. Both entities constitute the official Competition Organization and are jointly responsible for the planning, coordination, management and full development of the Competition.

A.1.2.2 | The Competition Organization holds the authority to interpret, adapt and modify the Competition Rules & Regulations, the official calendar and any other official document whenever deemed necessary for the proper development, safety or integrity of the Competition.

- A.1.2.3 | Any modification will be duly communicated to the participating teams through the official communication channels of the Competition. The version of the Competition Rules & Regulations published on the official website shall always be considered the only valid and binding version.

### A.1.3 Competition Categories

- A.1.3.1 | The MotoStudent International Competition is structured into two distinct and independent categories:
- MotoStudent eFuel
  - MotoStudent Electric
- A.1.3.2 | The **MotoStudent eFuel** category involves prototypes powered exclusively by an internal combustion engine operating with the designated competition fuel, in accordance with the technical requirements established in these Competition Rules & Regulations.
- A.1.3.3 | The **MotoStudent Electric** category involves prototypes powered exclusively by an electric propulsion system, in accordance with the technical requirements established in these Competition Rules & Regulations.
- A.1.3.4 | Each category will be conducted in parallel and independently, with its own classification, evaluation process, scoring system, tests and milestones throughout all phases of the Competition, including the Final Event.

### A.1.4 Phases of the Competition

- A.1.4.1 | The Competition is structured into two distinct and sequential phases: MS1 and MS2. During these phases, the projects and prototypes presented by the teams will be evaluated through a series of defined tests and assessments.
- A.1.4.2 | **MS1** is the evaluation phase focused on engineering, management and organizational capabilities. It assesses the teams' capacity to conceive, plan, develop and validate a competitive real-world project in accordance with the Competition Rules & Regulations. The evaluation emphasizes efficiency, innovation and effective use of available resources.
- A.1.4.3 | **MS2** is the dynamic and performance phase, consisting of a series of technical and on-track tests aimed at assessing the safety, functionality and performance of the manufactured prototype under competitive conditions.
- A.1.4.4 | Access to any phase or stage of the Competition shall be conditional upon compliance with the requirements, mandatory deliverables and minimum criteria



established for that phase, as defined in these Competition Rules & Regulations. Failure to meet such requirements may result in restriction or denial of access to subsequent phases.

A.1.4.5 Non-compliance with the Competition Rules & Regulations at any stage of the Competition may result in penalties in accordance with the applicable provisions. The Competition Organization reserves the authority to take any necessary decision to safeguard the safety, integrity and proper development of the Competition.

A.1.4.6 Safety constitutes a fundamental principle of the MotoStudent International Competition's mission. All participants, officials and stakeholders are required to prioritize safety in the design, development and operation of their projects, in accordance with the standards and requirements established in these Competition Rules & Regulations.

#### **A.1.5 Awards**

A.1.5.1 Awards will be granted independently in each category (MotoStudent eFuel and MotoStudent Electric) in accordance with the results obtained during the Competition, except where expressly stated otherwise.

##### **A.1.5.2 MS1 Phase Awards**

The following awards will be granted in each category based on the evaluation results of the MS1 phase:

- Best MS1
- Best Design
- Best Innovation

##### **A.1.5.3 MS2 Phase Awards**

The following awards will be granted in each category based on the official classification of the MS2 phase:

- 1<sup>st</sup> Classified of the MS2 Phase
- 2<sup>nd</sup> Classified of the MS2 Phase
- 3<sup>rd</sup> Classified of the MS2 Phase

##### **A.1.5.4 Overall Category Awards**

The following awards will be granted in each category based on the combined results of MS1 and MS2:

- Best MotoStudent: awarded to the team obtaining the highest total score from the combined results of MS1 and MS2.
- Best MotoStudent Rookie: awarded to the highest-ranked eligible team, based on the combined results of MS1 and MS2, representing an

institution that has never previously participated in any past edition of the MotoStudent Competition.

For the purposes of this award, eligibility shall be determined at institutional level. The Competition Organization reserves the right to verify institutional continuity in order to prevent artificial reclassification of teams.

#### A.1.5.5 Special Nominations

In addition to the category-based awards, the following Special Nominations may be granted by the Competition Organization:

- *MotoStudent Fair Play nomination*: granted to the team demonstrating exemplary integrity, respect and professional conduct throughout all phases of the Competition.
- *MotoStudent Media Hero nomination*: granted to the team demonstrating outstanding communication, media engagement and public outreach during the Competition.

#### A.1.5.6 Summary of the MotoStudent International Competition Awards

A summary of the awards structure is provided in Table 1 for reference purposes only:

| MotoStudent eFuel          |                            | MotoStudent Electric       |                            |
|----------------------------|----------------------------|----------------------------|----------------------------|
| MS1 Phase Awards           |                            |                            |                            |
| Best MS1                   |                            | Best MS1                   |                            |
| Best Design                | Best Innovation            | Best Design                | Best Innovation            |
| MS2 Phase Awards           |                            |                            |                            |
| 1 <sup>st</sup> classified |                            | 1 <sup>st</sup> classified |                            |
| 2 <sup>nd</sup> classified | 3 <sup>rd</sup> classified | 2 <sup>nd</sup> classified | 3 <sup>rd</sup> classified |
| Overall Category Awards    |                            |                            |                            |
| Best MotoStudent           |                            | Best MotoStudent           |                            |
| Best Rookie Team           |                            | Best Rookie Team           |                            |
| Special Nomination         |                            |                            |                            |
| MotoStudent Fair Play      |                            |                            |                            |
| MotoStudent Media Hero     |                            |                            |                            |

Table 1 - MotoStudent awards summary

A.1.5.7 The economic and/or material value of any award will be communicated by the Competition Organization through the official communication channels to all registered teams.

#### A.1.6 Entries and Participation

A.1.6.1 | MotoStudent is open to student teams officially representing universities worldwide. Participating universities may collaborate with officially accredited vocational training institutions or other equivalent academic entities. However, the representing university shall remain the sole institutional counterpart before the Competition Organization. Participation must be officially endorsed by the representing university.

A.1.6.2 | The registration period for the IX Edition of the MotoStudent International Competition will open on **20/04/2026** and shall be formalized exclusively through the official website of the Competition: [www.motostudent.com](http://www.motostudent.com). All applicable deadlines will be published through the official communication channels of the Competition.

A.1.6.3 | Registration and participation in the Competition shall be conditional upon full payment of the participation fee. For the IX Edition, the participation fee per team amounts to **€ 4,550** (VAT included). No registration shall be considered valid until full payment of the participation fee has been received by the Competition Organization.

The participation fee corresponds to the right to participate in the MotoStudent Competition and to access the elements, services and opportunities associated with the current edition. It shall not be interpreted as the purchase price of any specific component of the MotoStudent Kit.

A.1.6.4 | Team registration includes:

- The right to participate as an official team in the IX Edition of the MotoStudent International Competition.
- The right to use the official MotoStudent Kit during the current edition, including the mandatory common components to be installed in the prototype. Shipping costs of the MotoStudent Kit are included. Customs duties, import taxes or any similar charges applicable in the destination country shall not be included.
- Registration for up to ten (10) students (nine Team Members and one Team Leader) and two (2) Team Tutors to participate at the current edition.
- Access to the Competition phases, evaluations, official activities and Final Event.
- Access to official challenges, partner initiatives and opportunities offered by the Competition Organization and collaborating entities.
- Access to services, supplies, discounts and opportunities made available by the Competition Organization and its collaborating entities within the framework of the Competition.

A.1.6.5 | The fee for adding additional Team Members slots amounts to **€ 240** (VAT included). Additional Team Members may be registered until 30 days before the first day of the Final Event.

Requests to purchase additional Team Member slots must be made through the MotoStudent website, either during the initial team registration process or after the team has been registered, through the Store section in the team's private area.

A.1.6.6 | Registration as a Team Member includes:

- Passes for the Final Event, with exclusive access to areas designated for Team Members.
- The right to receive a certificate validating participation in the current edition of the International MotoStudent Competition.
- Access to specialized on-site or online training courses via the MotoStudent Platform, including the corresponding official certificate and possible discounts.
- Access to the MotoStudent Talent Roadshow to stay informed about job opportunities with leading mobility companies.

A.1.6.7 | The fee for adding additional Team Tutor slots amounts to **€ 115** (VAT included). Additional Team Tutors may be registered until 30 days before the first day of the Final Event.

Requests to purchase additional Team Tutor slots must be made through the MotoStudent website, either during the initial team registration process or after the team has been registered, through the Store section in the team's private area.

A.1.6.8 | Registration as a Team Tutor includes:

- Passes for the Final Event, with exclusive access to areas designated for Team Members.
- The right to receive a certificate validating participation in the current edition of the International MotoStudent Competition.
- Access to specialized on-site or online training courses via the MotoStudent Platform, including the corresponding official certificate.

A.1.6.9 | All Team Members and Team Tutors must be registered through the MotoStudent Platform once the team registration has been completed and confirmed by the Competition Organization.

Individual registration of Team Members and Team Tutors shall only be possible after the team has been officially registered.

The complete registration process is described in Annex 5.

- A.1.6.10 | A team, Team Member, Team Leader or Team Tutor shall only be considered registered upon receipt of official confirmation issued by the Competition Organization via email or any other electronic means.
- A.1.6.11 | The Competition Organization reserves the right to deny the registration of any team, Team Member or Team Tutor.
- A.1.6.12 | The number of available team entries is limited. Entries shall be confirmed in order of registration. Teams that have not completed payment of the applicable fees shall not be considered registered and shall not be entitled to reserve a place in the Competition.
- A.1.6.13 | Entry **fees are non-refundable**. Once a team member / tutor has been registered through the MotoStudent Platform, this team slot is blocked and can't be modify.
- In the event of interruption, cancellation, or modification of the Competition schedule or format (including changes to the dates of the Final Event) due to circumstances beyond the control of the Competition Organization, the Organization shall not be obliged to refund the registration fees, whether partially or in full.

#### **A.1.7 Official Publications and Final Event Information**

- A.1.7.1 | All official communications, regulations, annexes and related documents will be published on the official website of the Competition: [www.motostudent.com](http://www.motostudent.com)
- A.1.7.2 | Teams are required to read, understand and remain familiar with all official publications issued by the Competition Organization.
- A.1.7.3 | The Competition Organization may use email, private messaging within the MotoStudent Platform, or any other electronic means to distribute official information regarding the Competition.
- A.1.7.4 | Any official communication issued by the Competition Organization shall be binding within their scope.
- Clarifications and interpretations may be communicated through the official channels of the Competition. Any modification of the Competition Rules & Regulations or of the applicable technical or sporting requirements shall be formalized through the corresponding official amendment, Technical Note or updated version of the applicable document.

- A.1.7.5 | The Competition Organization reserves the right to designate additional communication channels as official, provided that prior notice is given to the teams.

#### **A.1.8 Official Languages**

- A.1.8.1 | The official languages of the MotoStudent International Competition shall be:
- English
  - Spanish
- A.1.8.2 | Official publications and documents shall be issued in English.
- A.1.8.3 | All communications between participating teams and the Competition Organization shall be conducted in one of the official languages.

#### **A.1.9 Official Time**

- A.1.9.1 | The official time of the MotoStudent International Competition shall be the time in force in mainland Spain:
- Winter Time: CET (UTC+1)
  - Summer Time: CEST (UTC+2)

### **ARTICLE 2: APPLICATION OF THE MOTOSTUDENT REGULATIONS**

#### **A.2.1 Ambiguities in the Regulations**

- A.2.1.1 | Any ambiguity, interpretation issue or question regarding the meaning or intent of these Competition Rules & Regulations shall be resolved exclusively by the Competition Organization.
- A.2.1.2 | Any matter not expressly permitted by these Competition Rules & Regulations shall be considered prohibited. Teams may submit a formal request for clarification or authorization to the Competition Organization. The corresponding decision will be communicated to all teams through an official communication.

#### **A.2.2 Validity of the Regulations**

- A.2.2.1 | The Competition Rules & Regulations of the MotoStudent International Competition published on the official website [www.motostudent.com](http://www.motostudent.com), including any subsequent official amendments or revisions, shall constitute the only valid regulatory framework for the current edition of the MotoStudent International Competition.
- A.2.2.2 | Regulations from any previous edition shall not apply to the current edition of the MotoStudent International Competition.

- A.2.2.3 | In the event of any sporting situation or infraction not expressly covered by these Competition Rules & Regulations or by any official communication issued by the Competition Organization, the RFME Sporting Regulations of the Spanish Speed Championship ([ESBK](#)) 2026 shall apply for procedural purposes.

### **A.2.3 Compliance with the Regulations**

- A.2.3.1 | By taking part in the MotoStudent International Competition, the team, Team Members, Team Leader, Team Tutors, Riders and any other university staff members, in their individual capacity, accept and shall comply with all rules established by the Competition Organization, including these Competition Rules & Regulations, subsequent official communications, and the rules of the sports complex where the Final Event takes place.
- A.2.3.2 | The Competition Organization reserves the right to sanction any team that fails to comply with the Competition Rules & Regulations, established schedules, or that demonstrates unsportsmanlike conduct or behaviour that may endanger persons or facilities.
- A.2.3.3 | During the MS2 phase, Race Direction may apply the relevant sporting penalties for non-compliance with applicable sporting regulations. Such penalties may be accompanied by score penalties where appropriate.
- A.2.3.4 | Disobedience of instructions issued by the Competition Organization or Race Direction, unsportsmanlike or aggressive behaviour, or acts prejudicial to persons attending the event may result in disqualification.
- A.2.3.5 | Teams, Riders, Team Members, Team Leaders and Team Tutors shall refrain from issuing public statements or press releases that may irresponsibly harm the legitimate interests of the MotoStudent International Competition, its Officials, or that may be contrary to the integrity of the Competition or the sport.
- A.2.3.6 | Public statements covered by this provision include, but are not limited to:
- Statements or comments to the media that attack, belittle, discredit, or irresponsibly harm Competition Officials.
  - Public comments made by Team Members, Team Leaders, Team Tutors or Riders who know, or should reasonably know, that such comments may harm the reputation, image, or interests of the Competition or its Officials.
  - Constructive and respectful feedback regarding Competition decisions or policies may be submitted through the official communication channels of the Competition.

#### **A.2.4 Penalty types**

- A.2.4.1 | A Non-Compliance (hereinafter NC) shall result in the penalized delivery scoring 0 points.
- A.2.4.2 | A Not Presented (hereinafter NP) shall result in the penalized delivery scoring 0 points.
- A.2.4.3 | A Minor Penalty (hereinafter MP) shall result in a deduction of 10 points from the total MotoStudent score.
- A.2.4.4 | A Severe Penalty (hereinafter SP) shall result in a deduction of 50 points from the total MotoStudent score.
- A.2.4.5 | A Disqualification penalty (hereinafter DQ) shall entail, automatically and independently of any other penalty, the invalidation of the results obtained in an event, session, race, milestone or delivery.
- During the MS2 phase, immediate disqualification from a practice session or race may be indicated by means of a black flag.
- A.2.4.6 | Any sporting-related penalty arising from the MS2 phase shall be determined by Race Direction.
- A.2.4.7 | The Competition Organization reserves the right to impose penalties not expressly defined in Article A.2.4 if deemed necessary.
- A.2.4.8 | A Warning may be issued either privately or publicly and may be imposed without a formal hearing.
- A.2.4.9 | In the case of a Long Lap Penalty (hereinafter LLP), the Rider shall complete the predefined Long Lap route.
- A.2.4.10 | A Grid Penalty shall consist of a drop of one or more grid positions or the obligation to start the race (either the Playoff Race or the Final Race) from the pit lane exit.
- A.2.4.11 | An Expulsion penalty shall result in a score of 0 in the relevant phase. Depending on the seriousness of the infringement, the Competition Organization may impose the automatic expulsion of the team and its members from the Competition.

#### **A.2.5 Right to Claims and Appeals**

- A.2.5.1 | Any team has the right to submit a protest or request for clarification.



- The Competition Organization shall review any doubt, alleged breach or penalty submitted through the official procedure.
- A.2.5.2 | A team may file a claim in case of disagreement with any action taken by the Competition Organization, including scoring decisions or any other official action that may be considered detrimental to the team.
- A.2.5.3 | A team may submit a protest against another participating team if it observes any breach of these Competition Rules & Regulations or improper conduct.
- A.2.5.4 | All protests and complaints shall be submitted to the Competition Organization in accordance with the format established in Section J, Annex 1 of these Competition Rules & Regulations, clearly stating the article considered to have been breached and providing sufficient evidence for verification.
- Protests and complaints submitted prior to the Final Event may be sent scanned or digitally signed by email to [faq@motostudent.com](mailto:faq@motostudent.com). Protests concerning the interpretation or application of the Competition Rules & Regulations must be submitted no later than sixty (60) days prior to the first day of the Final Event.
- During the Final Event, protests and complaints shall be submitted in written form to the Competition Organization through the Race Direction or the Technical Organization.
- A.2.5.5 | To submit a claim or protest, the applicant team shall deposit 50 points from the score of the relevant phase or test and € 800. If the decision is rendered in favour of the applicant team, the 50 points and € 800 will be returned. If the protest is rejected, the team shall forfeit the 50 points and the € 800 deposit.
- A.2.5.6 | Claims or protests related to tests conducted during the Final Event shall be submitted within a maximum of 30 minutes after publication of the results of the affected test.
- A.2.5.7 | The Competition Organization will issue the final decision on any protest. Such decision shall be final and binding and not subject to appeal, except as provided below.
- A.2.5.8 | In purely sporting matters, an appeal procedure may be requested in accordance with Article 119 of the RFME Sporting Regulations. Any participant may appeal penalties imposed or decisions taken by the Competition Organization or Race Direction before the Disciplinary Court of the RFME. The appeal must be submitted in writing, signed by the interested party and accompanied by a deposit of € 800. The right of appeal will expire fifteen days after notification of the decision.

- A.2.5.9 | In accordance with Article 119 of the RFME Sporting Regulations, appeals to the FIM may be lodged through the RFME. The RFME shall determine whether the appeal is admissible and may refuse to process it without being obliged to provide justification. The right of appeal shall not suspend the imposed penalty or sanction. However, where the RFME admits the appeal to the FIM, the appellant may request temporary suspension of the decision, subject to submission of the guaranteed deposit required by the FIM.

#### **A.2.6 Right of Confiscation**

- A.2.6.1 | The Competition Organization reserves the right, during the Final Event, to confiscate or retain any prototype, component, tool or rider equipment deemed non-compliant with these Competition Rules & Regulations.
- A.2.6.2 | Any material confiscated by the Competition Organization during the Final Event will be returned to the corresponding team at the end of the Final Event.
- A.2.6.3 | The Competition Organization reserves the right to confiscate, retain or recall any component of the MotoStudent Kit if it is used outside the MotoStudent International Competition without prior authorization from the Competition Organization.

#### **A.2.7 Changes to the Competition Rules & Regulations**

- A.2.7.1 | The Competition Organization reserves the right to amend the Competition Rules & Regulations, as well as the official Competition Calendar, if deemed necessary. Any amendment will be incorporated into Section I and published on the official website of the Competition.
- A.2.7.2 | The Competition Organization will communicate any amendments through the communication channels it considers appropriate. In all cases, the most updated version of the Competition Rules & Regulations will be the one published on the official website of the Competition.

### **ARTICLE 3: PARTICIPATION REQUIREMENTS**

#### **A.3.1 Team Eligibility**

- A.3.1.1 | The Competition has a purely academic objective and is conceived as a multidisciplinary educational project open to students from diverse fields of study, such as engineering, economics, marketing and project management. Although it involves the design and manufacture of a racing motorcycle prototype,

MotoStudent is not a conventional speed championship but an academic competition in which engineering, management, innovation and teamwork are the core pillars.

A.3.1.2 | Institutions whose primary activity is training in professional motor racing and / or motorsport competition are not eligible to participate in the MotoStudent Competition.

A.3.1.3 | There is no limit to the number of teams per university. Each university may register multiple teams in any category of the Competition.

A.3.1.4 | Each registered team may submit only one project and one prototype.

A.3.1.5 | Each team must represent a recognized higher education institution and must be officially endorsed by that institution. The name of the institution shall be clearly identifiable in the team registration and official documentation of the Competition.

A.3.1.6 | Each participating team must choose a bike number between 1 and 99. Numbers will be allocated according to the team's selection, subject to availability and in order of registration in the Competition.  
Numbers 1, 2 and 3 shall be exclusively reserved for the teams representing the universities ranked first, second and third respectively in the Best MotoStudent Award of the previous edition of the MotoStudent International Competition.

### **A.3.2 Participants**

A.3.2.1 | Teams, Team Members, Team Leaders and Team Tutors registered in the Competition shall be considered "Participants of the Competition" from the moment of registration until the end of the Final Event.

A.3.2.2 | Students from Vocational Training Centres (enrolled in the 2025–2026 and/or 2026–2027 academic year) may also be part of a team. In such cases, the team must provide proof of an agreement between the Vocational Training Center and the participating university.

A.3.2.3 | The consideration of the Competition and its compatibility with the academic curriculum shall be the decision of each university. The representing university is not obliged to include the MotoStudent International Competition in its curriculum, including as an elective course.

A.3.2.4 | To participate in the Competition, students must be enrolled in the 2025–2026 and/or 2026–2027 academic year, regardless of the percentage of subjects or credits passed.

- A.3.2.5 | All students must be at least 18 years old at the beginning of the Final Event.
- A.3.2.6 | If a team includes members from different Universities or Vocational Training Centres, in compliance with the rules indicated above, they must draft and formalize an agreement document in which both Educational Entities formally approve this collaboration.
- Additionally, a template of the agreement document is attached in the Annex 4 for your reference and use.

### **A.3.3 Liability Waiver**

- A.3.3.1 | All Participants, including Team Members, Team Leaders, Team Tutors and Riders, must sign a liability waiver upon entering the MotoStudent International Competition. Any other persons attending the Final Event with the teams shall be considered members of the general public.



*Picture 1 - Motorsport can be dangerous*

### **A.3.4 Team Tutor Duties**

- A.3.4.1 | Each team must designate at least one Team Tutor, who may advise the team on any aspect of the project.
- A.3.4.2 | A Team Tutor must have a contractual relationship with the representing university at the time of registration.
- A.3.4.3 | The Team Tutor shall be considered the highest responsible person of the team in the event of any major incident during the Competition.

- A.3.4.4 | The Team Tutor should accompany the team during the Final Event. If the Team Tutor is unable to attend, responsibility shall be delegated to the Team Leader, who shall act as the highest responsible person of the Team during the Final Event.
- A.3.4.5 | The Team Tutor shall not take part in the development, assembly or manufacture of any component, document or presentation of the prototype.
- A.3.4.6 | The Team Tutor shall represent the university in administrative matters throughout the Competition.
- A.3.4.7 | Any change of Team Tutor, for duly justified reasons, must be communicated as soon as possible to the Competition Organization. Such change shall be subject to an administrative fee.

### **A.3.5 Team Leader Duties**

- A.3.5.1 | Each team must designate one Team Member as its Team Leader.
- A.3.5.2 | The Team Leader shall have the same rights and obligations as the other enrolled students and shall act as the communication link with the Competition Organization and as the official representative of the team.
- A.3.5.3 | Any reassignment of the Team Leader role within a team must be communicated to the Competition Organization as soon as possible. The Team Leader role may only be reassigned to another registered Team Member of the same team and shall be subject to an administrative fee.

### **A.3.6 Replacements**

- A.3.6.1 | **Only role modifications** of Team Members or the Team Leader shall be allowed. Such modifications shall entail the payment of the corresponding administrative fee for management expenses, in accordance with Article A.3.8.1, and shall only be accepted up to 30 days before the first day of the Final Event.
- A.3.6.2 | No Team Member or Team Tutor registration fee shall be cancelled or reimbursed.
- A.3.6.3 | All permitted modifications concerning participating students must be communicated to the Competition Organization by the Team Tutor or the Team Leader through the MotoStudent Platform.

### **A.3.7 Insurances**

- A.3.7.1 | An insurance coverage is mandatory.

- The registered universities must ensure that their students are covered by valid institutional insurance for all activities arising from participation in the MotoStudent International Competition.
- A.3.7.2 Each registered team must hold a valid insurance policy covering the Final Event, including Team Members, Team Leader, Team Tutor and Rider, as well as civil liability for damages to third parties. Such insurance must be valid in Spain and may be subject to documentary verification by the Competition Organization at any time prior to or during the Final Event.
- A.3.7.3 The Final Event will provide the medical and safety services required in accordance with applicable sporting regulations. Such services shall not substitute the insurance obligations of the participating teams or universities.
- A.3.7.4 Each team is solely responsible for the design, manufacture, assembly, maintenance, handling and operation of its prototype. Technical inspection, verification, scrutineering or approval by the Competition Organization shall not be construed as certification, homologation or guarantee of safety, reliability or regulatory compliance beyond the scope of the Competition.
- A.3.7.5 The Competition Organization shall not be liable for any material damage, personal injury, indirect loss or third-party claims arising from the team's participation in the Competition, including but not limited to the development, construction, testing, racing, transport or post-event use of the prototype, including any modification or manipulation carried out by the team.
- A.3.7.6 The team shall indemnify and hold harmless the Competition Organization, its officials and partners against any third-party claims, damages, losses or liabilities arising from the team's participation in the Competition.
- A.3.7.7 Any components supplied as part of the MotoStudent Kit, including engines and/or motors provided under a lease regime, remain the property of the Competition Organization and are placed under the responsibility of the team during their possession and use.
- The team shall be responsible for the proper handling, storage, transport, installation and operation of such components.
- The team's insurance policy must provide adequate coverage for any damage, loss or liability arising from the handling, transport, modification, operation or misuse of these components while under the responsibility of the team.

For components supplied under a lease regime, including engines and / or motors, the insurance coverage must name Fundacion Moto Engineering Foundation as beneficiary or additional insured party.

- A.3.7.8 | The Competition Organization does not guarantee the fitness for purpose or commercial safety of any supplied component.

### **A.3.8 Administrative Procedures and Fees**

- A.3.8.1 | Administrative procedures requested by the teams to the Competition Organization, which are not part of the development of the official milestones and activities of MotoStudent, shall entail a standard administrative fee of € 45 (VAT included) for management expenses.

Examples of such administrative procedures include:

- Issuance of individual certificates or other justification documents (excluding those issued generically by the Competition Organization).
- Issuance of duplicate documentation.
- Modification of team data, including the assignment or change of the Team Leader role, Team Tutor, team name and/or representing university.
- Modification of Rider data not required by the Competition Organization (e.g. replacement due to injury).
- Management of parcels, shipments, customs interaction and storage of prototypes or other team equipment upon arrival at the Competition facilities may be provided upon prior written request. Such local handling services shall entail the standard administrative fee and may additionally be subject to the logistical and operational costs incurred. The Competition Organization shall evaluate the corresponding costs in advance and communicate them to the Team. The services shall only proceed upon written approval by the requesting Team.

## **ARTICLE 4: MANUFACTURING THE MOTORBIKE**

### **A.4.1 Prototype Manufacturing**

- A.4.1.1 | The motorbikes participating in MotoStudent must be conceived, designed and assembled by the students registered in the team, without the direct involvement of professional engineers, race engineers, professional mechanics or other external professionals. Participation in the Competition shall constitute full acceptance of this requirement and of all provisions established in these Competition Rules & Regulations.

#### A.4.2 MotoStudent Kit

A.4.2.1 | The Competition Organization will provide each registered team with a MotoStudent Kit including compulsory components to be installed in the prototype. The technical regulations applicable to the Kit are defined in Sections B, C and D of these Competition Rules & Regulations.

A.4.2.2 | The MotoStudent Kit for the teams registered in the MotoStudent Competition includes, as a minimum, the following components:

- An internal combustion engine (only for the MotoStudent eFuel category).
- A permanent magnet electric motor (only for the MotoStudent Electric category).
- A set of front and rear slick tyres.

The engines and motors supplied by the Competition Organization are provided under a lease-for-use scheme exclusively for the development of the prototype and participation in the MotoStudent Competition. Ownership of the engines shall remain at all times with the Competition Organization. Teams shall be responsible for their proper use, conservation and return in accordance with the procedures established by the Competition Organization (see Annex 6 and Annex 7 for a complete explanation of this procedures).

The tyres supplied to the teams are intended for use during the Competition and shall become property of the team once delivered.

Additional compulsory parts may be included in the MotoStudent Kit at the discretion of the Competition Organization depending on the technical configuration of the Competition. Where applicable, such components supplied by third parties (e.g. IMD) will be specified in the MotoStudent Accepted Parts Sheet.

A.4.2.3 | The Competition Organization will arrange the shipment of the MotoStudent Kit to the facilities indicated by the teams. The Competition Organization will normally cover the shipping cost of the MotoStudent Kit, subject to logistical feasibility and destination conditions.

However, depending on the destination region, logistical constraints or extraordinary transport costs, the Competition Organization reserves the right to establish alternative delivery arrangements or distribution points, which will be communicated to the teams in advance.

Each team shall be responsible for any customs duties, import taxes or similar charges applicable in the destination country, region or territory.



Shipping conditions may vary depending on the geographical location of the team.

- A.4.2.4 Teams shall inform the Competition Organization prior to shipment of any applicable tariff policies, import restrictions or logistical limitations related to the destination of the MotoStudent Kit in order to avoid customs retention of the material.
- A.4.2.5 In certain countries, shipment of specific components included in the MotoStudent Kit may be restricted. In such cases, the Competition Organization will contact the team to determine an alternative solution for the delivery or management of those components.
- A.4.2.6 The components included in the MotoStudent Kit shall be used exclusively for the development of the team's prototype and for participation in the MotoStudent Competition. Any other use is strictly prohibited.
- A.4.2.7 The components included in the MotoStudent Kit are listed in the corresponding articles of the category-specific sections of these Competition Rules & Regulations.
- A.4.2.8 The components included in the MotoStudent Kit will incorporate an identification mark. This marking must remain intact and visible, as it will be verified by the Technical Staff of the Competition Organization during the Final Event. In the event of damage or alteration of such markings, teams shall contact the Competition Organization to determine the appropriate procedure.
- A.4.2.9 It is prohibited to cover, remove or alter any identification mark of the components supplied by the Competition Organization.
- A.4.2.10 Any technical question regarding the components included in the MotoStudent Kit shall be addressed through the Competition Organization. Direct contact between participating teams and supplying companies regarding technical matters related to the MotoStudent Kit is prohibited.

## **ARTICLE 5: PARTICULARITIES FOR NON-EUROPEAN TEAMS**

### **A.5.1 Transport and Import of the Prototype**

- A.5.1.1 Shipments and imports of prototypes and team equipment must comply with all applicable transportation, import and export requirements established by Spanish laws and regulations, as well as the regulations of the country of origin of the shipment.

- A.5.1.2 Teams are responsible for arranging transport, customs clearance and all administrative procedures related to the shipment of prototypes and team equipment. Teams are strongly recommended to manage transport and customs procedures through a specialized logistics or customs company.
- A.5.1.3 Under no circumstances shall the Competition Organization or the management company of the Speed Circuit act as consignee, importer of record or customs representative for the transport of prototypes or team equipment.
- A.5.1.4 Teams participating from outside the European Union are strongly recommended to use temporary import procedures (such as ATA Carnet or equivalent customs mechanisms) for the transport of prototypes and equipment to and from the Final Event.
- A.5.1.5 The prototype and equipment, except the engine and/or motor, brought by the teams for participation in the Final Event shall normally be treated as temporarily imported goods for the duration of the event. Teams must ensure that the selected customs procedure allows the subsequent re-export of such equipment after the Final Event.
- A.5.1.6 Components of the MotoStudent Kit supplied under a lease-for-use scheme (such as engines) remain the property of the Competition Organization and must be returned to the Competition Organization in accordance with the procedures established in these Competition Rules & Regulations. Teams must ensure that the customs procedures used for their transport allow their return to the Competition Organization when required.
- A.5.1.7 Components of the MotoStudent Kit supplied for permanent use by the team (such as tyres or other non-returnable components) may be treated as standard exports or imports according to the applicable customs regulations.
- A.5.1.8 The Competition Organization shall not be responsible for customs procedures, duties, taxes, storage costs, delays, administrative requirements or any other issues related to shipments arranged by the teams.
- A.5.1.9 The Competition Organization may accept package deliveries for the Final Event up to fifteen (15) days prior to the start date of the Final Event, subject to prior coordination with the Organization.
- Such reception of shipments shall be considered an exceptional logistical facilitation and shall not imply that the Competition Organization acts as carrier, consignee, storage provider or logistics operator.

Teams remain fully responsible for the shipment, transport conditions, customs procedures and any associated risks. The Competition Organization shall not be liable for loss, damage, delays or storage costs related to such shipments.

Where applicable, the reception and handling of shipments may be subject to administrative or logistical management fees in accordance with the applicable provisions of these Competition Rules & Regulations.

The Competition Organization reserves the right to refuse or return any shipment that has not been previously coordinated.

A.5.1.10 For shipments arriving before the week of the Final Event, the delivery information shall be as follows:

- **Addressee:** Team or University name
- **Address:** TechnoPark MotorLand, Ed. Dr. Joaquin Repolles, Oficina 1, 44600 Alcañiz – Teruel (Spain)
- **Contact phone:** Team's contact phone

A.5.1.11 For shipments arriving during the week of the Final Event, the delivery information shall be as follows:

- **Addressee:** Team or University name
- **Address:** MotorLand Aragón – Circuito de Velocidad, Ctra. TE-V-7033 km. 1, 44600 Alcañiz – Teruel (Spain)
- **Contact phone:** Team's contact phone

## A.5.2 Visa

A.5.2.1 Registration documents issued within the MotoStudent Competition, as well as other participation accreditations provided by the Competition Organization, may be used by the teams as supporting documentation when applying for a visa, if required. Upon request, the Competition Organization may provide a certificate confirming the registration of the team in the MotoStudent Competition. The issuance of such certificates may entail the applicable administrative fees established in Article A.3.8 of these Competition Rules & Regulations.

The Competition Organization strongly recommends that non-EU teams verify visa requirements and arrange all travel documentation as early as possible in order to avoid delays or unforeseen issues.

Information regarding entry requirements for Spain can be found on the official website of the Ministry of Foreign Affairs of the Government of Spain: [www.exteriores.gob.es](http://www.exteriores.gob.es).

- A.5.2.2 The Competition Organization does not grant visas and will not intervene with any authority, embassy or consulate regarding visa applications of team members. The responsibility for obtaining the necessary travel documentation lies entirely with the participants.

## ARTICLE 6: COMPETITION CALENDAR

### A.6.1 Official Deadlines and Calendar

- A.6.1.1 The MotoStudent International Competition takes place over three academic semesters. During this time, several milestones must be completed by all teams as part of the evaluation process of the Competition. The following calendar presents the main deadlines of the Competition.

| Period                                                                       | DATE<br>Start - End                             | 2026 |     |     |     |     |     |     |     |     |     |     |     | 2027 |     |     |     |     |     |     |     |     |     |
|------------------------------------------------------------------------------|-------------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|                                                                              |                                                 | Jan  | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Jan  | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct |
| Team Registration                                                            | From XX April to 30 days before the Final Event |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| Additional team members and team tutors registration, and role modifications | From XX April to 30 days before the Final Event |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 1 <sup>st</sup> milestone, Team plan                                     | 01/11 - 30/11                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 2 <sup>nd</sup> milestone, Product Definition                            | 01/02 - 28/02                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MSE 1 <sup>st</sup> delivery, Electric and Battery Pack Schemes              | 01/03 - 31/03                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 3 <sup>rd</sup> milestone, Innovation                                    | 01/04 - 30/04                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 4 <sup>th</sup> milestone, Prototype CAD and drawings                    | 01/05 - 31/05                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 5 <sup>th</sup> milestone, Economical plan                               | 01/05 - 31/05                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 6 <sup>th</sup> milestone, Pitch presentation                            | 01/06 - 30/06                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MSE 2 <sup>nd</sup> delivery, Electrical Precheck                            | 01/07 - 31/07                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| Rider Eligibility Declaration                                                | 01/08 - 31/08                                   |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| MS1 7 <sup>th</sup> milestone, Final Event Presentation                      | Autumn 2027 during Final Event                  |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |
| Final Event (MS2 tests)                                                      | Autumn 2027 during Final Event                  |      |     |     |     |     |     |     |     |     |     |     |     |      |     |     |     |     |     |     |     |     |     |

|                   |  |
|-------------------|--|
| Rider Eligibility |  |
| MSE delivery      |  |
| MS1 milestone     |  |
| Registry process  |  |
| Final Event       |  |

Picture 2 – Current Edition calendar

### A.6.2 Team Registration Period

- A.6.2.1 The Team Registration period opens on the date defined in Article A.1.6.2 and will remain open until all available places are filled. If vacancies remain, registration will close 30 days before the Final Event.

### A.6.3 Additional Team Members and Team Tutors registration

- A.6.3.1 The registration period for additional Team Members and Team Tutors remains open from the opening of the Team Registration period defined in Article A.1.6.2 until one month prior to the first day of the Final Event.

#### A.6.4 MS1 Milestones

- A.6.4.1 | The *MS1 1<sup>st</sup> milestone, **Team Plan***, aims to provide a general overview of the team composition, team objectives and internal organization.
- The submission period for this milestone is from **1 November 2026** to **30 November 2026**.
- A.6.4.2 | The *MS1 2<sup>nd</sup> milestone, **Product Definition***, requires the teams to describe to the jury the different technical solutions selected for the prototype, including the calculations used to ensure the safety of the theoretical design.
- The submission period for this milestone is from **1 February 2027** to **28 February 2027**.
- A.6.4.3 | The *MS1 3<sup>rd</sup> milestone, **Innovation***, requires teams to introduce improvements or new concepts in the prototype through new methods, ideas or products.
- The submission period for this milestone is from **1 April 2027** to **30 April 2027**.
- A.6.4.4 | The *MS1 4<sup>th</sup> milestone, **Prototype CAD and Drawings***, requires the teams to submit a 3D CAD model of their prototype together with a defined set of technical drawings.
- The submission period for this milestone is from **1 May 2027** to **31 May 2027**.
- A.6.4.5 | The *MS1 5<sup>th</sup> milestone, **Economical Plan***, requires the teams to present the total amount of funds obtained, the project expenses, the estimated total value of the prototype and the projected financial sustainability of the project.
- The submission period for this milestone is from **1 May 2027** to **31 May 2027**.
- A.6.4.6 | The *MS1 6<sup>th</sup> milestone, **Pitch Presentation***, must showcase the development of the project during the MotoStudent International Competition.
- The submission period for this milestone is from **1 June 2027** to **30 June 2027**.
- A.6.4.7 | The *MS1 7<sup>th</sup> milestone, **Final Event Presentation***, takes place during the **Final Event (autumn 2027)**, where teams present their project to the jury. Each team must explain the main project decisions, the development process and the innovation presented. The jury may conduct a question round at the team box to further evaluate the project.

#### A.6.5 MSE Deliveries

- A.6.5.1 | The *MSE 1<sup>st</sup> delivery, **Electric and Battery Pack Schemes***, is a specific requirement for teams participating in the MotoStudent Electric category. Teams must submit the electrical schematics of the prototype together with the mechanical and electrical configuration of the battery pack.

The submission period for this delivery is from **1 March 2027** to **31 March 2027**.

- A.6.5.2 | The *MSE 2<sup>nd</sup> delivery, **Electrical Precheck***, requires teams to review each article of Chapter D of these Competition Rules & Regulations and provide a brief explanation, supported by graphical evidence, demonstrating compliance.

The submission period for this delivery is from **1 July 2027** to **31 July 2027**.

#### **A.6.6 Rider Eligibility Declaration**

- A.6.6.1 | The ***Rider Eligibility Declaration*** requires each team to submit the sporting background and achievements of their rider in order to allow the Organization to verify compliance with the sporting requirements established in the Rules & Regulations.

The submission period for this declaration is from **1 August 2027** to **31 August 2027**.

Submission of a valid Rider Eligibility Declaration within the defined period is a mandatory condition for participation in the dynamic activities of the Final Event.

**Teams that fail to submit this declaration will not be authorized to participate in any on-track activity.**

#### **A.6.7 Final Event**

- A.6.7.1 | The ***Final Event*** will take place during **autumn 2027** (dates to be confirmed by the Competition Organization and FIM) at the MotorLand Aragón Speed Circuit. The specific dates and timetable for tests and activities will be published by the Competition Organization.

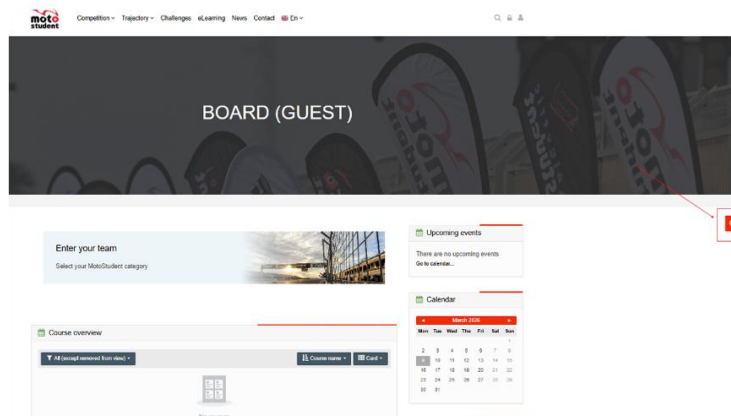
### **ARTICLE 7: COMMUNICATION BETWEEN TEAMS AND ORGANIZATION**

#### **A.7.1 Publication of questions: Frequently Asked Questions (FAQ)**

- A.7.1.1 | By submitting a question to the Competition Organization, teams accept that the Organization may partially or totally reproduce the question and the corresponding official response in the FAQ Database, which is available to all

registered teams through the official MotoStudent platforms. Accessible on the following URL or clicking in the following icon:

<https://www.motostudent.com/blocks/helpdesk/new.php>



A.7.1.2 The [FAQ Database](#) collects the most common questions regarding the MotoStudent Competition submitted by the teams and will be updated periodically by the Competition Organization.

A.7.1.3 Registered teams must regularly review the updates published in the FAQ Database. The clarifications and interpretations published through the FAQ Database shall be considered official and binding for the participating teams.

However, if a clarification implies a modification, addition or change to the content of these Competition Rules & Regulations, such change shall be formalized by the Competition Organization through the corresponding official amendment, Technical Note or updated version of the applicable document.

A.7.1.4 Teams are expected to consult the FAQ Database before submitting new questions in order to avoid duplicate queries already addressed by the Organization.

A.7.1.5 All technical or regulatory questions related to the Competition should be submitted through the official communication channels established by the Competition Organization.

## A.7.2 Types of Questions

A.7.2.1 The Competition Organization will answer questions that are not already addressed in the Competition Rules & Regulations or in the FAQ Database.

A.7.2.2 The Competition Organization reserves the right not to answer questions whose response is clearly reflected in the Competition Rules & Regulations, the FAQ Database, the registration documentation or any other official communication issued by the Competition Organization.

### A.7.3 Consultation Procedure

- A.7.3.1 All questions addressed to the Competition Organization must be submitted through the platform established by the Organization for this purpose and must include the following information:
- Full name of the person submitting the question.
  - Category (MotoStudent Electric / MotoStudent eFuel).
  - Bike Number.
  - Team Name.
  - Relevant article(s) of the Competition Rules & Regulations or FAQ reference number.
  - Question: Fill in the Summary and Message describing your question.
- A.7.3.2 Teams are recommended not to attach pictures, drawings or files exceeding 500 KB in size.
- A.7.3.3 The Competition Organization will not resolve questions by telephone or ordinary email. All questions must be submitted and addressed through the consultation procedure defined in this article (ticketing system through the MotoStudent Platform).
- A.7.3.4 The Competition Organization reserves the right to group similar questions and publish the corresponding clarifications through the FAQ Database.

### A.7.4 Response Time

- A.7.4.1 The Competition Organization will respond to questions as soon as reasonably possible. Depending on the complexity of the matter or the need for internal or external consultation, the estimated response time may extend up to fifteen (15) calendar days.

## ARTICLE 8: OTHER COMPETITIONS

### A.8.1 Other Competitions

- A.8.1.1 These Competition Rules & Regulations have been developed exclusively for the current Edition of the MotoStudent International Competition. Therefore, prototypes designed and manufactured in accordance with these regulations are not required to comply with the technical or sporting regulations of any other motorcycle racing competition outside MotoStudent. **The Competition Organization does not guarantee** that prototypes built under these regulations comply with the requirements of other competitions **and shall not be responsible** for the use that teams may make of their prototypes in any competition or activity outside the MotoStudent International Competition.



# **SECTION B**

## **GENERAL TECHNICAL REGULATIONS**

## ARTICLE 1: TECHNICAL REQUIREMENTS OF THE PROTOTYPE AND RESTRICTIONS

### B.1.1 Introduction to Technical Regulations

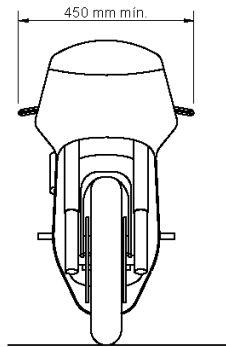
- B.1.1.1 **ANY MATTER NOT EXPRESSLY PERMITTED AND SPECIFIED IN THESE TECHNICAL REGULATIONS SHALL BE CONSIDERED PROHIBITED.** Teams may submit a request for clarification or authorization to the Competition Organization. Any corresponding decision will be communicated through an official Technical Note.
- B.1.1.2 The design, manufacture and assembly of the prototype must comply with these Competition Rules & Regulations in order to be eligible to participate in the Competition.
- B.1.1.3 Prototypes must comply with all applicable technical requirements throughout the entire Final Event. The Competition Organization reserves the right to verify compliance at any time.
- B.1.1.4 Any non-compliance must be corrected and re-inspected before the prototype is allowed to participate in any test or activity during the Final Event.
- B.1.1.5 The rules contained in this section apply equally to both MotoStudent eFuel and MotoStudent Electric categories, unless specific requirements are explicitly indicated.
- B.1.1.6 All eligible components are defined in the *MotoStudent Approved Parts List*, which also identifies mandatory components. Any component not included in the corresponding list shall not be permitted.
- B.1.1.7 Where no Approved Parts List is defined for a component or system, its design and selection shall be free, provided that it complies with these Rules & Regulations.
- B.1.1.8 Teams are solely responsible for the design, construction and safe operation of their prototype. Technical inspections carried out by the Competition Organization do not constitute certification of safety, reliability or regulatory compliance of the prototype.

## ARTICLE 2: GENERAL DESIGN REQUIREMENTS

### B.2.1 Dimensions

- B.2.1.1 The dimensions of the prototype are free, except for the requirements specified in Article B.2.1.

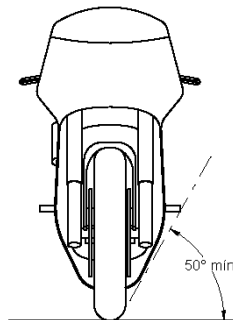
- B.2.1.2 The minimum width between the outer ends of the semi-handlebars must be 450 mm.



Picture 3 - Minimum prototype width

- B.2.1.3 The prototype must allow a minimum lean angle of 50° without any part of the motorcycle (except the tires) touching the ground.

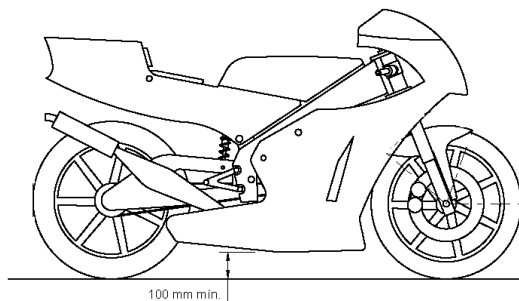
This measurement shall be performed with the prototype unloaded (without rider), but fully equipped and with all fluids required for normal operation.



Picture 4 - Minimum tilt angle

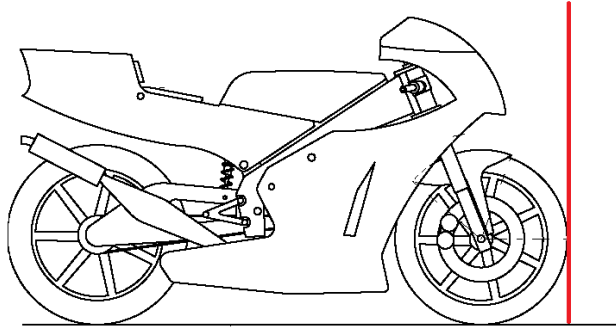
- B.2.1.4 The minimum ground clearance of the prototype in upright position must be at least 100 mm in static condition.

This measurement shall be performed with the prototype unloaded (without rider), fully equipped for operation and with the tires inflated to 2 bar.



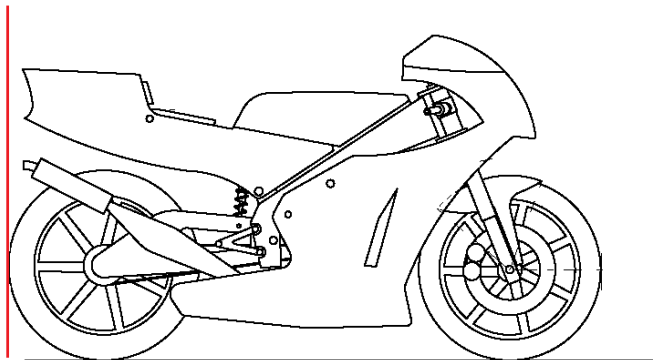
Picture 5 - Minimum ground clearance distance

- B.2.1.5 | No element of the prototype may protrude beyond the vertical plane tangential to the external circumference of the front tire.



Picture 6 - Front vertical line

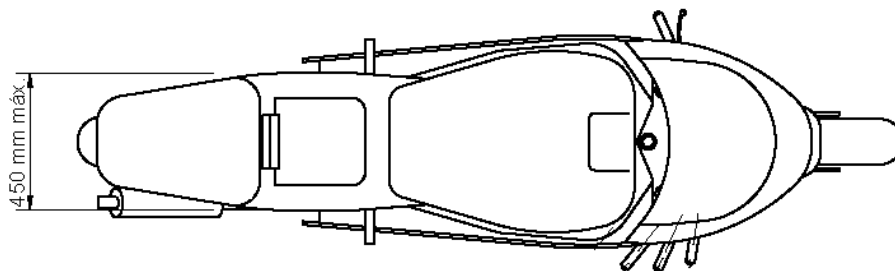
- B.2.1.6 | No element of the prototype may protrude beyond the vertical plane tangential to the external circumference of the rear tire.



Picture 7 - Rear vertical line

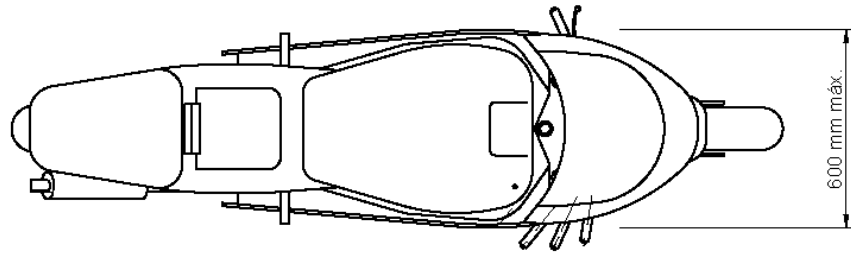
- B.2.1.7 | A minimum clearance of 15 mm must be maintained between the tire tread and any part of the prototype along the outer circumference of the tire, in any suspension position or geometry setup.

- B.2.1.8 | The maximum seat and tail section width must not exceed 450 mm.  
No other element of the prototype may exceed this width from the seat to the tail section, except for the exhaust system in MotoStudent eFuel category prototypes.



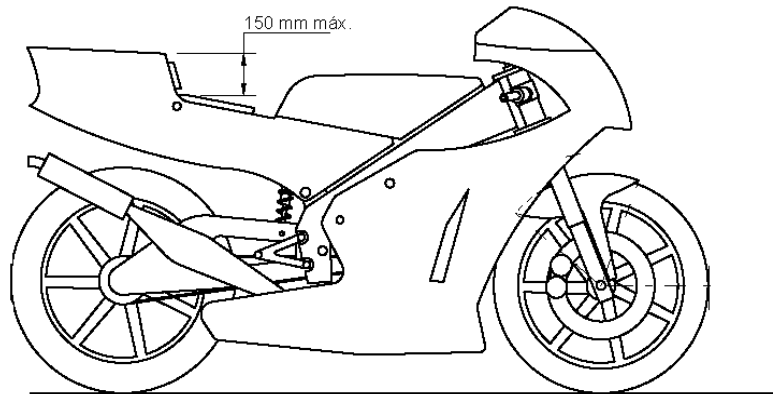
Picture 8 - Maximum seat width

- B.2.1.9 | The maximum fairing width must not exceed 600 mm.



Picture 9 - Prototype maximum fairing width

- B.2.1.10 | The maximum height difference between the seat surface and the highest point of the tail section must not exceed 150 mm.



Picture 10 - Prototype maximum seat height difference

## B.2.2 Weight

- B.2.2.1 | The minimum total weight of the prototype without the rider shall be 85 kg for both categories, including all fluids necessary for the correct operation of the prototype.
- B.2.2.2 | The weight of the complete prototype shall not fall below the minimum weight defined in Art. B.2.2.1 at any time during the Final Event. The Technical Staff may check the weight of any prototype at any time during the Final Event.
- B.2.2.3 | The use of ballast to reach the minimum prototype weight is permitted. If ballast is used, it must be declared to the Technical Staff during scrutineering.
- B.2.2.4 | Any ballast used must be securely fixed to the chassis in such a way that it cannot become detached in the event of a crash or fall (e.g. bolted fixation).
- Ballast must be attached using mechanical fasteners.

## ARTICLE 3: FRAME

### B.3.1 Design

B.3.1.1 | There are no restrictions on the design or materials of the frame, swingarm or subframe, provided that the resulting structure complies with these Competition Rules & Regulations.

Teams are solely responsible for the structural integrity of the frame and chassis components.

B.3.1.2 | In case of doubt regarding the structural safety of the proposed frame design, the Competition Organization may request a safety justification report. Such report may include finite element analysis (FEA), simulations or other equivalent validation tests.

B.3.1.3 | The whole structure of the prototype must be designed so that the engine, fuel system, intake system and all electrical and electronic control and/or safety systems, including battery packs, are adequately protected in the event of a crash or fall, particularly in the case of a lateral impact.

B.3.1.4 | The use of commercial chassis components is not permitted, including modified commercial main frames, subframes or swingarms.

The chassis must be a self-designed prototype.

For the purposes of these Technical Regulations, the term chassis refers to the main structural assembly of the prototype and includes, at minimum, the main frame, the subframe and the swingarm.

B.3.1.5 | Swingarm spindles, wheel spindles and the steering shaft must be manufactured only from steel alloys.

The use of aluminium alloys, magnesium alloys, titanium alloys, composite materials or any other non-steel material is not permitted for these components.

This restriction does not apply to bolts, nuts, washers, circlips or other standard fasteners.

B.3.1.6 | The engine / motor must not be used as a structural element of the chassis.

The chassis shall be designed so that the main frame, subframe and swingarm form the load-bearing structural system of the prototype.

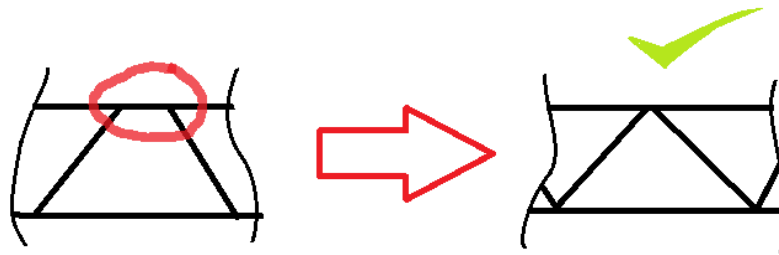
The engine / motor may be mounted to the chassis and may transmit its own functional loads through its mounting points during normal operation, but it must not be used to complete the structural continuity of the chassis or to replace any primary load path between the main structural elements of the prototype.

In particular, the engine / motor must not be used as the element that structurally links the steering area of the prototype to the swingarm pivot area or to the rear structural assembly.

### **B.3.2 Welding and Structural Joints**

**B.3.2.1** For trellis or lattice type structures, nodes must ensure proper triangulation and effective load transfer between structural members in order to provide adequate structural stiffness.

A small clearance between tubes may be allowed to facilitate welding or joining processes. However, the node must ensure proper structural continuity and load transmission between the connected members.



*Picture 11 - Example of correct triangulation*

**B.3.2.2** The joining of structural elements may be carried out using any suitable manufacturing method (e.g. welding, brazing, bolting, bonding or additive manufacturing), provided that the resulting structure ensures adequate strength and structural integrity.

All welding operations must ensure adequate structural integrity of the chassis. The team is solely responsible for the quality and safety of all welded joints. The Competition Organization reserves the right to request additional justification if the structural safety of a welded assembly is in doubt.

### **B.3.3 Crash Protectors**

**B.3.3.1** The use of chassis crash protectors (frame sliders) is mandatory on both sides of the prototype. These elements must protect the chassis and propulsion system from lateral contact with the ground in the event of a crash or fall.

B.3.3.2 | Crash protectors must be made of nylon, fibre-reinforced materials or other materials of similar hardness capable of absorbing impact without damaging the track surface.

They must be securely fixed to the frame or to a structural part of the chassis. Fixation using cable ties, adhesives or similar temporary fastening methods is not permitted.

Crash protectors may be located inside or outside the fairing provided they effectively protect the frame and propulsion system.



*Picture 12 - Example of crash protector*

B.3.3.3 | Additional crash protection may be installed on wheel axles, handlebar ends or other suitable locations, provided that these elements do not create sharp edges and contribute to protecting the motorcycle structure during a fall.

#### **ARTICLE 4: FAIRING**

##### **B.4.1 General Requirements**

B.4.1.1 | All edges and finishes of the fairing must be rounded with a minimum external radius of 1 mm.

B.4.1.2 | The fairing must not cover the rider laterally, except for the forearms when the rider is in the minimum aerodynamic resistance position.

B.4.1.3 | There are no restrictions regarding the manufacturing materials of the fairing.

B.4.1.4 | Aerodynamic devices (e.g. wings, fins, ducts or bulges) protruding from the fairing or bodywork are permitted, provided they do not exceed the maximum dimensions established in these Rules & Regulations and are securely fixed to the fairing or structural bodywork elements.



Aerodynamic ducts integrated into the fairing for cooling or airflow management are permitted.

- B.4.1.5 Any leading edge of an aerodynamic device must have a minimum external radius of 2.5 mm.

Aerodynamic devices must not create sharp edges or projections that could endanger the rider or other participants.

- B.4.1.6 Aerodynamic devices must not deflect more than 10 mm at any point under a vertical downward point load of 50 N.

For the purposes of this requirement, vertical downward means the direction of the vertical axis of the prototype relative to the ground.

The 50 N load may be applied at any relevant point of the aerodynamic device, particularly at the location where the greatest deflection is expected.

- B.4.1.7 Mobile aerodynamic devices, including active aerodynamic systems, are permitted, provided that their operation does not compromise rider safety or the safety of other participants.

If such devices require activation, they must be actuated only by direct rider input through a dedicated rider-operated control. Activation through the dashboard, brake controls, sensors, electronic logic or control algorithms is not permitted.

All dimensional and safety requirements established in these Rules & Regulations shall be respected in any position or configuration of the aerodynamic device during operation. In case of doubt, the decision of the Technical Staff of the Organization shall be final.

- B.4.1.8 Aerodynamic devices must be designed and positioned so that they cannot contact the ground when the motorcycle reaches the minimum lean angle defined in Art. B.2.1.3.

Aerodynamic devices must be securely mounted so that they cannot become detached from the prototype during normal operation or in the event of a crash.

## **B.4.2 Lower Fairing**

- B.4.2.1 The lower fairing must be designed and manufactured to retain, in the event of an incident, at least half of the total volume of liquids of the prototype (fuel excluded), with a minimum containment capacity of 2.5 liters.

B.4.2.2 | The lower fairing must include a drainage hole with a diameter of 25 mm located at the lowest point of the fairing. This hole must remain watertight closed with a plug in dry conditions and may only be opened in wet conditions.

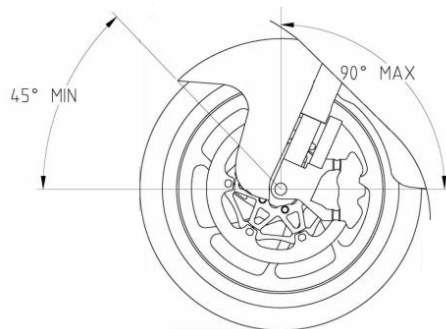
B.4.2.3 | The lower fairing may incorporate aerodynamic shaping, ducts or flow-deflecting features intended to manage airflow or water spray around the rear wheel or propulsion system, provided that these elements remain integrated within the fairing geometry and comply with all dimensional and safety requirements established in these Rules & Regulations.

### **B.4.3 Mudguards**

B.4.3.1 | The use of front and rear mudguards is mandatory.

B.4.3.2 | The front mudguard shall not cover more than 135° of the wheel circumference, measured from the rearmost point of the tyre. The origin of the angle shall be the horizontal line passing through the wheel axle.

Mudguard mounting points attached to the front fork, fork legs or brake disc covers may exceed this limit; however, under no circumstances shall they protrude more than 10 mm beyond the lateral projection of the front brake disc.



*Picture 13 - Prototype front mudguard*

B.4.3.3 | The rear wheel shall not be covered over more than 180° of its circumference by the rear assembly of the prototype.

B.4.3.4 | Mudguards may incorporate aerodynamic shaping, airflow channels or spray deflectors intended to manage airflow, cooling or water spray generated by the rotating wheels, provided that these elements remain integrated within the mudguard structure and comply with all dimensional and safety requirements established in these Rules & Regulations.

B.4.3.5 | The use of transparent materials for the purpose of circumventing the coverage limits established in this Article is prohibited.

#### **B.4.4 Protection Against Trapping**

- B.4.4.1 | If the swingarm design does not cover the lower section of the drive chain or transmission belt, a guard must be installed to prevent trapping between the lower drive chain run and the rear sprocket (or equivalent transmission element).
- B.4.4.2 | This protection device (commonly known as “shark fin”) must be securely fixed to the swingarm or rear structure and must be made of rigid material, capable of maintaining its shape and protective function during normal use and foreseeable contact.
- B.4.4.3 | The guard must extend at least to the rear axle height and cover the lower rear quadrant of the sprocket in order to effectively prevent trapping.



*Picture 14 - Example of “Shark fin”*

### **ARTICLE 5: HANDLEBARS AND FOOTRESTS**

#### **B.5.1 Handlebars and Manual Controls**

- B.5.1.1 | Handlebars or semi-handlebars must be manufactured from steel or aluminium alloys. Welded or repaired handlebars are strictly forbidden.  
  
The use of magnesium alloys, titanium alloys or composite materials is not permitted.
- B.5.1.2 | Handlebar or semi-handlebar mounting supports must be designed to minimize the risk of fracture or detachment in the event of a crash.
- B.5.1.3 | All exposed edges of the handlebars and their mounting components must have a minimum external radius of 2 mm.
- B.5.1.4 | The throttle control must incorporate an automatic return mechanism that closes the throttle when the rider releases the grip.
- B.5.1.5 | The clutch lever (if installed) is free in design, provided that the distance between the pivot point and the outer end does not exceed 200 mm. The lever must

terminate in a ball end with a minimum diameter of 16 mm. The ball may be flattened but must maintain rounded edges and a minimum thickness of 14 mm.

- B.5.1.6 The installation of a front brake lever protector is mandatory. This device must protect the brake lever from accidental activation in the event of contact with another prototype or external object.



*Picture 15 - Examples of front brake lever protections*

- B.5.1.7 The TSMS, electric ignition or start button must be located on the handlebar and must be easily accessible to the rider at all times. Additional rider control switches or actuators may also be installed on the handlebar, provided that they do not interfere with the safe operation of the motorcycle.

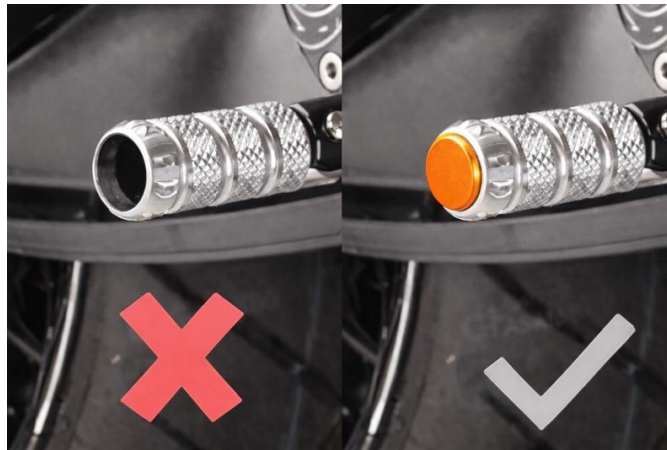
## **B.5.2 Footrests and Their Controls**

- B.5.2.1 Footrests may be fixed or folding type ("folding in case of crash"). If folding footrests are used, they must be fitted with a return device that restores them automatically to their normal position and prevents unintended folding during riding. In resting position without the rider, the footrests must remain in the unfolded position.

- B.5.2.2 The end of each footrest must have a continuous rounded end with a minimum spherical radius of 8 mm.

- B.5.2.3 A protective cap must be installed at the outer end of each footrest. This cap must be made of aluminium, plastic, nylon or another material with equivalent mechanical resistance and must be permanently fixed. The protective surface must cover the widest possible area of the footrest end.

Open-ended footrests without such protective cap are not permitted.



*Picture 16 Footrest ends. Left, footrest with hole (not allowed). Right, footrest with cap (allowed).*

- B.5.2.4 | Footrests must incorporate side protection devices designed to prevent the rider's boot from coming into contact with moving elements such as the drive chain, rear sprocket or rear tyre.

These devices may take the form of heel guards or protective plates integrated into the footrest assembly and must not introduce any safety risk. Such elements may also contribute to airflow management around the rear wheel area.

- B.5.2.5 | The rear brake pedal design is free.

- B.5.2.6 | The gear shift pedal design (if installed) is free.

- B.5.2.7 | Footrest mounting systems may include adjustable mechanisms intended to adapt the riding position to the rider's ergonomics, provided that such systems are securely fixed and do not introduce structural weakness or risk of unintended movement during riding.

## ARTICLE 6: BRAKE SYSTEM

### B.6.1 Allowed Type of Brake System

- B.6.1.1 | The prototype must be equipped with a hydraulic-driven disc brake system on both the front and rear wheel.

A single-disc or double-disc configuration is permitted on the front wheel.

- B.6.1.2 | Brake systems must be designed and installed following safe engineering practices and must provide reliable braking performance under all operating conditions.

### B.6.2 Command and Control

B.6.2.1 | Combined brake systems are not allowed. The front and rear braking systems must operate independently.

B.6.2.2 | The front brake must be operated by a hand lever located next to the throttle grip on the right handlebar.

B.6.2.3 | The rear brake actuation system is free.

The rear brake may be operated by:

- a foot pedal located near the right footrest, or
- a hand lever located on the left handlebar, or
- a thumb brake control located on the left handlebar,

but only one of these actuation systems may be used in each prototype.

The complete brake system must comply with all applicable safety and technical requirements of these Rules & Regulations.

### **B.6.3 Brake Discs**

B.6.3.1 | Brake discs are free choice provided they comply with the requirements of this Article.

B.6.3.2 | Brake discs must be manufactured from steel alloys. The use of carbon or ceramic brake discs is strictly prohibited.

B.6.3.3 | Brake disc carriers may be manufactured from aluminium, magnesium or steel alloys. Composite materials are not permitted.

B.6.3.4 | The use of internally ventilated brake discs is prohibited.

B.6.3.5 | Brake discs must be installed on the front wheel / rim and on the rear wheel / rim in accordance with the selected brake configuration.

B.6.3.6 | The installation of spacers between the brake disc and the wheel hub is permitted.

### **B.6.4 Brake Callipers**

B.6.4.1 | Brake callipers are free choice, provided that they are manufactured from aluminium or steel alloys.

The use of magnesium alloys, titanium or composite materials for brake callipers is not permitted.

- B.6.4.2 | The rear brake calliper must be mounted using at least one dedicated anchoring element directly fixed to the swingarm.
- B.6.4.3 | The rear calliper may be mounted on a floating support system positioned on the rear axle, provided that at least one dedicated fixing point positively connects the support to the swingarm, as indicated in Art. B.6.4.1.
- B.6.4.4 | Brake pads are free choice.
- B.6.4.5 | Brake pad retaining pins may be replaced and quick-change systems are permitted.
- B.6.4.6 | Supplementary cooling ducts directing fresh air to the brake callipers are not permitted.
- Brake disc covers, brake calliper covers or any additional device intended to cover or partially cover the brake discs or brake callipers for cooling or aerodynamic purposes are not permitted.

#### **B.6.5 Brake Master Cylinders**

- B.6.5.1 | When specified by the Competition Organization, the use of brake master cylinders included in the *MotoStudent Approved Parts List* shall be mandatory for both the front and rear wheel.
- B.6.5.2 | Replacement of the front brake master cylinder activation lever requires prior authorization from the Competition Organization. Authorized replacement parts will be specified in the *MotoStudent Approved Parts List*.
- B.6.5.3 | Modification of any approved brake lever for the front brake master cylinder is prohibited.

#### **B.6.6 Brake Lines**

- B.6.6.1 | Brake hoses are free choice.
- B.6.6.2 | Front brake hoses must pass in front of the lower steering yoke (lower triple clamp).
- B.6.6.3 | Quick connectors on brake hoses are permitted.
- B.6.6.4 | Sensors for research or telemetry purposes (e.g. brake pressure or temperature sensors) may be installed in the brake system provided that:
- they do not modify the hydraulic circuit layout beyond the use of manufacturer-approved fittings

- they are securely mounted and protected
- they do not introduce leakage risk or interfere with the safe operation of the braking system

In case of doubt, the decision of the Technical Staff of the Competition Organization shall be final.

#### **B.6.7 ABS System**

B.6.7.1 | The use of anti-lock braking systems (ABS) is prohibited.

#### **B.6.8 Brake Fluid**

B.6.8.1 | Brake fluid is free choice.

B.6.8.2 | Brake fluid specifications must comply with or exceed the recommendations of the brake system manufacturer.

### **ARTICLE 7: SUSPENSION AND DAMPERS**

#### **B.7.1 General aspects**

B.7.1.1 | Suspension systems are free in design and configuration, provided they comply with the requirements established in this Article.

The suspension system includes all components whose function is to provide wheel travel, damping, springing or ride-height variation of the prototype, including the fork, rear shock absorber, linkages and any auxiliary mechanism acting on them.

B.7.1.2 | Active suspension systems, semi-active suspension systems and any system capable of modifying the suspension behaviour or ride height automatically during riding by means of electronic, electromechanical, hydraulic or pneumatic control are not permitted.

Holeshot devices, launch devices, locking devices or any system intended to temporarily reduce suspension travel, lower the ride height or alter the normal suspension position for launch or riding purposes are not permitted.

B.7.1.3 | Suspension adjustments may only be performed manually by mechanical or hydraulic means (e.g. spring preload, damping adjustment or ride height adjustment). Automatic or electronically controlled adjustments during operation are not permitted.



- B.7.1.4 | Inertial or dynamic devices intended to influence suspension behaviour, such as mass dampers, tuned mass systems, inerters or equivalent mechanisms, are considered suspension systems and are not permitted unless explicitly authorized by the Competition Organization.
- B.7.1.5 | The Competition Organization reserves the right to reject the participation of any prototype whose suspension system is considered unsafe, structurally inadequate or incompatible with safe track operation.
- B.7.1.6 | Suspension systems must ensure adequate wheel control and structural integrity under normal racing conditions and during potential crash scenarios.

## **B.7.2 Front Suspension**

- B.7.2.1 | Any front suspension architecture is permitted, including but not limited to conventional telescopic forks, inverted forks, telelever, duolever, front swingarm systems or other alternative designs.
- Both commercially available and self-designed suspension systems are permitted.
- B.7.2.2 | The front suspension assembly may incorporate mechanical or hydraulic adjustment systems such as spring preload, compression damping, rebound damping or ride height adjustment. Suspension settings may only be adjusted when the prototype is stationary. No adjustment of suspension behaviour or ride height is permitted while riding.

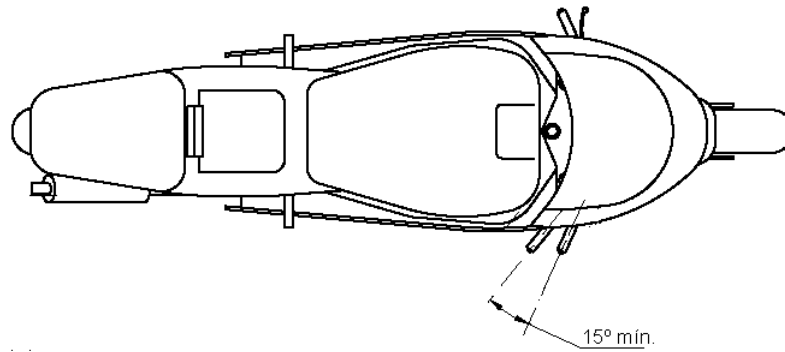
## **B.7.3 Rear Suspension**

- B.7.3.1 | Any rear suspension architecture is permitted. Both commercially available and self-designed suspension systems are allowed.
- B.7.3.2 | The rear suspension assembly may incorporate mechanical or hydraulic adjustment systems such as spring preload, compression damping, rebound damping or ride height adjustment. Suspension settings may only be adjusted when the prototype is stationary. No adjustment of suspension behaviour or ride height is permitted while riding.

# **ARTICLE 8: STEERING SYSTEM**

## **B.8.1 Restrictions and Geometry**

- B.8.1.1 | The steering system must allow a minimum steering angle of 15° on each side of the longitudinal axis of the prototype, measured at the steering axis.



Picture 17 - Minimum turning angle

- B.8.1.2** | The steering angle must be limited by dedicated steering stops installed on both sides of the steering system. These stops must be made of nylon, aluminium or materials of similar hardness.

Steering stops may contact the frame only through the dedicated stop elements designed for that purpose. No other structural element of the prototype shall act directly as a steering stop.

- B.8.1.3** | A clearance zone of at least 30 mm must be maintained around the rider's hands and control areas throughout the full steering range in order to prevent trapping or injury to the rider's fingers or hands in the event of a crash.

The steering system must operate smoothly and must not exhibit excessive play, binding or interference throughout its full range of motion.

## **B.8.2 Steering Dampers**

- B.8.2.1** | The use of steering dampers is permitted.
- B.8.2.2** | Steering damper adjustments may only be performed manually by mechanical or hydraulic means.
- B.8.2.3** | Steering dampers must not act as devices that limit the steering angle.

## **ARTICLE 9: RIMS AND TIRES**

### **B.9.1 Rims**

- B.9.1.1** | The rim sizes (measured at the tire mounting area) must be:
- Front rim: 2.5" x 17"
  - Rear rim: 3.5" x 17"

- B.9.1.2 | Rims made of composite materials (e.g. carbon fibre, glass fibre reinforced composites or similar) are not permitted. Rims must be manufactured from steel, aluminium or magnesium alloys.
- B.9.1.3 | The use of crash protectors at the ends of the wheel shafts is permitted. These protectors must be rounded and must have a diameter equal to or greater than the diameter of the corresponding shaft.
- B.9.1.4 | The ends of the wheel shafts must not protrude more than 30 mm from their housing. Crash protectors described in Article B.9.1.3 are not considered part of this measurement.
- B.9.1.5 | Rims may be freely selected by each team, including solid or closed rim designs, provided that such features form an integral part of the rim itself and comply with the dimensions, material requirements and safety provisions defined in this Article.
- B.9.1.6 | The use of additional wheel covers or devices intended to cover or partially cover the wheel for aerodynamic purposes is not permitted, except for the mudguards. In case of dispute regarding the interpretation of a wheel cover, the decision of the Competition Organization shall be final.
- B.9.2      Tires**
- B.9.2.1 | Each team will receive one set of dry-condition slick tires (one front and one rear) supplied by the Official Tire Supplier as part of the MotoStudent Kit.
- B.9.2.2 | Additional dry tires may be purchased through the Competition Organization. Wet-condition tires will only be available for purchase during the Final Event and will be limited to two sets per team. Tires may be supplied randomly from the available stock.
- B.9.2.3 | A maximum of two sets of dry tires (two front tires and two rear tires) must be declared for use during the Final Event. The first set must be declared before dynamic scrutineering. The second set may be declared during the Final Event at the Technical Control Area. The tire declaration template can be found in Annex 8.
- B.9.2.4 | Only declared tires may be used during dynamic scrutineering, Test 1, Test 2, Test 3, free practice sessions, qualifying sessions, the Play-off Race and the Final Race. Non-compliance with this rule will result in the cancellation of the results obtained in the corresponding session.
- B.9.2.5 | Only tires supplied by the Official Tire Supplier of the respective Competition Edition may be used during MotoStudent.

- B.9.2.6 | Official tires may incorporate an identification label or marking provided by the Competition Organization for control purposes. The presence and integrity of this identification may be verified by the Technical Staff at any time during the Final Event.

The technical specifications of the official tire set will be communicated to all teams participating in the Competition.

- B.9.2.7 | The use of tire warmers is permitted.

## ARTICLE 10: ELECTRONIC SYSTEMS

### B.10.1 Riding assistance electronic systems

- B.10.1.1 | The use of electronic control or assistance systems that provide a clear riding advantage is forbidden. This includes, but is not limited to, traction control, ABS, anti-wheelie systems, launch control or any similar rider assistance system.

- B.10.1.2 | Engine management systems intended for normal engine operation are permitted, provided they do not incorporate any of the rider assistance systems prohibited in Art. B.10.1.1.

### B.10.2 Rider information systems

- B.10.2.1 | Rider dashboard, information and warning systems are freely configurable, provided they are limited to the display of information and warning signals. The dashboard must not be used to command, activate controls or modify any system affecting vehicle dynamics.

### B.10.3 Data Acquisition and Operational Monitoring Systems

#### B.10.3.1 General requirements

Each prototype must be equipped with a data acquisition system capable of monitoring and recording operational parameters of the motorcycle during track operation.

The data logging system may be integrated in the ECU, motor controller, Battery Management System (BMS), or in any independent onboard data acquisition system.

Recording of powertrain operational parameters is mandatory in order to monitor the correct functioning, safety and reliability of the prototype.

The data acquisition system must automatically record data whenever the vehicle is powered and capable of motion.

B.10.3.2 Minimum required parameters – General

At minimum, the data acquisition system must record the following parameters:

- Vehicle speed or wheel speed
- Throttle position
- Powertrain rotational speed (engine RPM or motor RPM depending on category)
- Engine / motor temperature
- Coolant temperature at engine / motor inlet and outlet

The recorded channels must be sampled at a frequency sufficient to allow meaningful technical analysis of the operation of the prototype. The Competition Organization reserves the right to reject data of insufficient quality or resolution for verification purposes.

B.10.3.3 Additional required parameters – MotoStudent eFuel

For prototypes competing in the MotoStudent eFuel category, the following parameters must additionally be monitored and recorded:

- Lambda value
- Oil pressure or oil switch status (if available)

B.10.3.4 Additional required parameters – MotoStudent Electric

For prototypes competing in the MotoStudent Electric category, the following parameters must additionally be monitored and recorded:

- Battery pack temperature (see article D.3.5)
- Battery pack voltage
- Battery current
- Motor controller and / or inverter temperature

B.10.3.5 Sensors and installation

Teams may record additional parameters related to motorcycle dynamics, rider behaviour or system performance. The use of all types of sensors is allowed, provided that their installation does not affect compliance with these Technical Regulations. All sensors, wiring and components of the data acquisition system must be properly secured and installed in safe locations.

B.10.3.6 Telemetry

Live telemetry systems transmitting data from the motorcycle to external receivers during track operation are prohibited.

B.10.3.7 Data access and download

Data acquisition systems may only be accessed when the motorcycle is stopped in the pit box area.

Data download is permitted through wired or wireless connections (e.g. USB, Bluetooth or Wi-Fi), provided that the motorcycle is stationary in the pit area.

#### B.10.3.8

##### Data availability

The Competition Organization may request access to recorded data from any team at any time during the Competition, especially during the Final Event, for technical verification, safety monitoring, reliability analysis or incident investigation purposes.

Such request shall be made by the Technical Director, the Technical Staff or any person expressly authorized by the Competition Organization.

Teams must provide the requested data as soon as reasonably possible and, in any case, before the next on-track activity of the prototype unless another deadline is expressly authorized by the Competition Organization.

Data must be provided in a readable digital format that allows the analysis of the recorded channels. When applicable, raw data files or export files in common formats (e.g. CSV, TXT or equivalent) shall be provided.

For categories where a standard ECU is defined, data shall be provided in the native format generated by the ECU or in an exported format that preserves the recorded channels.

#### B.10.3.9

##### Data integrity and retention

The data logging system must operate automatically and continuously whenever the vehicle is powered.

Recorded data may be stored in the motorcycle, in the ECU, in the motor controller, in the Battery Management System or in another dedicated recording device, provided that the data remain under the control of the team and can be made available to the Competition Organization upon request during the Final Event.

Recorded data must not be intentionally altered, manipulated or deleted during the Final Event.

Teams must retain the recorded data for the entire duration of the Final Event and make it available to the Competition Organization if requested during technical inspections.

If a team is temporarily unable to provide the requested data due to a justified technical issue, the team must immediately inform the Competition Organization and explain the cause of the issue. The Competition Organization will determine whether the situation is acceptable and what additional measures, deadline or consequences may apply.

Failure to provide the requested data without justified cause may be considered a breach of these Technical Regulations and may result in penalties at the discretion of the Competition Organization.

#### B.10.3.10 Use of data by the Competition Organization

The Competition Organization may use the recorded operational data for:

- technical verification
- safety monitoring
- reliability analysis
- incident investigation
- technical evaluation purposes within the MotoStudent program

### B.10.4 Transponder

B.10.4.1 Prior to the MS2 tests during the Final Event, the Competition Organization will supply a timing transponder to all teams to be installed on the prototype.

B.10.4.2 The transponder must be securely mounted in the location specified by the Competition Organization to ensure proper signal reception.

Unless otherwise instructed by the Organization, the transponder shall be mounted on the left fork leg of the front suspension, facing outward from the prototype and in an easily accessible position. It must be properly secured using zip ties or equivalent fastening elements.

B.10.4.3 If the design of the prototype does not allow installation in the recommended location, the Technical Staff of the Organization will indicate an alternative mounting position.

B.10.4.4 The transponder must be installed in an easily accessible area and must remain properly fixed throughout the Final Event.

B.10.4.5 The transponder will be provided under a rental fee, € 35, not included in the MotoStudent registration fee.

### B.10.5 On-board Cameras

B.10.5.1 The use of on-board cameras by teams is not permitted during the MS2 phase.

- B.10.5.2 | The Competition Organization may install on-board cameras on selected prototypes for broadcast, safety or promotional purposes.

#### **B.10.6 Rear Safety Light**

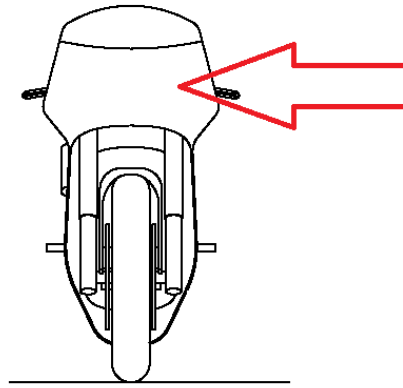
- B.10.6.1 | All motorcycles must have a functioning red safety light mounted at the rear of the prototype for rain or low visibility conditions. The team must ensure that the light is switched on any time the motorcycle is on track or being ridden in the pit lane whenever rain tyres are fitted on the motorcycle. All lights must comply with the following requirements:
- Lighting direction must be parallel to the motorcycle running direction and clearly visible from the rear at least 15° to both the left and right sides of the prototype centre line.
  - The rear light must be mounted near the end of the seat or rear fairing and approximately on the prototype centre line, in a position approved by the Competition Organization. In case of dispute over mounting position or visibility, the decision of the Organization will be final.
  - Power output or luminosity equivalent to approximately: 10–15 W (incandescent) or 0.6–5 W (LED).
  - The light output must be continuous while the motorcycle is on track. Flashing lights are not permitted during track operation, although flashing may be allowed in the pit lane if required by specific procedures, such as pit limiter active.
  - The safety light power supply may be independent from the motorcycle main electrical system.
  - The Competition Organization reserves the right to refuse any lighting system that does not satisfy the intended safety purpose.

### **ARTICLE 11: IDENTIFICATION, ADVERTISING AND COMPETITION NUMBERS**

#### **B.11.1 Bike Numbers**

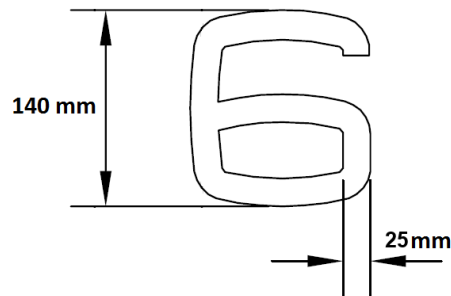
- B.11.1.1 | Each prototype must display three (3) competition numbers on the fairing corresponding to the number assigned to the team as defined in Art. A.3.1.6.
- B.11.1.2 | The front number must be placed on the front fairing. It may be located centrally or slightly offset to either side of the longitudinal centreline of the prototype, provided that it remains clearly visible and readable.





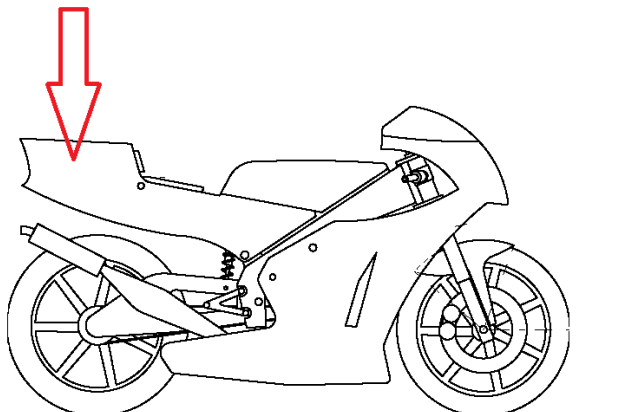
Picture 18 - Front number prototype position

- B.11.1.3 | On the front number area, each character must have a minimum height of 140 mm and a minimum stroke width of 25 mm. The minimum spacing between characters must be 10 mm.



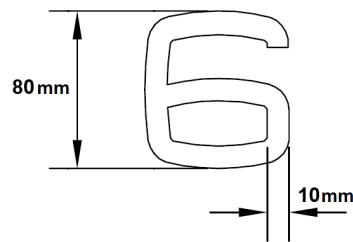
Picture 19 - Prototype front number minimum dimensions

- B.11.1.4 | Two additional numbers must be placed on both sides of the motorcycle. These numbers must be located either on the lower rear portion of the main fairing or on the rear section of the fairing (tail).



Picture 20 - Rear bike number placement

- B.11.1.5 For the side numbers, each character must have a minimum height of 80 mm and a minimum stroke width of 10 mm. The minimum spacing between characters must be 5 mm.



*Picture 21 – Prototype rear number minimum dimensions*

- B.11.1.6 Competition numbers must be displayed in a solid colour providing clear contrast with the background. Gradients, patterns or graphic effects that reduce readability are not permitted.

If the Competition Organization considers that the numbers are not sufficiently visible or readable, the team must modify them as instructed. The Competition Organization may require replacement or apply temporary numbers to ensure proper identification during the Competition.

- B.11.1.7 The font used for competition numbers is free, provided that the numbers remain clearly readable. The Competition Organization reserves the right to require modification of the font if readability is considered insufficient. The inclusion of graphics, logos or decorative elements inside the number characters is not permitted.

- B.11.1.8 The background behind the competition numbers must be a continuous homogeneous area providing clear contrast with the numbers and extending at least 25 mm around their full perimeter. Graphic elements that reduce readability are not permitted in this area.

## **B.11.2 Identification**

- B.11.2.1 The name, logo or initials of the represented university must appear on all prototypes with a minimum dimension of 100 mm in height or width.

- B.11.2.2 A reserved space must be available on the external left side of the frame (considering the riding direction) for scrutineering stickers. These stickers must remain visible with the fairing mounted and must be applied directly to the frame. The Competition Organization will place up to three (3) rectangular scrutineering labels with a maximum size of 35 x 35 mm.



Picture 22 - MotoStudent scrutineering stickers

B.11.2.3 Every prototype must have an identifying code engraved on both the frame and the swingarm in a visible location during scrutineering. The code format will be as follows:

**MSCXXYYZZ**  
e.g.: MSF012701

Where:

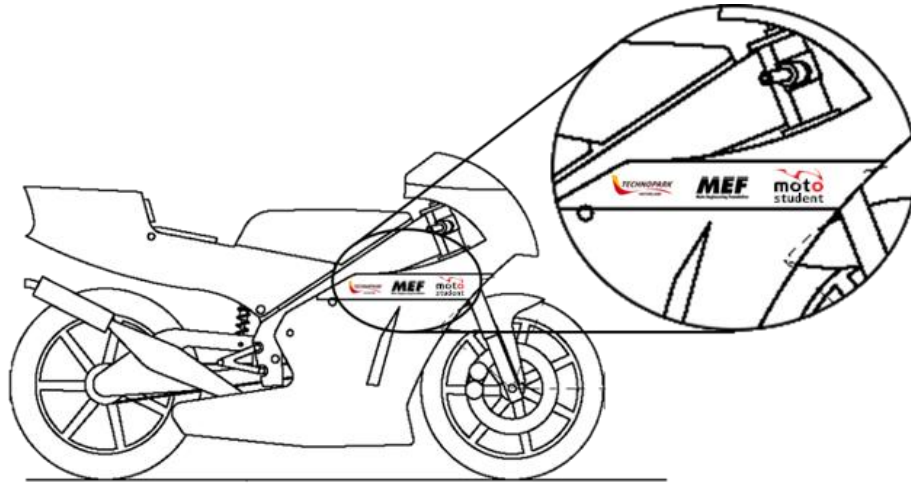
- MS is the acronym of MotoStudent.
- C is the category in which the team is competing, whereas F for MotoStudent eFuel category and E for MotoStudent Electric category (the example shows a MotoStudent eFuel prototype).
- XX is the bike number (the example shows the bike number 01).
- YY represents the last two digits of the Final Event year of the frame and / or swingarm (the example shows the part produced in the year 2027).
- ZZ represents the unit number of the frame and / or swingarm if more than one unit has been produced (the example 01 shows the 1<sup>st</sup> produced part).

### B.11.3 Competition Logos and Advertising

B.11.3.1 In the final design of the prototype, all teams must include on the upper part of the side fairing a reserved horizontal white strip complying with the following specifications:

- The strip must be white.
- The strip must extend from the rear part of the side fairing towards the front.
- The strip must be approximately horizontal to the ground with a tolerance of  $\pm 15^\circ$ .
- The minimum length of the strip must be 400 mm.
- The minimum height of the strip must be 80 mm.

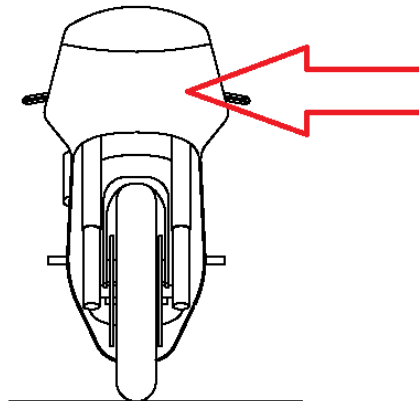
This strip is reserved for the placement of official MotoStudent and partner logos.



Picture 23- White strip example

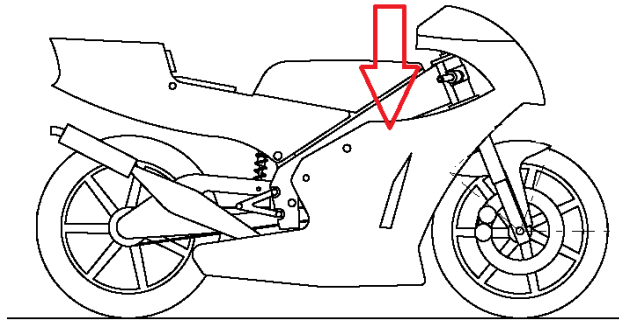
B.11.3.2 | In addition to the strip defined in Art. B.11.3.1, three spaces must be reserved on the prototype to include category stickers that will be fixed by the Competition Organization during the Final Event. These category stickers will have a maximum size of 150 mm wide by 100 mm high.

B.11.3.3 | One category sticker will be placed on the front part of the fairing near the front number, under the windscreen.



Picture 24 - Front category sticker placement

B.11.3.4 | Two category stickers will be placed on both sides of the prototype on the upper part of the front fairing.



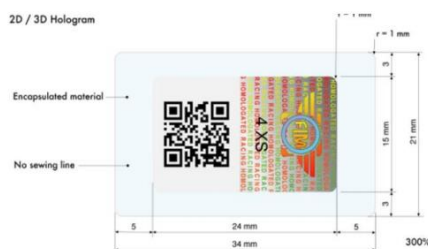
Picture 25 - Lateral category sticker placement

- B.11.3.5 | The Competition Organization may require space for its brand logo or name, or for the logos of official partners or sponsors, on both sides of the swingarm and on the front mudguard. Maximum dimensions will be:
- 200 x 70 mm on the swingarm
  - 60 x 80 mm on the front mudguard
- B.11.3.6 | Teams must reserve the areas described in this article during the design phase of the prototype. If these spaces are not respected, the Competition Organization reserves the right to place the official stickers in the most suitable location, even if this results in partial coverage of existing sponsor logos, promotional stickers or fairing graphics. Teams will not have the right to appeal or request relocation of these stickers.
- B.11.3.7 | It is strictly forbidden to include advertising for alcoholic beverages, gambling and / or tobacco on the prototype or in any official team environment related to the Competition.
- B.11.3.8 | It is strictly forbidden to include advertising or messages that violate human dignity or promote violence, intolerance, xenophobia, racism or any other offensive content. The Competition Organization reserves the right to review all advertising content and prohibit any element it considers inappropriate.

## ARTICLE 12: RIDER EQUIPMENT

### B.12.1 Helmet

- B.12.1.1 | The rider must wear a homologated full-face helmet complying with at least one of the following standards:
- a) FIM FHRPhe-01 or FHRPhe-02



Picture 26 - FHRPhe-01 homologation example

- b) ECE 22-05 "P" or ECE 22.06 "P"



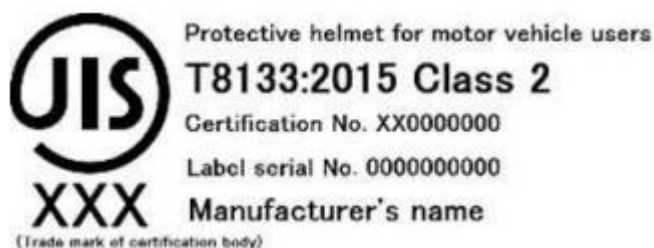
Picture 27 - ECE 22-05 homologation example

- c) SNELL M 2015 o MD2020D o 2020R



Picture 28 - SNELL M 2015 homologation example

- d) JIS T 8133:2015



Picture 29 - JIS T 8133:2015 homologation example

Helmets homologated according to the FIM FRHPhe standard are strongly recommended.

The Competition Organization reserves the right to update or modify the list of accepted helmet homologations in these Rules & Regulations or in future editions of the Competition.

Teams are encouraged to progressively adopt FIM homologated helmets in preparation for future regulatory updates.

- B.12.1.2 | The use of disposable visor screen covers (tear-offs) is permitted.
- B.12.1.3 | Helmet visors must be made of shatterproof material.
- B.12.1.4 | Riders must wear a helmet that is in good condition, correctly fitted and properly fastened at all times while the prototype is operating on track and during all MS2 Tests.
- B.12.1.5 | No cameras, sensors, communication devices or electronic equipment may be attached to the helmet unless expressly authorized by the Competition Organization.  
  
The Competition Organization may require the removal of any device considered unsafe or non-compliant.

## **B.12.2 Safety Clothing**

- B.12.2.1 | The rider must wear a complete one-piece suit made of leather or another material with equivalent abrasion resistance, covering the full torso and limbs and including integrated protection for elbows, shoulders and knees complying with EN 1621-1 or equivalent standards.



*Picture 30 - Rider equipment*

- B.12.2.2 | The use of a back protector complying with EN 1621-2 and a chest protector complying with EN 1621-3 is mandatory.
- B.12.2.3 | Riders must wear boots in good condition made of leather or equivalent protective material and specifically designed for motorcycle competition.
- B.12.2.4 | Riders must wear gloves in good condition made of leather or equivalent material and incorporating protection for fingers and knuckles.

- B.12.2.5 | The use of fire-resistant or competition-grade underwear is recommended. Undergarments made of synthetic materials that may melt and cause injury in the event of a crash should be avoided.
- B.12.2.6 | The use of an airbag system approved for motorcycle racing is strongly recommended. Riders should consult the latest FIM or equivalent self-certified airbag compatibility lists when selecting such systems.
- B.12.2.7 | The use of wearable sensors, monitoring devices or electronic equipment attached to the rider's body or equipment is not permitted unless expressly authorized by the Competition Organization for safety or technical reasons.
- B.12.3 Long hair safety**
- B.12.3.1 | It is the rider's responsibility to ensure that long hair is securely contained inside the helmet or suit during track activities so that it does not present a safety risk to the rider or others.



# **SECTION C**

## **SPECIFIC TECHNICAL REGULATIONS FOR THE CATEGORY "MOTOSTUDENT eFUEL"**

## ARTICLE 1: ENGINE

### C.1.1 Preliminary information

C.1.1.1 The Competition Organization will supply a standardized internal combustion engine to all teams registered in the MotoStudent eFuel category.

The engine remains the exclusive property of the MotoStudent Organization (MEF – Fundacion Moto Engineering Foundation) and is provided to the teams under a temporary loan system for the sole purpose of developing their prototype and participating in the MotoStudent International Competition.

Participation in the Competition and acceptance of these Rules & Regulations implies acceptance of the conditions governing the use, care and return of the supplied engine (see Annexes 6 and 7).

C.1.1.2 The supplied engines have undergone quality control checks prior to shipment and are based on series-production engines.

However, as the engines will be installed and operated within prototypes designed and manufactured by the Teams, and may be used on external test benches and/or motorcycles without the original OEM components of the production vehicle, the Competition Organization assumes no responsibility for any damage the engine may suffer during installation, operation, testing or competition activities.

Teams shall be responsible for the correct installation, handling and use of the engine, as well as for covering the costs of any repair required due to damage occurring while the engine is under their custody. This responsibility also includes any theft, loss or other incidents affecting the engine during such period.

C.1.1.3 The supplied engines may only be used within the framework of the MotoStudent Competition and its related development activities, including prototype development, testing and track sessions.

Any use of the engine outside activities directly related to the MotoStudent Competition is strictly prohibited.

Any breach of this rule may result in sanctions imposed by the Competition Organization.

C.1.1.4 Teams are responsible for the correct installation, operation and maintenance of the engine according to the technical instructions provided by the Competition Organization and the manufacturer. Any operation outside the specified operating limits may be considered improper use.

- C.1.1.5 | The engine is delivered sealed by the Competition Organization.
- The engine must not be opened, disassembled, modified or altered in any way by the teams.
- Breaking, removing or manipulating the engine seals is strictly prohibited unless expressly authorized in writing by the Competition Organization.
- C.1.1.6 | Any engine with damaged, missing or manipulated seals may be considered non-compliant and may be subject to sanctions imposed by the Competition Organization.
- C.1.1.7 | Teams are not allowed to open, modify, alter or attempt to reverse engineer the internal components of the engine.
- Any modification to the engine is strictly prohibited unless explicitly authorized in these Technical Regulations.
- Only those external interfaces and components expressly permitted in the relevant articles of these Rules and Regulations may be modified or adapted for installation purposes.
- C.1.1.8 | Teams must immediately notify the Competition Organization of any malfunction, abnormal behaviour or suspected damage to the engine. The Competition Organization may require the engine to be returned for inspection, maintenance or repair.
- C.1.1.9 | Normal wear resulting from proper use of the engine during development and competition activities will not be charged to the team. Engines must always be operated in a responsible manner.
- However, any damage resulting from misuse, abusive operation, incorrect installation, operation outside the specified limits, lack of maintenance or unauthorized modification will be the financial responsibility of the team. Examples of improper or abusive operation may include, but are not limited to, excessive engine revving without load, burnouts, prolonged operation outside the recommended operating conditions, or any other use that may cause abnormal stress or damage to the engine.
- C.1.1.10 | The Competition Organization shall determine whether engine damage results from normal wear or improper use. The decision of the Competition Organization shall be final.

- C.1.1.11 | The engine must be returned to the Competition Organization at the end of the current edition.
- Teams present at the Final Event are required to return the engine on-site during the Final Event, in accordance with the instructions of the Technical Organization.
- C.1.1.12 | Teams that will not attend the Final Event must return the engine to the Competition Organization no later than thirty (30) days before the first day of the Final Event.
- C.1.1.13 | All transportation costs for sending engines between the team's facilities and MEF headquarters, including for inspection, repair, replacement or return, shall be borne by the team.
- C.1.1.14 | If an engine is lost, not returned within the specified period, or returned with damage beyond normal wear, the team shall be responsible for covering the repair costs or replacement value of the engine.
- C.1.1.15 | The Competition Organization may establish maintenance procedures, inspection intervals and service operations in order to ensure reliability and equal performance of the engines during the Competition.
- C.1.1.16 | The engine sealing system, inspection procedures and maintenance operations will be defined in the relevant MotoStudent technical documentation and communicated to the teams through official channels.
- C.1.1.17 | The price of replacement engines, standard maintenance procedures and repair operations will be defined in the official MotoStudent eFuel Engine Information document and / or in these Rules & Regulations.
- C.1.1.18 | In the event that an engine requires non-standard repair following inspection by the Competition Organization, the team will be notified of the corresponding repair cost through a proforma invoice prior to the intervention. The cost of the repair shall be borne by the team.
- If the team declines the repair or fails to assume the corresponding costs, the Competition Organization may retain the engine and restrict the team's continued participation in the Competition.
- Any outstanding payments related to engine repair, replacement or loss must be settled before the end of the Competition edition.
- Failure to settle the obligations described in this Article C.1.1 may result in the Competition Organization declaring the team, the represented university or any

related entity ineligible to participate in future editions of the MotoStudent Competition until all outstanding obligations have been fulfilled.

The Competition Organization reserves the right to refuse participation to teams or institutions with outstanding obligations from previous editions.

### **C.1.2 Sealing**

**C.1.2.1** The use of the engine supplied by the Competition Organization is compulsory. All engines will be delivered sealed by the Competition Organization.

The engine seals must remain intact at all times. It is strictly forbidden for teams to open, disassemble, manipulate or alter any internal component of the engine.

Any missing, broken or damaged seal may result in the engine being declared non-compliant and may lead to sanctions as determined by the Competition Organization.

**C.1.2.2** If a team detects a malfunction, internal failure or suspected issue affecting the engine, the team must immediately inform the Competition Organization.

Teams are not allowed to attempt any repair, disassembly or intervention affecting sealed parts of the engine.

The Competition Organization will determine the appropriate inspection, repair or replacement procedure.

**C.1.2.3** The inspection and verification of engine seals may be performed by the Competition Organization at any moment during the Competition, including scrutineering, practice sessions, races or technical inspections.

The technical assessment of the engine condition and sealing status performed by the Competition Organization shall be final.

### **C.1.3 Engine Technical Specifications**

**C.1.3.1** The detailed technical specifications of the engine supplied by the Competition Organization will be published in the official Engine Technical Information document and provided to all registered teams.

**C.1.3.2** The engine supplied by the Competition Organization will be a single-cylinder four-stroke internal combustion engine with a displacement between 240 cc and 340 cc.

- C.1.3.3 | The engine supplied by the Competition Organization is equipped with a liquid cooling system.

#### **C.1.4 Crankcase, cylinder head and engine block**

- C.1.4.1 | External or internal modifications to the engine assembly are not permitted.
- C.1.4.2 | The crankcase, cylinder head and cylinder block must remain in their original configuration as supplied by the Competition Organization. Machining, polishing, material removal, surface treatment, structural modification or replacement of internal components is strictly prohibited.
- C.1.4.3 | No modifications to the engine mounting points on the engine block are permitted. The prototype frame must be designed to match the original engine mounting geometry.
- C.1.4.4 | The original cooling passages and cooling interfaces of the engine must not be modified. Teams must design the external cooling system in accordance with the original cooling connections provided by the engine.
- C.1.4.5 | The intake and exhaust interfaces of the engine must not be modified. The intake and exhaust systems developed by the teams must connect to the engine using the original geometry and connection points provided by the engine. No machining, modification or adaptation of the engine intake or exhaust mounting surfaces is permitted.
- C.1.4.6 | The installation of external protective engine covers is recommended, provided that their installation does not involve any modification of the engine structure. Such protectors may use the original engine cover bolts as mounting points.
- Protective covers must not hide or obstruct any official seal, identification mark or label placed by the Competition Organization. Only metallic protectors or FIM-homologated protectors are permitted.
- C.1.4.7 | The original oil pump integrated into the engine must not be modified or replaced.
- C.1.4.8 | The original water pump integrated into the engine must not be modified or replaced.

#### **C.1.5 Modifications**

- C.1.5.1 | Any modification to the engine or its components is prohibited unless it is expressly permitted in these Technical Regulations.

C.1.5.2 | Any modification intended to circumvent the spirit of these Rules & Regulations may be considered a non-compliance by the Competition Organization.

C.1.5.3 | In the event of any doubt or dispute regarding the interpretation of these provisions, the decision of the Competition Organization shall be final.

#### **C.1.6 Engine operating limits**

C.1.6.1 | The engine must be operated at all times within safe operating limits regarding temperature, engine speed and air-fuel mixture.

C.1.6.2 | The maximum coolant temperature during operation must not exceed the limit defined by the Competition Organization in the official MotoStudent eFuel Engine Information document.

C.1.6.3 | The air-fuel mixture must remain within safe limits as defined in Article C.6. Lambda values outside the permitted range may be considered unsafe engine operation.

C.1.6.4 | Engine speed must remain within the limits defined by the engine manufacturer or by the Competition Organization. Excessive engine revving without load is strictly forbidden.

C.1.6.5 | Operation practices that may cause abnormal thermal or mechanical stress to the engine are prohibited. These include, but are not limited to:

- excessive engine revving without load
- burnouts or intentional rear wheel spinning
- prolonged operation outside recommended temperature limits
- operation without adequate cooling airflow
- operation with clearly unsafe air-fuel mixture conditions

C.1.6.6 | The Competition Organization reserves the right to review engine operating data, including but not limited to engine speed, coolant temperature and lambda values, in order to determine whether an engine failure may have resulted from operation outside safe limits.

C.1.6.7 | If engine damage is determined to be the result of operation outside these limits or abusive use of the engine, the repair or replacement costs shall be charged to the team.

## ARTICLE 2: AIR INTAKE SYSTEM

### C.2.1 Intake pipes

- C.2.1.1 | The design, dimensions and position of the air intake ducts are free, provided that they comply with the overall dimensional limits of the prototype defined in these Rules & Regulations. The intake system may incorporate filters, airboxes or aerodynamic intake devices freely designed by the teams.
- C.2.1.2 | Mobile devices intended to vary the effective length of the intake duct are permitted, provided that they are located upstream of the intake valves.
- C.2.1.3 | Only air and recirculated engine gases (if applicable) may flow through the intake ducts. The injection or introduction of additional substances such as ethanol, methanol, water or any other chemical compound into the intake system is not permitted.

### C.2.2 Throttle control

- C.2.2.1 | Only one throttle valve or an equivalent mechanical system is permitted. The throttle must be operated exclusively by a mechanical control (e.g. cable) directly actuated by the rider.
- C.2.2.2 | Electronic throttle control systems ("ride-by-wire") are not permitted.
- C.2.2.3 | In the event of a control system failure, the throttle valve must automatically return to the closed position.
- C.2.2.4 | A bypass system for idle speed control may be implemented.

### C.2.3 Overpressure Systems

- C.2.3.1 | Any system intended to artificially increase the pressure of the intake air is prohibited. The use of turbochargers, superchargers or any other forced induction system is strictly forbidden.
- C.2.3.2 | Only passive pressure effects generated by the vehicle motion (ram-air effect) through aerodynamic air intakes are permitted.

### C.2.4 Airbox and Air Filtering

- C.2.4.1 | The design and construction of the airbox are free, provided that it complies with these Technical Regulations.



- C.2.4.2 | The use of an airbox incorporating an air filter is mandatory.
- C.2.4.3 | The internal volume of the airbox must not be less than 2.4 liters.
- C.2.4.4 | The air filter must be specifically intended for engine air intake applications. Its design, dimensions and filtering media are free, provided that it ensures adequate filtration and protects the engine from dust, debris or foreign particles that could cause damage.
- The use of commercial air filters or equivalent self-designed filtration solutions is permitted.
- C.2.4.5 | The intake system must be securely mounted and designed to prevent the ingestion of foreign objects.
- C.2.5      Recycling of gases**
- C.2.5.1 | Recycling of engine breather gases is permitted.
- C.2.5.2 | A closed breather system directing crankcase gases to the intake system is allowed. These breather lines may include filtration devices and oil catch tanks.
- C.2.5.3 | If the recycling system described in Art. C.2.5.2 is not implemented, a breather duct must be installed to discharge oil residues into a dedicated oil catch tank with a minimum capacity of 200 cc.
- C.2.5.4 | Under no circumstances may breather gases or oil residues be discharged directly onto the track or onto any part of the prototype not intended for this purpose.

### ARTICLE 3: FUEL SUPPLY SYSTEM

#### C.3.1      Fuel Tank

- C.3.1.1 | Fuel tanks, regardless of their design or manufacturing material, must be **fully filled** with flame-retardant mousse-type safety foam (preferably Explosafe, ExploStop, Baffle Foam ATL, Fuel Safe or an equivalent motorsport safety foam compatible with gasoline fuels containing renewable components, ethanol, other alcohols and oxygenated compounds) intended to reduce fuel movement and minimize explosion risks in case of impact.



*Picture 31 - Example of flame-retardant foam*

- C.3.1.2

It is mandatory to install a breather duct in the fuel tank to prevent possible overpressures. This duct must be fitted with a retention valve that allows the free circulation of air (fuel vapours) but prevents the circulation of liquids (fuel), in order to avoid fuel spills in case of an accident. The outlet of the breather line must be connected to a recovery container specifically installed for this purpose, with a minimum capacity of 250 cc.
- C.3.1.3

The main structure of the fuel tank must be made entirely of steel or aluminium.

The use of composite materials (such as carbon fibre, aramid fibre or glass fibre) or polymeric materials is permitted only as secondary structural elements, external protection or reinforcement of the tank.

Composite or polymeric materials must not constitute the primary fuel containment structure.
- C.3.1.4

The lowest part of the fuel tank must be located above the height of the engine intake valves.
- C.3.1.5

The fuel tank cap must be fitted with a threaded opening and closing system. Lockable caps are not permitted.
- C.3.1.6

The fuel tank cap must ensure a watertight seal that prevents any fuel leakage in the event of a fall or accident.
- C.3.1.7

All materials and components used in the fuel system, including fuel tank, internal components, hoses, connectors, pump, filters, seals and fittings, must be compatible with gasoline fuels containing renewable components ethanol, other alcohols and oxygenated compounds as specified by the Competition Organization.

## **C.3.2 Fuel Pipes**

- C.3.2.1

All fuel pipes running from the tank to the carburettor or to the injection system must be fitted with at least one self-closing valve (quick connector). This connector must automatically close in the event of disconnection, so that if the tank becomes

detached from the motorcycle it is the connector that disconnects rather than another section of the fuel line.

The force required to disconnect the connector must be, at maximum, 50% of the force required to loosen any other connection or to cause breakage of the pipe material.



*Picture 32 - Example of self-closing quick connector*

C.3.2.2 | Fuel pipes must be securely mounted and routed in such a way that they are protected from mechanical damage, excessive heat exposure or accidental disconnection during operation.

C.3.2.3 | Fuel hoses, seals, connectors and related components must be compatible with gasoline fuels containing renewable components and oxygenated compounds.

### **C.3.3 Injection system**

C.3.3.1 | Replacement or modification of the original fuel supply system of the supplied engine is allowed. Both carburetion and electronic fuel injection systems are permitted.

C.3.3.2 | The fuel injector must be installed upstream of the intake valves of the cylinder. Direct fuel injection into the combustion chamber is not allowed.

C.3.3.3 | The injector supplied with the MotoStudent kit must be used in the throttle body and must remain unmodified (same brand and reference).

C.3.3.4 | Article C.1.1.1 also applies to the injector body delivered with the MotoStudent Kit as part of the internal combustion engine.

### **C.3.4 Fuel Cooling**

C.3.4.1 | Artificial cooling of the fuel is not allowed. Only passive cooling resulting from the aerodynamic design of the prototype may be used for the fuel system.

### **C.3.5 Fuel Pressure**

C.3.5.1 | Only one fuel pump may be installed. There are no restrictions regarding the type of pump used.

C.3.5.2 | Fuel pressure must not exceed 3.5 bar at any point of the supply circuit.

- C.3.5.3 | Refuelling operations must be carried out using a non-pressurized container.
- C.3.5.4 | Artificial pressurization of the fuel tank is strictly forbidden.
- C.3.5.5 | The fuel supply system must be designed and installed in a way that prevents fuel leakage during normal operation, vibration or accidental impacts.

#### ARTICLE 4: FUEL AND LUBRICANTS

##### C.4.1 Fuel

- C.4.1.1 | The fuel used during the Final Event of the Competition shall be the Official Fuel specified and supplied by the Competition Organization.

During the Final Event, at all times and for any use related to the prototype, only the Official Fuel distributed by the Competition Organization may be used. The use, storage or refuelling of any other fuel by the teams during the Final Event is not permitted.

The Official Fuel supplied during the Final Event is not provided free of charge. Teams must acquire it through the official MotoStudent Store during the Final Event.

The technical specifications, commercial conditions and relevant information of the Official Fuel will be communicated to all registered teams through the official documentation of the Competition. Such publication is intended to assist the teams, but does not exempt them from their responsibility to ensure fuel-system compatibility and correct engine calibration.

- C.4.1.2 | During the development phase of the prototype, teams may use commercially available unleaded gasoline for testing and calibration purposes.

The Official Fuel used during the Final Event has combustion characteristics comparable to commercially available gasoline fuels typically available in international markets.

Teams are responsible for ensuring the compatibility of their fuel system, engine calibration and materials with the Official Fuel.

- C.4.1.3 | Any alteration of the fuel by means of additives, blending, chemical treatments or any other modification is strictly forbidden.

- C.4.1.4 | At any time during the Competition, the Organization may require fuel samples from any participating team for verification purposes. If the fuel used is found not to correspond to the Official Fuel supplied by the Organization, the team may be disqualified from the test in progress or from the Competition.
- C.4.1.5 | The Official Fuel complies with the applicable chemical requirements established within the European Union. Teams from non-EU countries should take into account that fuels available in their countries may have different chemical compositions and should consider this during engine calibration and prototype development.
- C.4.1.6 | The Competition Organization will keep a reference sample of the Official Fuel used during the Competition. This reference sample may be used as the baseline for any fuel verification, comparison or dispute regarding fuel compliance. The decision of the Competition Organization based on this reference fuel shall be final.
- C.4.2 Engine lubricating oil**
- C.4.2.1 | The lubricating oil is free choice. However, the selected oil must be suitable for high-performance four-stroke motorcycle engines and compatible with the specifications and viscosity grades recommended by the engine manufacturer.
- C.4.2.2 | The installation of radiators to cool the engine oil is not allowed.
- C.4.2.3 | The oil filler cap, oil drainage plug and oil filter cap bolts must be safety wired in order to prevent oil leakages during the MS2 phase. Engines are delivered to the teams with the required bolts already safety wired as a reference.

## ARTICLE 5: COOLING SYSTEM

### C.5.1 Cooling systems design

- C.5.1.1 | The design and manufacturing of the external cooling system is free.
- C.5.1.2 | The number, position, size and material of the cooling radiators and associated cooling components are free, provided that they comply with the dimensional limits of the prototype defined in these Rules and Regulations.
- C.5.1.3 | The cooling system must be securely installed and designed to prevent coolant leaks during normal operation or in the event of a crash.
- C.5.1.4 | Any cooling system components must not present sharp edges or protrusions that may create safety hazards for the rider or other participants.

- C.5.1.5 | Cooling airflow may be improved by aerodynamic design of ducts, fairings or radiator positioning.

## **C.5.2 Cooling fluids**

- C.5.2.1 | Only distilled water may be used as cooling fluid.  
Recommended operating range: 80–90 °C  
Maximum allowed: 105 °C
- C.5.2.2 | The use of additives, glycol, antifreeze agents or any other chemical products in the cooling fluid is forbidden.
- C.5.2.3 | Any coolant overflow outlet must discharge into a recovery container in order to prevent coolant being released onto the track. The recovery container must be securely mounted on the prototype.

## **ARTICLE 6: EXHAUST SYSTEM**

### **C.6.1 Exhaust system design**

- C.6.1.1 | The design and manufacturing of the exhaust system is free, provided that it complies with the dimensional requirements of the prototype and the noise limits defined in these Rules & Regulations.
- C.6.1.2 | No mobile or variable geometry devices are allowed in the exhaust system downstream of the engine exhaust valves (e.g. exhaust valves, movable deflectors or similar devices).
- C.6.1.3 | The exhaust system must be securely mounted and designed to avoid contact with the rider, tyres or other safety-critical components.
- C.6.1.4 | Exhaust gases must not be directed towards the tyres, brakes or rider in a way that may create safety risks.
- C.6.1.5 | The use of a lambda (oxygen) sensor is compulsory. The lambda value must be recorded at any time the engine is running.  
  
The data acquisition system must record lambda values, and the Competition Organization may request these data at any time during the Competition.

C.6.1.6 | Lambda sensors may only be used for monitoring and engine calibration purposes. The use of lambda signals for active engine control strategies such as traction control or similar rider-assistance systems is not allowed.

C.6.1.7 | For safe engine operation, the recommended lambda operating range is between 0.85 and 0.88 under normal engine load conditions.

The lambda value must not exceed 0.95 during normal engine operation. Short transient conditions occurring during gear shifts or throttle transitions may be disregarded.

The Competition Organization reserves the right to review recorded engine data in order to determine whether an engine failure may have resulted from operation outside safe limits.

## **C.6.2 Noise**

C.6.2.1 | The maximum noise level allowed for the exhaust is 105 dB(A) measured statically at 6,000 rpm.

No tolerance will be applied during the initial technical inspection. After a race session, a tolerance of +3 dB(A) may be applied.

## **ARTICLE 7: TRANSMISSION SYSTEM**

### **C.7.1 Clutch**

C.7.1.1 | The original clutch type must be maintained.

C.7.1.2 | Clutch discs may be replaced.

C.7.1.3 | Clutch springs may be replaced.

C.7.1.4 | The clutch basket may be replaced provided that the replacement does not require any modification of the engine crankcase or internal engine components.

C.7.1.5 | The installation of clutch systems with limited slip (anti-skid or anti-bounce type) is allowed and recommended.

### **C.7.2 Gearbox**

C.7.2.1 | The original gearbox integrated in the supplied engine must be used and shares the lubrication system with the engine.

C.7.2.2 | The original gearbox cannot be replaced or internally modified.

C.7.2.3 | External quickshifter systems are allowed provided that they operate only by interrupting ignition during gear shifts.

C.7.2.4 | The shift lever may be replaced.

### **C.7.3 Secondary transmission**

C.7.3.1 | Only chain-driven secondary transmission systems are allowed.

C.7.3.2 | The gearbox output pinion may be replaced. Variations in geometry and number of teeth are allowed, provided that the modification does not require any alteration of the gearbox or engine components.

C.7.3.3 | The rear wheel sprocket and the chain are free choice.

C.7.3.4 | The secondary transmission system must include adequate protection devices in order to prevent injuries to the rider or damage in case of chain failure.

## **ARTICLE 8: ELECTRIC INSTALLATION**

### **C.8.1 ECU**

C.8.1.1 | Only the Engine Control Unit (ECU) specified and published by the MotoStudent Organization in the *MotoStudent Approved Parts List* may be used.

C.8.1.2 | Under no circumstances may the signals from sensors integrated in the engine be modified, intercepted or distorted. This includes any attempt to bypass or exceed the engine speed limit defined by the ECU (13,500 rpm). The engine speed limit may be verified by the Technical Staff of the Organization at any time during the Final Event.

Any device, software modification or wiring alteration intended to modify the original ECU engine speed limitation is strictly prohibited.

C.8.1.3 | Ignition and injection adjustments are limited to those permitted by the ECU tuning software under the official user license. Base maps must not be modified. Any team modifying parameters not permitted by the official software license will be disqualified from the test in which the team has participated since the last ECU verification.

C.8.1.4 | The MotoStudent Organization may request to connect to the ECU and download its configuration at any time during the Competition. Teams must ensure that the necessary means are available to allow the Organization to perform this



verification. Failure to provide access will be considered non-compliance with Article C.8.1.3.

## **C.8.2 Battery**

C.8.2.1 | It is compulsory to install a battery with an operating voltage between 8 V to 18 V.

## **C.8.3 Electric installation**

C.8.3.1 | The configuration of the electrical installation is free.

C.8.3.2 | The use of commercial electrical systems or components is allowed.

C.8.3.3 | Cable types, wiring design and routing are free provided that the installation is properly insulated and securely mounted.

C.8.3.4 | The connector type is free. Teams must ensure that the electrical system operates correctly under wet conditions.

C.8.3.5 | Commercial components such as coils, batteries, regulators and connectors may be used.

C.8.3.6 | All prototypes must be equipped with an engine kill switch located on the left-hand side of the handlebar. The switch must be clearly marked in red and easily accessible in case of emergency. The rider must be able to activate the switch while keeping both hands on the handlebar grips. The switch must immediately stop the engine when activated.

C.8.3.7 | Only one ignition coil may be used.

C.8.3.8 | The electrical installation must be properly integrated into the prototype structure. All wiring must be securely fixed and protected.

C.8.3.9 | Excess wiring must not be coiled or loosely installed. Cable lengths must be appropriate for their installation.

C.8.3.10 | Electrical wiring must be routed away from hot components such as the engine, exhaust system and cooling system. Electrical cables and components must not be in direct contact with parts that may reach high temperatures.

## **C.8.4 Start System**

C.8.4.1 | The electric starter system integrated in the official engine must remain operational and may not be removed.

- C.8.4.2 | In the event of a malfunction of the electric starter during the start procedure of a race session, external starting devices are not allowed. In case of failure, the motorcycle may only be started by pushing.

#### **C.8.5 Data Logging**

- C.8.5.1 | The use of a data logging system is compulsory. Commercial loggers are recommended.

- C.8.5.2 | Engine speed, engine temperature and lambda value must be monitored and recorded whenever the engine is running, from the first engine start until the end of the Final Event.

- C.8.5.3 | Logged data must be stored by the team and must be made available to the Competition Organization at any time upon request during the Competition and particularly during the Final Event.

- C.8.5.4 | The Competition Organization may require the download and submission of data logs at any time during the Final Event, including after any session or test.

- C.8.5.5 | Teams must ensure that the necessary hardware, software and connection means are available to allow the Competition Organization to access the recorded data. Failure to provide access to the data may be considered a technical non-conformity.

- C.8.5.6 | Logged data should preferably be provided in CSV or another widely accessible format. The Competition Organization may also request the original native data format generated by the ECU or data logging system.

#### **C.8.6 Launch Control and Traction Control**

- C.8.6.1 | Launch Control and Traction Control functions integrated in the homologated MotoStudent eFuel ECU may be used as the only exception to Article B.10.1.1. No additional Launch Control or Traction Control systems are allowed.

# **SECTION D**

## **SPECIFIC TECHNICAL REGULATIONS FOR THE CATEGORY “MOTOSTUDENT ELECTRIC”**

**ARTICLE 1: DEFINITIONS AND GENERAL ASPECTS****D.1.1 High Voltage (HV), Low Voltage (LV), Volts Direct Current (VDC)**

D.1.1.1 Any circuit with a potential difference equal to or greater than 60 VDC is considered High Voltage (HV). The maximum voltage of the prototype at any time must not exceed 126 VDC.

D.1.1.2 The High Voltage System (HVS) is defined as all electrical components operating at High Voltage. A component may belong simultaneously to both the HVS and the LVS.

D.1.1.3 Any circuit with a potential difference between 0 VDC (included) and 60 VDC (not included) is considered Low Voltage (LV).

D.1.1.4 The Low Voltage System (LVS) is defined as all electrical components operating at Low Voltage. A component may belong simultaneously to both the HVS and the LVS.

**D.1.2 High Voltage System – HVS**

D.1.2.1 The High Voltage System (HVS) must be galvanically isolated from the prototype chassis, ground and any conductive part accessible from the outside of the prototype.

D.1.2.2 All High Voltage System components must be enclosed or protected to prevent accidental contact.

D.1.2.3 The use of software elements to control the maximum and minimum voltage of the HVS is allowed.

D.1.2.4 Clearly visible warning labels indicating electrical danger must be placed on housings or areas near components operating at High Voltage (HV). These labels must include the text “HIGH VOLTAGE”.



*Picture 33 - High Voltage mandatory label*

- D.1.2.5 | All HVS components must be properly insulated and protected against direct or indirect contact, including protection for team members working inside or outside the prototype.
- D.1.2.6 | HVS connections must be protected so that they cannot be accessed using a cylindrical probe of 100 mm length and 6 mm diameter.
- D.1.2.7 | All HVS cables or conductors must be orange or clearly marked with orange along their entire length and must comply with non-flammable standards such as UL-94 V0, FAR25, ISO 6722 or equivalent.
- D.1.2.8 | Insulation materials must be rated for the expected operating temperatures and voltages and must resist mechanical damage such as scratches or abrasion. The use of only insulating tape, rubber-like paint or anodization as insulation is prohibited.
- D.1.2.9 | Any cable or conductor that is not part of the HVS must not be orange.
- D.1.2.10 | All HVS components and containers must be protected against moisture and water exposure such as rain or puddles.

#### **D.1.3 Low Voltage System - LVS**

- D.1.3.1 | The use of DC/DC converters for supplying the Low Voltage System is allowed.
- D.1.3.2 | The use of a separate battery as a Low Voltage source is allowed, provided that its voltage remains within the Low Voltage limits defined in Article D.1.1.  
  
Such battery shall not be considered part of the traction battery pack for the purposes of these Technical Regulations.
- D.1.3.3 | Any battery used exclusively as a Low Voltage source must be installed in a separate container and must not supply traction power.  
  
If the voltage of such battery exceeds the Low Voltage limits at any time, it shall be considered part of the battery pack and shall comply with the requirements of Article D.3.

#### **D.1.4 Insulation between HVS and LVS**

- D.1.4.1 | The High Voltage System and the Low Voltage System must be physically or galvanically separated.

- |         |                                                                                                                                                                                                                                                                                      |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D.1.4.2 | The minimum physical separation between two conductive parts inside a container with a voltage difference lower than 100 VDC must be 10 mm.                                                                                                                                          |
| D.1.4.3 | The minimum physical separation between two conductive parts inside a container with a voltage difference equal to or higher than 100 VDC must be 20 mm.                                                                                                                             |
| D.1.4.4 | The minimum distances specified in Art. D.1.4.2 and Art. D.1.4.3 may be reduced if a humidity-resistant and electrically insulating barrier with a minimum temperature resistance of 150 °C is present.                                                                              |
| D.1.4.5 | <p>The minimum separation between any HVS and LVS conductive parts located on the same PCB must be:</p> <ul style="list-style-type: none"> <li>- 6.4 mm over the surface</li> <li>- 3.2 mm through the air</li> <li>- 2 mm if the PCB is protected by a conformal coating</li> </ul> |
| D.1.4.6 | Any optocoupler, transceiver, sensor or integrated circuit rated for voltages equal to or higher than the HVS voltage is exempt from the minimum distances defined in Art. D.1.4.5.                                                                                                  |
| D.1.4.7 | All conductive parts of the prototype that may become electrically energized must be electrically referenced to the LVS ground.                                                                                                                                                      |

## ARTICLE 2: ELECTRIC MOTOR AND POWER DEMAND

### D.2.1 Preliminary information

- |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D.2.1.1 | <p>The Competition Organization will supply a standardized electric motor to all teams registered in the MotoStudent Electric category.</p> <p>The motor remains the exclusive property of the MotoStudent Organization (MEF – Fundación Moto Engineering Foundation) and is provided to the teams under a temporary loan system for the sole purpose of developing their prototype and participating in the MotoStudent International Competition.</p> <p>Participation in the Competition and acceptance of these Rules &amp; Regulations implies acceptance of the conditions governing the use, care and return of the supplied motor (see Annexes 6 and 7).</p> |
| D.2.1.2 | The supplied motors have undergone quality control checks prior to shipment and are based on motors designed for motorsport applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

However, since the motor will be installed and operated within prototypes designed and manufactured by the teams, the Competition Organization assumes no responsibility for any damage the motor may suffer during installation, operation, testing or competition activities.

Teams shall therefore assume responsibility for the correct installation and use of the motor and for covering the costs of any repair required due to damage occurring during its use.

**D.2.1.3** | The supplied motor may only be used within the framework of the MotoStudent Competition and its related development activities, including prototype development, testing and track sessions.

Any use of the motor outside activities directly related to the MotoStudent Competition is strictly prohibited.

Any breach of this rule may result in sanctions imposed by the Competition Organization.

**D.2.1.4** | Teams are responsible for the correct installation, operation and maintenance of the motor according to the technical instructions provided by the Competition Organization and the manufacturer.

Operation outside the specified operating limits or misuse of the motor may be considered improper use.

**D.2.1.5** | The motor is delivered sealed by the Competition Organization.

The motor must not be opened, disassembled, modified or altered in any way by the team under any circumstances.

Breaking, removing or manipulating the motor seals is strictly prohibited unless expressly authorized in writing by the Competition Organization.

The delivery of the motor will take place once the team has successfully completed its registration in the Competition and the motors from previous editions have been fully inspected and prepared by the Competition Organization.

As delivery timelines may be affected by logistical factors beyond the control of the Competition Organization, including international transport conditions, no specific delivery deadline can be guaranteed. Teams are advised to take this into account when planning their development activities.

- D.2.1.6 | Any motor with damaged, missing or manipulated seals may be considered non-compliant and may be subject to sanctions imposed by the Competition Organization.
- D.2.1.7 | Teams are not allowed to open, modify, alter or attempt to reverse engineer the internal components of the motor.
- Any modification to the motor is strictly prohibited unless explicitly authorized in these Technical Regulations.
- Only those external interfaces and components expressly permitted in the relevant articles of these Rules and Regulations may be modified or adapted for installation purposes.
- D.2.1.8 | Teams must immediately notify the Competition Organization of any malfunction, abnormal behaviour or suspected damage to the motor.
- The Competition Organization may require the motor to be returned for inspection, maintenance or repair.
- D.2.1.9 | Normal wear resulting from proper use of the motor during development and competition activities will not be charged to the team.
- However, any damage resulting from misuse, abusive operation, incorrect installation, operation outside the specified limits, lack of maintenance or unauthorized modification will be the financial responsibility of the team.
- D.2.1.10 | The Competition Organization shall determine whether motor damage results from normal wear or improper use. The decision of the Competition Organization shall be final.
- D.2.1.11 | The motor must be returned to the Competition Organization at the end of the current edition.
- Teams present at the Final Event are required to return the motor on-site during the Final Event, in accordance with the instructions of the Technical Organization.
- D.2.1.12 | Teams that will not attend the Final Event must return the motor to the Competition Organization no later than thirty (30) days before the first day of the Final Event.
- D.2.1.13 | All transportation costs for sending motors between the team's facilities and MEF headquarters, including for inspection, repair, replacement or return, shall be borne by the team.



- D.2.1.14 | If a motor is lost, not returned within the specified period, or returned with damage beyond normal wear, the team shall be responsible for covering the repair costs or replacement value of the motor.
- D.2.1.15 | The Competition Organization may establish maintenance procedures, inspection intervals and service operations in order to ensure reliability and equal performance of the motors during the Competition.
- D.2.1.16 | The motor sealing system, inspection procedures and maintenance operations will be defined in the relevant MotoStudent technical documentation and communicated to the teams through official channels.
- D.2.1.17 | The price of replacement motors, standard maintenance procedures and repair operations will be defined in the official MotoStudent Electric Motor Information document and / or in these Rules & Regulations.
- D.2.1.18 | In the event that a motor requires non-standard repair following inspection by the Competition Organization, the team will receive a quotation for the repair before any intervention is carried out.
- The repair will only be performed after the team has formally accepted the quotation.
- If the team declines the repair or fails to assume the corresponding costs, the Competition Organization may retain the motor and restrict the team's continued participation in the Competition.
- Any outstanding payments related to motor repair, replacement or loss must be settled before the end of the Competition edition.
- Failure to settle the obligations described in this Article D.2.1 may result in the Competition Organization declaring the team, the represented university or any related entity ineligible to participate in future editions of the MotoStudent Competition until all outstanding obligations have been fulfilled.
- The Competition Organization reserves the right to refuse participation to teams or institutions with outstanding obligations from previous editions.

## **D.2.2 Motor Sealing**

- D.2.2.1 | The use of the electric motor supplied by the Competition Organization is compulsory. All motors will be delivered sealed by the Competition Organization.

The motor seals must remain intact at all times. It is strictly forbidden for teams to open, disassemble, manipulate or alter any internal component of the motor.

Any missing, broken or damaged seal may result in the motor being declared non-compliant and may lead to sanctions as determined by the Competition Organization.

D.2.2.2 | If a team detects a malfunction, internal failure or suspected issue affecting the motor, the team must immediately inform the Competition Organization.

Teams are not allowed to attempt any repair, disassembly or intervention affecting sealed parts of the motor.

The Competition Organization will determine the appropriate inspection, repair or replacement procedure.

D.2.2.3 | The inspection and verification of motor seals may be performed by the Competition Organization at any moment during the Competition, including scrutineering, practice sessions, races or technical inspections.

The technical assessment of the motor condition and sealing status performed by the Competition Organization shall be final.

### **D.2.3 Electric motor technical datasheet**

D.2.3.1 | Technical information about the supplied motor will be published and distributed by the Competition Organization to all registered teams.

D.2.3.2 | The electric motor supplied by the Competition Organization will provide at least the following nominal performance values:

- 30 Nm nominal torque
- 15 kW nominal power

D.2.3.3 | The electric motor supplied by the Competition Organization will provide at least the following peak performance values:

- 60 Nm peak torque
- 30 kW peak power

D.2.3.4 | The electric motor supplied by the Competition Organization will be liquid cooled.

D.2.3.5 | The technical motor datasheet provided by the Competition Organization is intended exclusively for teams participating in the Competition.

Distribution of this information to third parties without authorization is prohibited and may result in sanctions imposed by the Competition Organization.

#### **D.2.4 Inverter**

D.2.4.1 | The inverter is defined as the electronic hardware responsible for controlling the torque and speed of the electric motor. The inverter and its control software may be either commercial or self-developed.

D.2.4.2 | The inverter must be installed between the accumulator (battery pack) and the electric motor. Direct electrical connection between the accumulator and the motor without an inverter is not allowed.

D.2.4.3 | Active inverters capable of controlling torque and speed are allowed.

D.2.4.4 | In case of an internal inverter fault or failure, the inverter must automatically command zero torque and zero speed demand to the motor.

D.2.4.5 | In case of a critical internal fault, the inverter shall trigger the opening of the HVS disconnection circuit by means of analog electrical logic. Digital communication alone shall not be accepted for this function.

#### **D.2.5 Throttle Control**

D.2.5.1 | The prototype must be equipped with a throttle potentiometer or equivalent throttle sensor including an automatic return feature and an integrated normally open on/off switch.

The switch must act as a safety enable device (dead-man function). When the switch is released, opened or the signal becomes invalid, the inverter must immediately command zero torque demand to the motor.

The throttle control must always return to zero torque demand when released.

D.2.5.2 | The throttle control must be located on the right-hand side of the handlebar.

D.2.5.3 | The throttle signal must be used to determine the torque or speed demand of the electric motor.

D.2.5.4 | The on/off switch signal must be directly connected to the inverter. Whenever the switch state is open, the inverter torque demand must immediately be set to zero.

D.2.5.5 | The throttle signal may be transmitted using analog or digital communication.

- D.2.5.6 | The inverter must command zero torque demand whenever any of the following failures are detected:
- Open circuit
  - Short circuit to ground or to supply voltage
  - Out-of-range throttle signal
  - Data corruption
  - Loss and/or delay of messages
  - Any other fault affecting the integrity of the control signal

## **D.2.6 Regenerative Braking**

- D.2.6.1 | It is allowed to configure regenerative braking or motor braking through the inverter control system.
- D.2.6.2 | An additional hand lever or thumb lever located on the left handlebar may be used to control motor braking.
- D.2.6.3 | If an additional lever is installed, a lever protection device must be fitted according to the requirements defined in Article B.5.1.6.
- D.2.6.4 | The use of regenerative braking to recover energy during deceleration is allowed.

## **D.2.7 Modifications**

- D.2.7.1 | Any modification to the supplied electric motor or its internal components is prohibited unless it is expressly permitted in these Technical Regulations.
- D.2.7.2 | Any modification intended to circumvent the spirit or intent of these Rules & Regulations may be considered a non-compliance by the Competition Organization.
- D.2.7.3 | In the event of any doubt or dispute regarding the interpretation of these provisions, the decision of the Competition Organization shall be final.

## **D.2.8 Electric Powertrain Operating Limits**

- D.2.8.1 | The electric powertrain must be operated at all times within safe operating limits regarding motor speed, electrical current, voltage and component temperatures.
- D.2.8.2 | The applicable operating limits for the supplied motor and the associated electric powertrain, motor speed, current, voltage and temperatures, shall be those established by the Competition Organization in the official technical documentation for the Electric Motor.

- D.2.8.3 | These operating limits are mandatory and binding for all teams. Their purpose is to ensure the safe, reliable and equal use of the supplied motor during development, testing and competition activities.
- D.2.8.4 | The inverter control system must ensure that torque demand remains within safe operating limits of the electric motor and associated components.
- D.2.8.5 | Teams are responsible for correctly monitoring the operation of the electric powertrain during all development, testing and Competition activities, and for ensuring that the motor, inverter and associated components are used within the applicable limits.
- D.2.8.6 | Operating practices that may cause abnormal electrical, thermal or mechanical stress to the electric powertrain are prohibited. These include, but are not limited to:
- operating the motor beyond its rated speed limits
  - excessive torque demand causing abnormal current peaks
  - operation outside the specified voltage limits
  - operation with insufficient cooling of the motor or inverter
  - operating the system with clearly unsafe electrical conditions
- D.2.8.7 | Exceeding the applicable operating limits or using the motor in a manner that may cause permanent damage shall not be considered a valid form of technical development or innovation within the MotoStudent Competition.
- D.2.8.8 | The Competition Organization reserves the right to review powertrain operating data, including but not limited to motor speed, electrical current, voltage and temperature values, in order to verify compliance with the applicable operating limits and to determine whether a failure may have resulted from operation outside such limits.
- D.2.8.9 | If damage to the motor or associated components is determined to be the result of operation outside these limits or abusive use of the system, the repair or replacement costs may be charged to the team.
- D.2.8.10 | Failure to comply with the applicable operating limits may be considered a technical non-compliance and may result in the corresponding measures or penalties established by the Competition Organization in accordance with these Competition Rules & Regulations.

### ARTICLE 3: ENERGY STORAGE

#### D.3.1 Permitted storage systems

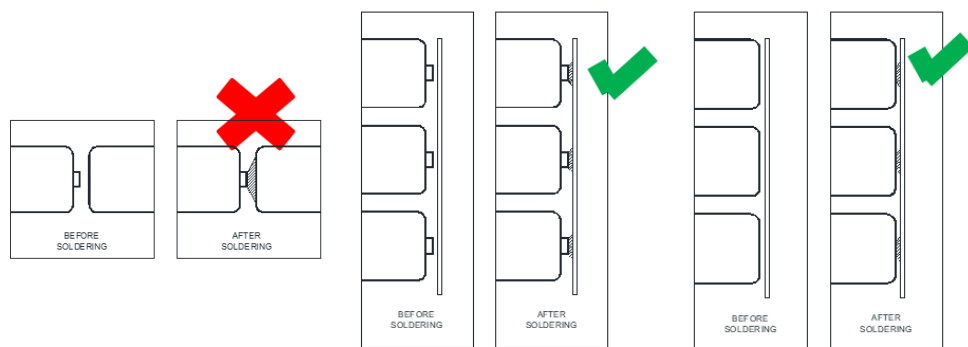
- D.3.1.1 | Any type of battery may be used as an energy storage system (including supercapacitors), except for molten salt batteries (thermal batteries) and fuel cells.
- D.3.1.2 | For the purposes of these Technical Regulations, the battery pack is the electric energy storage system used to supply traction power to the prototype, including its cells and / or supercapacitors, monitoring electronics, protective devices and associated connections.
- D.3.1.3 | Only one battery pack for traction purposes shall be installed and used in each prototype during the Competition.
- The battery pack presented at Static Technical Inspection shall be the only battery pack permitted for that prototype during the Final Event and may be identified, marked or sealed by the Competition Organization.
- The use of additional battery packs or the replacement of the declared battery pack during the Competition is not allowed unless expressly authorized by the Competition Organization for safety or technical reasons.
- Any repair or intervention on the declared battery pack shall require prior authorization from the Technical Director. Before any intervention is carried out, the team shall explain to the Competition Organization the intended repair actions and the applicable safety and verification procedure.
- D.3.1.4 | All energy storage systems must comply with the High Voltage limits and electrical safety requirements defined in Article D.1 of these Rules & Regulations.

## **D.3.2 Battery Container**

- D.3.2.1 | A battery container is any receptacle containing cells or supercapacitors that form part of the battery pack. All cells or supercapacitors belonging to the battery pack must be located inside one or more battery containers.
- A receptacle containing only contactors, fuses, busbars, electronics or other electrical components, but no cells or supercapacitors, shall not be considered a battery container.
- D.3.2.2 | A battery pack may be composed of multiple battery containers. Such containers may be installed separately or within a common external enclosure, provided that each battery container complies with the requirements established in these Technical Regulations.
- D.3.2.3 | Any battery container shall be capable of being opened during the Final Event for inspection purposes at the request of the Competition Organization.

### D.3.3 Electrical configuration of the battery pack

- D.3.3.1 | The terminals of the cells or supercapacitors must be properly protected and insulated using electrically insulating materials.
- D.3.3.2 | If the battery container is made of an electrically conductive material, the internal wall in contact with the battery cells must be properly insulated and protected against possible punctures or scratches. Prismatic cells with rigid electrically insulated housings are excluded from this requirement.
- D.3.3.3 | Any electrical connection between the battery pack and external systems (e.g. power cables, signal cables and / or charging connections) must include appropriate protection devices such as fuses. The fuse rating must be lower than the maximum allowable current of the circuit it is protecting.
- D.3.3.4 | The use of electronic fuses (eFuses) for High Voltage applications is forbidden.
- D.3.3.5 | Whenever the HVS disconnection circuit or the Charging disconnection circuit is set to open, no High Voltage shall be present in any element located outside the battery container(s), except as permitted in Article D.7.3.
- D.3.3.6 | Direct connection between cell terminals by welding or soldering is not allowed. Welding or soldering must be performed indirectly through conductive elements such as busbars, plates, cables or fusible links. Welding or soldering of BMS sensing wires to the terminals is allowed. In the case of pouch-type cells, direct welding between their tabs is allowed when permitted by the cell manufacturer.



Picture 34 - Correct battery cell welding

### D.3.4 Mechanical configuration of the battery container

- D.3.4.1 | Any battery container must be built using mechanically resistant materials and must be properly anchored to the prototype frame.

- D.3.4.2 | If the battery container is used as part of the structural frame of the prototype, it must comply with the appropriate stiffness and resistance requirements.
- D.3.4.3 | The battery container must be protected against impacts, punctures and external mechanical damage.
- D.3.4.4 | The battery cells must be securely fixed and protected against any relative movement inside the container.
- D.3.4.5 | Ventilation openings shall not cover an entire side of the battery container and shall incorporate a permanent filtering element to prevent the ingress of dust, particles and liquids. Cable pass-throughs shall not be considered ventilation openings. Any cable pass-through through the battery container shall be made using commercial cable glands, bulkhead connectors or equivalent commercial sealing elements with a minimum degree of protection of IP66D.
- D.3.4.6 | If the container is completely sealed, it must include a pressure relief or gas escape valve in order to prevent internal gas accumulation from reaching critical pressure levels.
- D.3.4.7 | The use or adaptation of commercially available containers or cases is allowed, provided that they comply with the requirements established in this article.
- D.3.4.8 | All battery container joints must be properly sealed to prevent the entry of dust, particles and liquids.
- D.3.4.9 | Any filtering element used must be properly dimensioned and protected to withstand rain, dust and possible projections (such as stones or debris). The removal or modification of these filtering and protective elements is strictly prohibited.
- D.3.4.10 | Battery containers and cells must be adequately protected against vibrations and mechanical stresses generated during prototype operation.

### **D.3.5 Battery Management System – BMS**

- D.3.5.1 | The Battery Management System (BMS) is defined as any system that monitors and / or controls every cell of the battery pack as well as the operating conditions of the battery system itself.
- D.3.5.2 | The installation of a Battery Management System (BMS) is compulsory.



- D.3.5.3 External forced cooling of the battery pack is allowed while the prototype is stationary, provided that the High Voltage System is disconnected and the prototype is in a safe state.
- Such external cooling may be used during charging or while the prototype is not in operation.
- Cooling systems integrated into the prototype and intended for normal operation of the motorcycle are permitted, provided that they comply with Article D.9.
- D.3.5.4 The BMS shall monitor, at minimum, the highest internal temperature within each battery container.
- D.3.5.5 A failure state of the BMS is defined as any overtemperature, overcurrent, overvoltage and / or undervoltage condition affecting any cell, component of the battery pack or battery container. At no time may these parameters exceed the limits specified by the cell manufacturer.
- D.3.5.6 The BMS must ensure cell balancing within the battery pack. Cell balancing may be performed either inside the battery pack by the BMS or externally by a dedicated balancing device. In both cases, the charging disconnection circuit must be active.
- D.3.5.7 Both active and passive cell balancing systems are allowed, provided that the components used are correctly rated and properly protected.
- D.3.5.8 When cell balancing or charging operations are performed outside the prototype, the HVS disconnection circuit shall ensure that no High Voltage is accessible outside the battery container(s), including through any balancing or charging connector.
- Any balancing connector providing access to the battery system shall be considered part of the HVS and shall be disconnected on all pins by the HVS disconnection circuit whenever it is set to open, as required by Article D.3.3.5.
- D.3.5.9 The BMS must be capable of detecting fault conditions and initiating appropriate protective actions, including disabling the High Voltage System when necessary.

#### **ARTICLE 4: DISCONNECTION SYSTEMS AND CIRCUITS**

##### **D.4.1 Emergency Shutdown Button (ESDB)**

- D.4.1.1 The ESDB must be a red mushroom-type push-rotary button (press to open and turn to close).

The minimum actuator diameter of the ESDB shall be 25 mm.



Picture 35 - ESDB button type

D.4.1.2 The ESDB shall act through an analogue LVS signal on the chain of the HVS disconnection circuit and shall immediately disconnect the High Voltage System.

D.4.1.3 The ESDB must be installed on a rigid and non-removable part of the prototype, such as a support fixed to the frame or another structural element of the motorcycle.

The ESDB must be located so that it is accessible to a track marshal in the event of a crash and reachable by the rider in case of system malfunction.

D.4.1.4 The ESDB must be protected against accidental activation caused by rider movement, impacts or contact with external elements, while remaining immediately accessible in an emergency.

The protection of the ESDB may be achieved by means of a mechanical guard, recessed installation, commercial protective housing or any equivalent design solution providing the same safety function.

D.4.1.5 Any protection system used for the ESDB must:

- minimize the risk of accidental activation during normal riding conditions
- keep the ESDB clearly visible and easily identifiable
- allow direct manual activation of the button by a track marshal wearing protective gloves
- not obstruct or delay the activation of the button in an emergency

As a minimum guideline, the protection should preserve free access to the upper part of the button and should not create obstacles that prevent direct activation with the palm of the hand.

A typical acceptable solution may consist of two side protective elements or an equivalent commercial housing that protects the button mainly from lateral contact while keeping the top area free for direct activation.

#### **D.4.2 Insulation Monitoring Device (IMD)**

D.4.2.1 | The insulation monitoring device (IMD) ensures the proper electrical isolation between the High Voltage System (HVS) and the chassis of the prototype.

D.4.2.2 | The installation of an IMD is mandatory.

D.4.2.3 | The IMD failure states are those defined in the datasheet published by the Competition Organization.

D.4.2.4 | If the IMD has any direct electrical connection to the High Voltage System, the IMD shall be installed inside the battery container. If the IMD is installed outside the battery container, all terminals and conductors connected to the High Voltage System shall be automatically de-energized by the HVS disconnection circuit whenever it is set to open.

#### **D.4.3 Traction System Master Switch (TSMS)**

D.4.3.1 | The Traction System Master Switch (TSMS) must be a mechanical switch.

D.4.3.2 | The TSMS must be located within reach of the rider and protected against accidental contact.

D.4.3.3 | The TSMS may be an analogue LVS switch acting on the HVS disconnection circuit chain. It shall not switch High Voltage directly.

#### **D.4.4 Line contactor**

D.4.4.1 | The line contactor consists of a normally open mechanical relay controlled by the HVS disconnection circuit.

D.4.4.2 | As minimum, the line contactor must have:

- A common terminal connected to the positive terminal of the battery pack.
- A normally open terminal directly connected to the powertrain terminal of the inverter.

- A coil terminal connected to LVS ground.
- A coil terminal connected to the HVS disconnection circuit output.

D.4.4.3 | The line contactor shall be installed inside one of the battery containers of the battery pack.

#### **D.4.5 Other HVS contactors**

D.4.5.1 | Normally open mechanical relays may be used for supplying low-power High Voltage components located outside the battery container (e.g. auxiliary power supply of the inverter).

For the purposes of the HVS disconnection circuit and the charging disconnection circuit, only normally open mechanical relays shall be accepted as safety switching devices. Solid-state relays, MOSFET-based switching devices or equivalent electronic switching solutions shall not be accepted as a substitute for the required mechanical relays.

D.4.5.2 | These contactors must be physically located inside the battery container.

D.4.5.3 | These contactors must be controlled by the HVS disconnection circuit so that they automatically open whenever the High Voltage System is disconnected.

D.4.5.4 | As minimum, the contactor must have:

- A common terminal connected to the positive terminal of the battery pack.
- A normally open terminal connected directly to the positive supply of the HVS component.
- A coil terminal connected to the HVS disconnection circuit output.

#### **D.4.6 Precharge circuit**

D.4.6.1 | It is mandatory to install a precharge circuit for the inverter. A precharge circuit is defined as any system that gradually introduces electrical charge to the motor inverter, equalizing voltages and preventing high inrush currents when the line contactors close.

D.4.6.2 | The minimum precharge level must reach at least 90% of the real battery pack voltage or reduce the voltage difference between terminals to less than 10 V.

D.4.6.3 | The precharge circuit must include, at minimum, a normally open mechanical relay referred to as the precharge contactor.

D.4.6.4 | It is permitted to use the precharge system integrated in the inverter if available. In such case, an additional precharge contactor is not required.

D.4.6.5 | The maximum time between closing the precharge circuit and closing the line contactor must not exceed 1 minute.

#### **D.4.7 HVS disconnection circuit**

D.4.7.1 | The HVS disconnection circuit manages the opening and closing of the precharge circuit (if present) and the line contactor(s). Its control must be implemented using analog electrical logic.

Digital systems may be used for monitoring or supervisory purposes, but they must not prevent, delay or override the shutdown of the High Voltage System.

D.4.7.2 | The HVS disconnection circuit must consist of at least:

- A Traction System Master Switch (TSMS)
- An Emergency Shutdown Button (ESDB)
- A normally open mechanical relay controlled directly by the Insulation Monitoring Device (IMD)
- A normally open mechanical relay controlled directly by the Battery Management System (BMS)

For the purposes of the HVS disconnection circuit and the charging disconnection circuit, only normally open mechanical relays shall be accepted as safety switching devices. Solid-state relays or equivalent electronic switching devices shall not be accepted as a substitute for the required mechanical relays.

D.4.7.3 | Whenever the HVS disconnection circuit opens, the line contactors and the precharge contactor (if present) must open immediately.

D.4.7.4 | The IMD relay must remain open whenever a failure state is detected by the IMD.

D.4.7.5 | The BMS relay must remain open whenever a failure state is detected by the BMS.

D.4.7.6 | When either the TSMS, ESDB, IMD relay or BMS relay changes to an open state, the HVS disconnection circuit must immediately open and disconnect the High Voltage System.

D.4.7.7 | Any additional components added to the HVS disconnection circuit must guarantee fail-safe behaviour. In case of malfunction, loss of signal or internal failure, the circuit must default to an open state that disconnects the High Voltage System.

The Competition Organization reserves the right to request modifications if any additional element compromises the safety or reliability of the shutdown system.

D.4.7.8 | In case of a failure in the BMS signal or IMD signal, the corresponding relays must default to an open state.

D.4.7.9 | The TSMS and the ESDB must be connected in series within the HVS disconnection circuit to ensure immediate shutdown.

D.4.7.10 | Whenever the LVS disconnection circuit state is set to open, the BMS and IMD relays must also default to an open state.

#### **D.4.8 LVS disconnection circuit**

D.4.8.1 | The LVS disconnection circuit must consist of at least:

- A Low Voltage System Master Switch (LVSMS).
- Any LV source present in the prototype (e.g. DC/DC converter or external LV battery)

D.4.8.2 | The LVSMS must be a switch designed to prevent self-consumption of the Low Voltage source when the system is switched off.

#### **D.4.9 Charging disconnection circuit**

D.4.9.1 | The charging disconnection circuit may use part or all of the HVS disconnection circuit and the LVS disconnection circuit. Its control must be implemented using analog logic.

D.4.9.2 | The charging disconnection circuit must consist of at least:

- Two Emergency Shutdown Buttons (ESDB)
- A normally open mechanical relay controlled directly by the Insulation Monitoring Device (IMD)
- A normally open mechanical relay controlled directly by the Battery Management System (BMS)
- A normally open mechanical relay controlled directly by the charger.

The charging disconnection function shall not rely solely on switching devices located outside the prototype. At least the required relay function affecting the prototype-side charging path shall be integrated in the prototype electrical architecture in accordance with these Technical Regulations.

D.4.9.3 | One ESDB must be located on the prototype.

D.4.9.4 | One ESDB must be located on the charger.

- D.4.9.5 | The IMD relay must open whenever a failure state is detected by the IMD.
- D.4.9.6 | The BMS relay must remain open whenever a failure state is detected by the BMS.
- D.4.9.7 | The charger relay must remain open whenever a failure state is detected by the charger or whenever the HVS disconnection circuit is open.
- D.4.9.8 | When either the ESDB, IMD relay, BMS relay or charger relay changes to an open state, the charging disconnection circuit must immediately open.
- D.4.9.9 | Any additional elements added to the charging disconnection circuit must guarantee immediate shutdown of the charging contactor whenever any of the protection devices is activated. The Competition Organization reserves the right to request modifications if any additional element compromises safety.
- D.4.9.10 | In the event of a failure in the BMS, IMD and / or charger signal, the charging disconnection circuit must default to an open state.
- D.4.9.11 | The charging circuit must ensure functional separation between the charging path and the traction powertrain. During charging, the traction powertrain, including the inverter / motor controller and motor supply path, must remain de-energized and incapable of generating torque.  
  
This requirement may be satisfied either by a dedicated charging relay / contactor or by a topology that physically disconnects the traction powertrain during charging.

## ARTICLE 5: BATTERY PACK CHARGING

### D.5.1 Charger types

- D.5.1.1 | The charger must have a rated input power less than or equal to 22 kW (maximum 32 rated amps in three-phase network side configuration).
- D.5.1.2 | The charger must be fitted with a respective ground conductor which must be duly protected and connected to the case of the charger.
- D.5.1.3 | The main connector can be single-phase (230 VAC, 50 Hz) or three-phase (400 VAC, 50 Hz).
- D.5.1.4 | No parallel configuration of different chargers is permitted. Series configuration is allowed.

- D.5.1.5 | It is allowed to store energy for charging purposes during the Final Event. These energy storages systems must comply with these regulations.

## **D.5.2 Connection to the prototype**

- D.5.2.1 | The charger, its connector and its cables are considered part of the High Voltage System (HVS).
- D.5.2.2 | The charging connector present on the prototype must be inaccessible when closed.
- D.5.2.3 | The charging connector must have a tightness degree of IP-65 when closed. When it is not used, a protective cap must be correctly installed.
- D.5.2.4 | The charging connector must be located in a protected area of the prototype to prevent damage due to crashes, contacts or impacts.
- D.5.2.5 | Charging must only be performed through the charging connector of the prototype. Direct connection to the battery pack or bypassing the charging connector is strictly forbidden.

## **D.5.3 Charging process**

- D.5.3.1 | A Team Member must be present during the entire charging process and must be prepared to react immediately to any abnormal condition.
- D.5.3.2 | Unattended charging of the prototype is strictly prohibited. At least one trained Team Member must supervise the charging process at all times.
- During the Final Event, charging operations are only permitted after the prototype has successfully passed the Static Technical Inspection and only between 08:00 and 20:00, in order to ensure that the Organization, medical services and fire safety personnel are present in the paddock in case of a safety incident.
- D.5.3.3 | It is mandatory to have a fire extinguisher with a minimum capacity of 2 kg suitable for electrical fires (CO<sub>2</sub>, F500 or equivalent). The extinguisher must be located at a maximum distance of 2 m from the prototype while charging. The use of extinguishers suitable for lithium battery fires is strongly recommended.
- It is mandatory to have a pair of High Voltage insulating gloves, class 0 CEI 60903-2002 or CEI 60903-2003 (or equivalent standard), available in the charging area during the charging process.



- D.5.3.4 | Charging may be carried out in the team's designated paddock or box area unless otherwise specified by the Organization.
- D.5.3.5 | During charging, the prototype must be located in a clearly identifiable charging area with a minimum free radius of 1 m around the motorcycle. This area must remain free of tools, equipment, fuel containers and non-essential personnel.
- While the prototype is connected to the charger, any manipulation of the High Voltage System, battery container or powertrain components is strictly forbidden.
- D.5.3.6 | Charging must not take place inside closed rooms, partitions or enclosed structures within the box that could limit ventilation, visibility or emergency access.
- D.5.3.7 | During the entire charging process, the prototype must remain stationary and the traction powertrain must remain de-energized. The entire HV system related to the powertrain (motor, motor controller, any HV signal and power wire connected to the motor or motor controller) must be at zero voltage during the entire charging process. The system configuration must ensure that no torque can be generated by the motor under any circumstance.
- The charging architecture must ensure functional separation between the charging path and the traction powertrain during the entire charging process.
- D.5.3.8 | The prototype, the charging connector, the charger and the charging cables must remain clearly visible and accessible to the Organization and scrutineers at all times during the charging process.
- The use of panels, covers or structures that obstruct visibility of the prototype or the charging process is forbidden. In particular, the LED status indicators of the High Voltage System must remain visible at all times.
- D.5.3.9 | Charging equipment must be positioned so that cables do not create tripping hazards or obstruct evacuation paths within the paddock or box.
- D.5.3.10 | Charging must be immediately stopped if any abnormal condition is detected, including smoke, unusual smell, abnormal temperature, noise, charger alarms or warnings from the Battery Management System (BMS).
- A failure state from the charger is defined as any current leak, internal component malfunction, battery pack under or over voltage, open circuit, overcurrent, undercurrent, or charger under or over temperature. If any failure state is detected, the charging process must be immediately stopped.
- D.5.3.11 | The Organization reserves the right to suspend or relocate charging operations if safety conditions are not met.

## ARTICLE 6: GENERAL INSTALLATION AND WIRING

### D.6.1 Wiring

- D.6.1.1 | Electrical wiring must be correctly dimensioned for the current, voltage and environmental conditions of its intended use.
- D.6.1.2 | The wire length must be appropriate for the installation. Excess cable length must not be coiled or loosely arranged and must be properly secured.
- D.6.1.3 | Any wiring routed near hot components must be adequately protected and rated for the expected temperatures.
- D.6.1.4 | Any wiring located near moving or rotating parts of the prototype must be securely fixed and protected against abrasion, contact or entanglement.
- D.6.1.5 | All wiring must be properly supported along its routing. Loose or unsupported wiring is forbidden.
- D.6.1.6 | High Voltage System (HVS) wiring must be clearly separated from Low Voltage System (LVS) wiring wherever possible.
- D.6.1.7 | Wiring insulation must not show visible damage, wear or deterioration.

### D.6.2 Overcurrent protection

- D.6.2.1 | All electrical systems must include appropriate overcurrent protection devices.
- D.6.2.2 | The current rating of any overcurrent protection device must not exceed the maximum current rating of the smallest conductor or device it protects.
- D.6.2.3 | The maximum permitted voltage rating of any overcurrent protection device must be higher than the highest voltage present in the electrical system it protects.
- D.6.2.4 | Overcurrent protection devices must be easily accessible for inspection by the Organization.

## ARTICLE 7: CONTROL AND CONTROL DEVICES

### D.7.1 LEDs Cluster

- D.7.1.1 | A cluster of LEDs (hereinafter “LED cluster”) is a group of light-emitting diodes intended to indicate the operational state of the prototype (safe, engaged, not safe, charging).

- D.7.1.2 | The LED cluster must indicate the following states:
- Safe: whenever the TSMS, line contactor and precharge contactor are open and the IMD relay, BMS relay, charger relay and all ESDB are closed.
  - Engaged: whenever the line contactor or precharge contactor are closed.
  - Not Safe: whenever the IMD relay, BMS relay, charger relay or ESDB are open.
  - Charging: whenever the charging contactor is closed.
- D.7.1.3 | The LED cluster must show a continuous green light whenever the Engaged state is active.
- D.7.1.4 | The LED cluster must show a blinking green light (1 Hz frequency) whenever the Safe state is active.
- D.7.1.5 | The LED cluster must show a continuous red light whenever the Not Safe state is active.
- D.7.1.6 | The LED cluster must show a continuous blue light whenever the Charging state is active.
- D.7.1.7 | Two states cannot be active at the same time.
- D.7.1.8 | All LEDs within a cluster must be located close to each other with a maximum separation of 10 mm.
- D.7.1.9 | Three LED clusters must be installed on the prototype:
- One on each side of the prototype, securely mounted to the frame or battery pack.
  - One on the motorcycle dashboard.
- D.7.1.10 | The LED clusters must remain clearly visible from a distance of at least 5 m by any third party or track marshal in normal conditions, in the starting grid and in the event of a crash.
- For the two LED clusters located on the sides of the prototype, visibility must be ensured when the rider is seated in the normal riding position and when standing on the motorcycle.
- D.7.1.11 | The LED clusters must have sufficient luminous intensity to remain clearly visible in daylight conditions. The Technical Director of the Electric category reserves the right to require modifications if visibility is deemed insufficient.

D.7.1.12 | In case of failure in the signal, wiring or logic of any LED within the cluster, the LED must default to the OFF state.

D.7.1.13 | If no LED from the cluster is active, the prototype state shall be considered Not Safe.

## **D.7.2 Voltage Display**

D.7.2.1 | A display (hereinafter “Voltage Display”) showing the total battery pack voltage must be installed on the prototype.

D.7.2.2 | The Voltage Display must be visible to the rider in the normal riding position.

D.7.2.3 | The Voltage Display must also be clearly visible to scrutineers or track marshals from outside the prototype, including in the event of a crash.

D.7.2.4 | If the Voltage Display reads the voltage value from a digitally transmitted signal, appropriate measures must be implemented to prevent data corruption, message loss and / or communication delays.

D.7.2.5 | The Voltage Display must measure the voltage upstream of the line contactor and charging relay so that it always displays the battery pack voltage.

## **D.7.3 Voltage Measuring Point – VMP**

D.7.3.1 | The Voltage Measuring Point (VMP) must consist of a panel connector with at least four (4) pins, with a minimum protection rating of IP66D and a voltage rating above 500 VDC.

D.7.3.2 | The VMP must be connected to the following points:

- The normally open terminal of the line contactor, referred to as “B+”.
- The normally open terminal of the charging relay, referred to as “C+”.
- The HVS ground, referred to as “B-”.
- The common terminal of the line contactor, referred to as “HV+”.

If a design reuses the main line contactor for charging purposes, the team must demonstrate that the traction powertrain is physically disconnected during charging and that the measurement points B+ and C+ still represent the functional nodes defined in this article.

D.7.3.3 | The VMP must be clearly marked with its acronym.

D.7.3.4 | The VMP must be located in each battery container and must be accessible by the Organization or scrutineers without the use of tools.

- D.7.3.5 | The VMP connector shall be wall-mounted to the battery container(s), provided with a protective cover, and positioned in an area not likely to be damaged or rendered inaccessible in the event of a fall or impact.
- D.7.3.6 | The “HV+” pin must be protected through a 120 kΩ resistor.
- D.7.3.7 | In order to allow scrutineers to perform voltage measurements, each team must provide the mating connector of the VMP connected to a 4 mm shrouded banana plug.



*Picture 36 - 4mm shrouded banana plug example*

- D.7.3.8 | Each cable from the mating connector must be clearly marked indicating the corresponding VMP pin. The cable assembly must also be labelled with the Team name.
- D.7.3.9 | The cable length from the mating connector must be at least 500 mm.
- D.7.3.10 | The VMP mating connector cable must be available to the Organization at any time during the Final Event for voltage measurement and verification procedures. At any time, the organization could test if the voltage measured at the connector is the same that one obtained at the HV system.

## ARTICLE 8: TRANSMISSION SYSTEM

### D.8.1 Types of transmission

- D.8.1.1 | Any type of primary transmission system is allowed (gearbox, CVT variator, reduction gear, etc.).
- D.8.1.2 | Direct transmission between the output shaft of the motor and the rear wheel is allowed.
- D.8.1.3 | There are no limitations regarding the installation of clutch elements between components of the transmission system.

D.8.1.4 | All transmission components must be securely mounted and designed to withstand the expected mechanical loads during operation.

D.8.1.5 | Any transmission element mounted directly on the motor output shaft (such as sprockets, pulleys or couplings) must be securely fixed using an appropriate mechanical locking system. The design must prevent loosening, axial displacement or detachment during operation.

## **D.8.2 Secondary transmission**

D.8.2.1 | There is no limitation regarding the type of secondary transmission used (chain, belt or equivalent systems).

D.8.2.2 | Any exposed transmission component that may present a trapping or contact risk for the rider must be protected by a rigid guard securely fixed by mechanical means.

D.8.2.3 | Guards must remain securely fixed during operation and must prevent accidental contact with moving transmission components.

D.8.2.4 | The design of the transmission system must prevent components such as chains or belts from contacting the rider in case of failure or derailment.

## **ARTICLE 9: COOLING SYSTEM**

### **D.9.1 Overview of the cooling systems**

D.9.1.1 | The design of the cooling system for the different components is freely configurable.

D.9.1.2 | Cooling of components may be achieved by air or liquid cooling systems.

D.9.1.3 | Multiple independent cooling loops are permitted.

D.9.1.4 | External cooling devices may be connected to the prototype only while the prototype is stationary and not in operation.

D.9.1.5 | The use of dry ice or equivalent external cooling media is permitted only for external cooling while the prototype is stationary. Such systems must not release liquids or slippery substances onto the ground or the track and must not create a safety risk for persons or equipment.

### **D.9.2 Liquid cooling systems**

- D.9.2.1 | Only distilled water may be used as cooling liquid.
- D.9.2.2 | The use of glycol, antifreeze and / or any slippery additives in the cooling liquid is strictly forbidden.
- D.9.2.3 | The cooling system must be designed and installed so as to prevent coolant leakage onto the track.
- D.9.2.4 | Each liquid cooling loop must include an expansion tank or recovery container with sufficient capacity to safely contain the expected coolant expansion and overflow of that loop under all operating conditions.
- The capacity of the expansion tank or recovery container is free, provided that the team demonstrates that it is adequate for the design of its cooling system.
- External cooling devices used only while the prototype is stationary are not required to include an expansion tank, provided that they do not release coolant or any other liquid onto the ground.
- If multiple liquid cooling loops are installed, each loop must independently comply with the applicable requirements of this Article.

## ARTICLE 10: DOCUMENTATION OF THE PROTOTYPE

### D.10.1 Documents to be submitted

- D.10.1.1 | Teams participating in the MotoStudent Electric category must submit additional technical documentation related to the electric prototype as required by the Organization.
- The documentation to be submitted and the corresponding deadlines are defined in Article A.6.5.
- D.10.1.2 | For safety reasons, the Technical Staff of the Competition Organization may request additional technical information or documentation beyond that described in this article if deemed necessary for the inspection or verification of any prototype.

# **SECTION E**

## **SCRUTINEERING**



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## ARTICLE 1: OBJECTIVE AND METHODOLOGY

### E.1.1 Objective

- E.1.1.1 | The objective of the Scrutineering is to verify that the prototypes presented by the participating teams comply with the performance and safety requirements established in the Technical Regulations (Sections B, C and D), and to verify that the rider designated by the team is eligible to participate in the MS2 Phase of the Competition.
- E.1.1.2 | Scrutineering does not award points in the MotoStudent International Competition. However, it may result in the exclusion of a prototype or rider if any deviation from the Rules & Regulations is detected or if the prototype is not considered sufficiently safe to participate in the tests.
- E.1.1.3 | Scrutineering will be carried out by experienced scrutineers and qualified Technical Staff of the Competition Organization.
- E.1.1.4 | In the event of any dispute regarding compliance with the Technical Regulations or the safety of a prototype on track, the decision of the Technical Staff of the Competition Organization shall be final.
- E.1.1.5 | The values measured using the Organization's instruments shall be considered final.
- E.1.1.6 | Passing Scrutineering does not constitute technical approval, certification or homologation of the prototype design by the Competition Organization.
- Scrutineering only verifies compliance with the Technical Regulations based on the inspections and tests carried out at the moment of the check.
- E.1.1.7 | The responsibility for the design, construction, safety and operation of the prototype remains entirely with the team at all times.
- The fact that a prototype has passed Scrutineering or has been allowed to participate in any activity of the Competition does not transfer any responsibility to the Competition Organization.
- E.1.1.8 | Scrutineering approval only applies to the condition of the prototype at the time of inspection.
- Any modification, adjustment, repair or replacement of components carried out after the inspection may require a new verification by the Technical Staff of the Organization.

- E.1.1.9 | The Competition Organization reserves the right to perform additional inspections at any time during the Competition if it considers that the safety or regulatory compliance of a prototype may have been affected.

### **E.1.2 Responsibility of the teams**

- E.1.2.1 | Teams are solely responsible for ensuring that their prototype complies with all the requirements established in the Technical Regulations of the Competition.

- E.1.2.2 | By presenting a prototype for Scrutineering, the team declares that it has verified that the prototype complies with the Rules & Regulations established by the Competition Organization.

- E.1.2.3 | Teams must present the prototype in a condition suitable for inspection and must facilitate access to all areas, systems and components that may need to be checked during Scrutineering.

- E.1.2.4 | Teams must provide any technical information, documentation, tools, connectors or support elements that may be required by the Competition Organization to carry out the inspection.

### **E.1.3 Procedure**

- E.1.3.1 | Scrutineering consists of the following stages:
- Administrative Check
  - Static Scrutineering
  - Dynamic Safety Check

- E.1.3.2 | A prototype may be rejected at any stage of Scrutineering if it does not comply with the Rules & Regulations or if it is considered unsafe.

- E.1.3.3 | If a non-conformity is detected, the Competition Organization may allow the team to correct the issue and present the prototype again for reinspection, provided that the official schedule allows it.

- E.1.3.4 | The order, scope and detailed methodology of the Scrutineering may be defined by the Competition Organization in the corresponding official documentation.

## **ARTICLE 2: ADMINISTRATIVE CHECK**

### **E.2.1 Process**

- E.2.1.1 | At the beginning of the Final Event, teams must perform the Administrative Check, during which the Competition Organization will verify that the rider and the team

comply with the administrative and federative requirements established for participation in the Competition.

- E.2.1.2 | Once these requirements have been verified and approved, the corresponding Administrative Check sticker will be placed on the prototype, allowing the team to proceed to the next Scrutineering stage.



*Figure 37 - Administrative Check sticker example*

- E.2.1.3 | The Administrative Check will be carried out at the Paddock General Office.
- E.2.1.4 | Teams will not be allowed to access the Static Scrutineering stage without having successfully passed the Administrative Check.
- E.2.1.5 | During the Administrative Check, the Organization may verify the following documentation:
- Rider Identification Document (ID card or passport)
  - Rider's federative license
  - "Start permission" issued by the rider's national federation when required
  - Any additional documentation required by the Competition Organization
- E.2.1.6 | If the rider arrives at the Final Event later than the rest of the team, the team may perform the Administrative Check by presenting the rider's identification document, federative license and start permission (if required).
- In such case, the rider must personally attend the Administrative Check location upon arrival to confirm participation and complete the required signatures.
- E.2.1.7 | The Competition Organization reserves the right to request additional verification of the rider eligibility or documentation if necessary.
- E.2.1.8 | Teams that fail to comply with the administrative or federative requirements may not proceed to the Static Scrutineering stage until the situation has been resolved.
- E.2.1.9 | The schedule, time slots and operational procedure for the Administrative Check will be defined in the official documentation of the Final Event.

## ARTICLE 3: STATIC SCRUTINEERING

### E.3.1 Process

- E.3.1.1 | Static Scrutineering consists of a series of checks performed on the prototype to verify its compliance with the Competition Technical Regulations and to assess whether the prototype is safe to participate in the MS2 Phase.
- E.3.1.2 | Rider protective equipment may also be verified during Static Scrutineering to confirm that it is appropriate for participation in the Competition.
- E.3.1.3 | The scrutineering checks described in this article apply to prototypes presented in both the MotoStudent eFuel and MotoStudent Electric categories.

### E.3.2 Development

- E.3.2.1 | Static Scrutineering will be carried out in the Technical Control Area designated by the Competition Organization.
- E.3.2.2 | The prototype must be presented in a condition ready for inspection and suitable for participation in the MS2 Phase, complying with all the requirements established in the Competition Rules & Regulations.
- E.3.2.3 | The prototype must be fully assembled and capable of being inspected without requiring major disassembly operations. Scrutineers may request the removal of fairings or other components if required for inspection.
- E.3.2.4 | The use during the Competition of components or configurations that have not been verified during Static Scrutineering may result in exclusion from the Competition.
- E.3.2.5 | Only two (2) Team Members are allowed to access the Technical Control Area during Static Scrutineering. These members will be responsible for transporting the prototype and presenting it in a stable static position suitable for inspection.
- E.3.2.6 | The team must bring all tools necessary to remove the prototype fairings and to provide access to the components required for inspection.

### E.3.3 Visual check

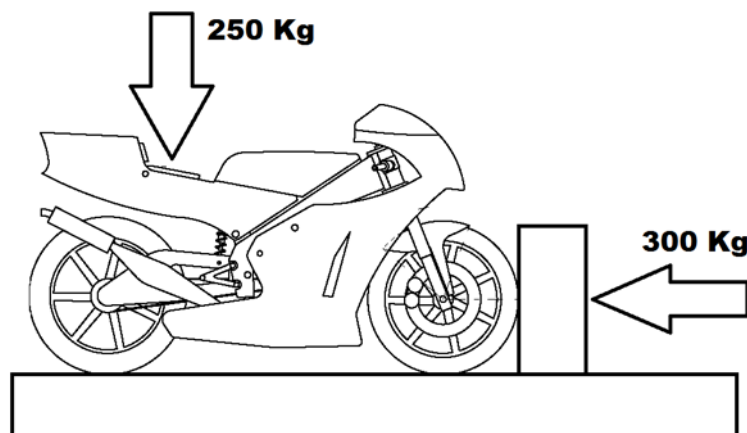
- E.3.3.1 | During this check, scrutineers will visually inspect the prototype in order to verify compliance with the Technical Regulations defined in Section B and other applicable sections of the Rules & Regulations.

#### **E.3.4 Weight check**

- E.3.4.1 | During this check, the prototype will be weighed using two scales, one placed under the front wheel and the other under the rear wheel.
- E.3.4.2 | The total weight of the prototype will be determined as the sum of the values measured on both scales.
- E.3.4.3 | The measured weight must be equal to or greater than the minimum weight defined in Article B.2.2.1.
- E.3.4.4 | The weight measured using the Organization's scales will be considered final.

#### **E.3.5 Load check**

- E.3.5.1 | During this check, the prototype will be subjected to horizontal and vertical loads in order to verify the structural rigidity of the frame, the correct assembly of the main structural elements and the behaviour of the suspension systems.
- E.3.5.2 | The following loads will be progressively applied to the prototype in order to evaluate structural behaviour:
  - 250 kg applied on the saddle area
  - 300 kg applied on the front wheel



*Picture 38 - Load test forces diagram*

- E.3.5.3 | These loads will be applied progressively and alternately while verifying that no structural failure, interference between components or unsafe deformation occurs.  
  
If any unsafe structural behaviour is detected during this test, the scrutineers may stop the test and declare the prototype non-compliant.

#### **E.3.6 Brake check**

- E.3.6.1 | During this check, the braking performance of the front and rear brake systems will be tested using a roller brake tester.



Picture 39 - Brake check

- E.3.6.2 | The braking force measured must exceed the following minimum values:

- Front axle: 0.3 kN
- Rear axle: 0.25 kN

If abnormal behaviour of the braking system is detected during the test (such as inconsistent force, excessive lever travel or system malfunction), the scrutineers may declare the prototype non-compliant even if the minimum values are reached.

- E.3.6.3 | The values measured using the Organization's equipment will be considered final.

### E.3.7 Approval

- E.3.7.1 | If the prototype successfully passes all the static checks described in this article (including those specific to each category), the corresponding Static Scrutineering sticker will be placed on the prototype.

Once approved, the prototype will be allowed to proceed to the Dynamic Safety Check stage.



Picture 40 - Static Scrutineering sticker

## ARTICLE 4: STATIC SCRUTINEERING SPECIFIC TO “MOTOSTUDENT eFUEL” CATEGORY

### E.4.1 Visual check

- E.4.1.1 | During this check, scrutineers will visually inspect the prototype in order to verify compliance with the Technical Regulations defined in Section C.
- E.4.1.2 | This inspection will be performed as part of the visual inspection described in Article E.3.3.

### E.4.2 Noise check

- E.4.2.1 | During this check, the exhaust noise emitted by the prototype will be measured.
- E.4.2.2 | The maximum permitted exhaust noise level is defined in Article C.6.2 of the Technical Regulations.
- E.4.2.3 | The sound level meter will be positioned approximately 50 cm from the end of the exhaust outlet at an angle of 45° relative to the centreline of the exhaust pipe and at the same height as the exhaust outlet, but at least 20 cm above the ground.  
  
If this position cannot be achieved, the measurement may be taken at a 45° upward angle from the exhaust outlet.
- E.4.2.4 | The noise test must be carried out in an open area adjacent to the Technical Control Area, at least 5 meters away from any large reflecting surface such as walls, vehicles or structures.
- E.4.2.5 | The engine must run out of gear. The rider or a team member must increase the engine speed until the specified test RPM defined in the Technical Regulations is reached. The sound measurement will be taken once the engine speed stabilizes at the specified RPM.
- E.4.2.6 | During the test, the ambient noise level must not exceed 90 dB(A) within a radius of 5 meters around the measurement area.
- E.4.2.7 | The sound level meter used by the Competition Organization must be equipped with a calibrator for verification and adjustment during the test period.
- E.4.2.8 | The sound level meter must be configured to use the “slow response” setting and A-weighted measurement scale.

- E.4.2.9 | The value measured using the Organization's sound level meter will be considered final.
- E.4.2.10 | If the prototype exceeds the permitted noise level, the team may be allowed to perform corrective actions and repeat the test if the event schedule allows it.

## ARTICLE 5: STATIC SCRUTINEERING SPECIFIC TO "MOTOSTUDENT ELECTRIC" CATEGORY

### E.5.1 Electric visual check

- E.5.1.1 | During this check, scrutineers will visually inspect the prototype in order to verify compliance with the Technical Regulations defined in Section D.
- E.5.1.2 | This inspection will be performed as part of the general visual inspection described in Article E.3.3.

### E.5.2 Isolation resistance measurement check

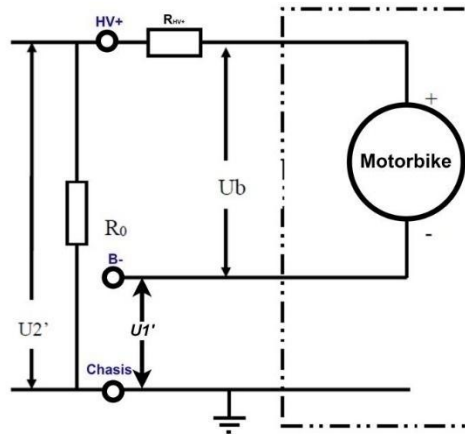
- E.5.2.1 | During this check, the scrutineers will verify the electrical isolation between the High Voltage System (HVS) and the prototype chassis (also the Low Voltage System - LVS).
- E.5.2.2 | The isolation resistance between the High Voltage System and the prototype chassis must be at least 100  $\Omega/V$  of the maximum battery pack voltage.
- E.5.2.3 | The isolation measurement will be performed using the Voltage Measuring Point (VMP) defined in Article D.7.3.
- E.5.2.4 | The measurement method will consist of the following steps:
- The battery pack voltage will be recorded (hereinafter  $U_b$ ) between the "HV+" and "B-" terminals of the VMP.
  - The voltage between the "B-" terminal of the VMP and the prototype chassis will be recorded (hereinafter  $U_1$ ).
  - The voltage between the "HV+" terminal of the VMP and the prototype chassis will be recorded (hereinafter  $U_2$ ).
  - A resistor  $R_o$  will be connected between the terminal presenting the highest voltage value relative to the prototype chassis and the prototype chassis.
  - The voltage will then be measured again after the resistor is connected ( $U_1'$  or  $U_2'$ ).
  - The insulation resistance  $R_i$  will then be calculated according to the following formulas:



$$R_i = R_o * U_b * \left( \frac{1}{U_2'} - \frac{1}{U_2} \right) \text{ if } U_2 > U_1$$

$$R_i = R_o * U_b * \left( \frac{1}{U_1'} - \frac{1}{U_1} \right) \text{ if } U_1 > U_2$$

- The isolation value will be obtained by dividing  $R_i$  by  $U_b$ .



Picture 41 - Test example for measuring  $U_2'$  and  $U_1'$

- E.5.2.5 | The resistor  $R_o$  used for this test must have a value of 15 k $\Omega$ .

### E.5.3 IMD check

- E.5.3.1 | During this check, the scrutineers will verify the correct operation of the Insulation Monitoring Device (IMD) and the High Voltage System disconnection circuit.
- E.5.3.2 | The test will consist of connecting the terminal with the lowest voltage recorded during the previous isolation test ( $U_1$  or  $U_2$ ) to the prototype chasis through a 50 k $\Omega$  resistor.
- E.5.3.3 | When this condition is applied, the IMD must detect the insulation fault and trigger the opening of the HVS disconnection circuit.
- E.5.3.4 | The HVS disconnection circuit must open within a maximum time of 30 seconds.
- E.5.4.5 | If the IMD does not trigger the shutdown of the High Voltage System within the specified time, the prototype will be considered non-compliant.
- E.5.3.6 | The Emergency Shutdown Button (ESDB) and the Traction System Master Switch (TSMS) will also be tested to verify that they correctly open the HVS disconnection circuit.
- E.5.3.7 | A disconnection circuit will be considered open when both the line contactor and the charging relay are open.

- E.5.3.8 | This condition will be verified by measuring the voltage at the VMP terminals:
- The voltage between “B+” and “B–” must be equal to 0 V.
  - The voltage between “C+” and “B–” must be equal to 0 V.
- E.5.4 Rain check**
- E.5.4.1 | This check applies only to prototypes participating in the MotoStudent Electric category.
- E.5.4.2 | The purpose of this test is to verify that the electrical insulation of the prototype remains safe under rain or high humidity conditions.
- E.5.4.3 | During this test, the High Voltage System must be activated and the drive wheel must be lifted off the ground using an appropriate stand.
- E.5.4.4 | Water will be sprayed over the prototype to simulate light rain conditions from different directions for a minimum duration of 60 seconds. High-pressure water jets must not be used.
- E.5.4.5 | The test will be considered successful if the IMD does not trigger during the 60 seconds of spraying and during the following 60 seconds after the spraying has stopped.
- The minimum total duration of the test will therefore be 120 seconds.
- E.5.4.6 | The Technical Staff may also verify that no water accumulates in areas that could compromise the safety of the High Voltage System.
- E.5.4.7 | Verification that the High Voltage System remains active during the test may be performed by:
- confirming that the motor can operate normally
  - measuring the voltage between “B+” and “B–” at the VMP

## ARTICLE 6: DYNAMIC SAFETY CHECK

- E.6.1 Process**
- E.6.1.1 | The Dynamic Safety Check consists of a series of on-track verifications performed by Official Riders appointed by the Competition Organization.
- E.6.1.2 | The purpose of this stage is to verify that the prototype operates correctly under basic dynamic conditions and that it can safely circulate on track.

E.6.1.3 | The checks described in this article apply to prototypes presented in both the "MotoStudent eFuel" and the "MotoStudent Electric" categories.

E.6.1.4 | The Dynamic Safety Check is not intended to evaluate the performance of the prototype but only to verify its basic operational safety.

## **E.6.2 Procedure**

E.6.2.1 | An Official Rider appointed by the Organization will perform a series of controlled dynamic tests on the track in order to verify the correct functioning of the prototype.

E.6.2.2 | During these tests the following basic aspects will be verified:

- The prototype must start and move under its own power.
- The prototype must respond correctly to throttle input.
- The prototype must brake correctly under both moderate and higher braking conditions.
- The prototype must be able to complete several turns at different speeds with stable and predictable behaviour.

E.6.2.3 | If the Official Rider detects any malfunction, instability or unsafe behaviour, the test may be stopped immediately.

The Official Rider will determine whether the prototype meets the minimum safety requirements necessary to participate in the Competition.

## **E.6.3 Lap around the track**

E.6.3.1 | Once the basic checks have been completed, the Official Rider may perform a complete lap of the Speed Racing Track in order to evaluate the general behaviour of the prototype under real track conditions.

The completion of this lap is at the discretion of the Official Rider and the Technical Staff of the Organization.

## **E.6.4 Approval**

E.6.4.1 | If the prototype successfully passes the Dynamic Safety Check, it will be marked with the corresponding approval sticker.

Once approved, the team will be allowed to participate in the MS2 Phase of the Competition.



Picture 42 - Example of approval sticker

## ARTICLE 7: CHANGES AND REPAIRS

### E.7.1 Rectification time

- E.7.1.1 | If a prototype is considered "Not Approved" after any Scrutineering stage, the Organization may grant the team a rectification period in order to correct the detected non-compliances.
- E.7.1.2 | During this period, the team must perform the required corrections in their designated paddock or box area.
- E.7.1.3 | Once the rectification has been completed, the prototype must be presented again at the scrutineering area during the next assigned inspection slot in order to verify compliance with the Rules & Regulations.
- E.7.1.4 | The Organization reserves the right to limit the number of re-inspection attempts depending on the event schedule.
- E.7.1.5 | No prototype will be allowed to participate in the MS2 Phase without successfully passing all required Scrutineering stages within the established timetable.
- E.7.1.6 | The MotoStudent Organization reserves the right to refuse access to scrutineering for any prototype once the official scrutineering schedule has concluded.

### E.7.2 Changes and repairs

- E.7.2.1 | Once a prototype has successfully passed scrutineering and has been approved for participation in the MotoStudent Competition, no modification or replacement of components may be carried out without the authorization of the Technical Director.
- E.7.2.2 | If a team intends to perform any modification or repair, the Technical Staff of the Organization must be informed beforehand.

- E.7.2.3 | The Organization may require the prototype to be re-inspected before it is allowed to continue participating in the Competition.
- E.7.2.4 | The following operations are allowed after scrutineering without requiring prior approval from the Organization, provided that they do not alter the safety or structural characteristics of the prototype:
- Secondary transmission adjustments
  - Brake system adjustments
  - Data acquisition or telemetry adjustments
  - Tire pressure adjustments
  - Refilling of fluids
  - General setup adjustments
  - Engine or powertrain setup adjustments
- E.7.2.5 | Any repair affecting structural, safety-critical or electrical High Voltage components must be approved by the Technical Director before the prototype may return to the track.
- E.7.2.6 | The Technical Director reserves the right to require a partial or full re-inspection of the prototype after any repair.

## ARTICLE 8: ADDITIONAL SCRUTINEERING ROUNDS

### E.8.1 Additional inspection rounds

- E.8.1.1 | The Organization may perform additional scrutineering inspections at any time during the Final Event in order to verify continued compliance with the Technical Regulations.
- E.8.1.2 | These inspections may be carried out in the paddock, in the Technical Control Area or on track.
- E.8.1.3 | Teams must cooperate with the Technical Staff during these inspections and provide access to the prototype if requested.
- E.8.1.4 | If a non-compliance is detected during an additional inspection round, the Organization may require corrective actions or suspend the prototype from participating in track activities until the issue has been resolved.

## ARTICLE 9: TIME CONTROL PROCEDURE

### E.9.1 Inspection scheduling

E.9.1.1 | At the beginning of the Final Event, each team will be assigned a time slot for their first scrutineering inspection.

E.9.1.2 | The initial inspection schedule will be determined by the Organization and may be based on accreditation time or other operational criteria defined by the Organization.

E.9.1.3 | Teams must present their prototype at the scrutineering area at the assigned time slot.

#### **E.9.2 Missed inspection slots**

E.9.2.1 | If a team fails to present its prototype at the assigned inspection time, the team may lose its priority and will be reassigned a new time slot depending on the availability of the scrutineering schedule.

E.9.2.2 | The Organization is not responsible for delays caused by teams missing their assigned inspection slots.

#### **E.9.3 Re-inspection scheduling**

E.9.3.1 | If a prototype does not pass an inspection stage, the team will be assigned a new inspection time slot for the next verification round.

E.9.3.2 | Re-inspection slots will be assigned by the Organization according to the availability of the scrutineering schedule.

#### **E.9.4 Publication of schedules**

E.9.4.1 | The scrutineering schedules and re-inspection time slots will be published, if needed, by the Organization through the official communication channels of the event.

E.9.4.2 | Teams are responsible for checking the assigned times and presenting their prototype accordingly.

# **SECTION F**

## **COMPETITION STRUCTURING**

## ARTICLE 1: OBJECTIVES

### F.1.1 Competition Objectives

F.1.1.1 The MotoStudent International Competition is an advanced educational program based on the complete development of a real racing prototype.

Its objective is to prepare young talent for their future professional careers, especially in the mobility and motorsport sectors, by requiring them to work in multidisciplinary teams, make decisions under technical, economic and time constraints, and justify those decisions in a competitive international environment.

Through MotoStudent, participants are expected to develop practical skills in engineering, project management, communication, innovation, economic planning, resource management and teamwork, in a context that resembles the challenges of a real industrial project.

### F.1.2 MS1 Phase objectives

F.1.2.1 The objective of the MS1 Phase is to evaluate the quality, maturity and coherence of the team's project development process before the Final Event.

F.1.2.2 MS1 is intended to guide teams through the full path of a real prototype development project, from the initial team organization and concept definition to design justification, innovation, technical documentation, economic viability, marketing and final communication of the project.

F.1.2.3 The MS1 Phase is specially focused on assessing how teams think, justify, plan, develop and communicate their project. It is not about which team has the greatest means, but about which team makes the best use of the means available to it.

F.1.2.4 A strong MS1 project should help the team arrive better prepared for the MS2 Phase, reducing uncertainty, improving decision-making and increasing the consistency between the original concept, the manufactured prototype and its final on-track performance.

F.1.2.5 Teams must therefore be able to demonstrate to the jury their full project development process, including the technical, organizational, marketing, economic and strategic decisions that support the prototype presented in the Competition.

### F.1.3 MS2 Phase objectives



- |         |                                                                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.1.3.1 | The objective of the MS2 Phase is to evaluate the dynamic behaviour, performance, reliability and competitive consistency of the manufactured prototype under real track conditions.                                  |
| F.1.3.2 | The MS2 Phase is intended to validate, under racing conditions, the ideas, decisions and solutions developed during the MS1 Phase.                                                                                    |
| F.1.3.3 | Through the different MS2 tests, teams must demonstrate that their prototype is not only theoretically well-conceived, but also effectively built, safe, functional and competitive in a real motorsport environment. |

## ARTICLE 2: PHASES OF THE COMPETITION

### F.2.1 MS1 Milestones

- |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.2.1.1 | <p>The MotoStudent Phase 1 (<b>MS1</b>) is structured as a sequence of project milestones intended to guide teams through the development of their prototype and its associated project.</p> <p>Each milestone evaluates a specific aspect of the project development, including team organization, engineering design, innovation, technical documentation, economic viability, communication and marketing.</p>                                                                                                                    |
| F.2.1.2 | <p>The MS1 Phase is divided into the following milestones:</p> <ul style="list-style-type: none"><li>- MS1 1<sup>st</sup> milestone: Team plan.</li><li>- MS1 2<sup>nd</sup> milestone: Product definition.</li><li>- MS1 3<sup>rd</sup> milestone: Innovation.</li><li>- MS1 4<sup>th</sup> milestone: Prototype CAD and drawings.</li><li>- MS1 5<sup>th</sup> milestone: Economical plan.</li><li>- MS1 6<sup>th</sup> milestone: Pitch presentation.</li><li>- MS1 7<sup>th</sup> milestone: Final event presentation.</li></ul> |

### F.2.2 MSE Deliveries

- |         |                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.2.2.1 | <p>Teams competing in the MotoStudent Electric (<b>MSE</b>) category must complete the following additional technical deliveries in order to verify the electrical design and safety compliance before the Final Event:</p> <ul style="list-style-type: none"><li>- MSE 1<sup>st</sup> delivery: Electric and battery pack scheme.</li><li>- MSE 2<sup>nd</sup> delivery: Electrical precheck.</li></ul> |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ARTICLE 3: MS1 1<sup>st</sup> MILESTONE, TEAM PLAN

### F.3.1 Section 1: Executive abstract

- F.3.1.1 | The Executive Abstract must provide a clear and structured overview of the entire milestone submission.
- F.3.1.2 | Its objective is to allow the jury to understand, in a very short time, the key aspects of the team, its organization, the main project objectives and the overall development strategy for the MotoStudent edition.
- F.3.1.3 | The Executive Abstract must synthesize the most relevant information contained in the following sections of this milestone, including:
- The purpose and motivation of the team.
  - The structure and organization of the team.
  - The main goals of the project, including their main indicators where applicable
  - The general development timeline, the main project milestones and their overall feasibility
  - The main decision-making and responsibility structure of the team
- F.3.1.4 | The Executive Abstract must be written in a clear, structured and professional way, focusing on the key elements of the project rather than detailed explanations.
- F.3.1.5 | The recommended length of the Executive Abstract is approximately one page and must not exceed two pages.
- F.3.1.6 | The Executive Abstract must prioritize clarity and synthesis of the project rather than detailed technical explanations, which should be developed in the subsequent sections.

### F.3.2 Section 2: Team structure

- F.3.2.1 | The objective of this section is to explain the organizational structure of the team and how the work is distributed among its members.
- F.3.2.2 | The team must describe the different departments or functional areas of the project (for example: engineering, manufacturing, electronics, marketing, management, etc.), their responsibilities and how they interact with each other.
- F.3.2.3 | The team must also explain how decisions are taken within the project, including the decision-making structure and coordination between departments.
- F.3.2.4 | Teams are encouraged to briefly present the academic background of the members in order to show the multidisciplinary nature of the project.

### **F.3.3 Section 3: Team goals**

- F.3.3.1 | This section must define the main objectives that the team intends to achieve during the current MotoStudent edition.
- F.3.3.2 | The goals must be clear, realistic and measurable and must contribute to the successful development of the project.
- F.3.3.3 | Goals should not only focus on performance results but also on relevant project aspects such as engineering development, reliability, innovation, prototype completion, team organization or communication.
- F.3.3.4 | Each goal must include:
- A short description of the objective.
  - A measurable indicator allowing progress tracking.
  - The responsible department or team member.
  - An estimated timeframe for completion.
- F.3.3.5 | Goals must be presented in a structured table format in order to facilitate tracking and evaluation.

### **F.3.4 Section 4: Milestones, deliveries and project timeline**

- F.3.4.1 | This section must present the main development phases of the team during the competition.
- F.3.4.2 | The team must include a time-driven project chart (for example a Gantt chart) showing the main project phases and their relation with the official MotoStudent milestones and deliveries.
- F.3.4.3 | The timeline should include the key phases of prototype development such as:
- Concept and design
  - Engineering development
  - Manufacturing
  - Assembly
  - Testing and validation
  - Preparation for the Final Event
- F.3.4.4 | It is recommended that the team defines internal review points in order to monitor project progress and detect possible delays or risks in advance.

### **F.3.5 Evaluation and Scoring**

- F.3.5.1 | The maximum score for the 1<sup>st</sup> milestone is **75** points.

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| F.3.5.2 | The milestone will be evaluated as a whole, considering the clarity, coherence and overall quality of the submission.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F.3.5.3 | Although teams are free to structure their submission as they consider appropriate, the evaluation will consider several key dimensions of the project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| F.3.5.4 | <p>The evaluation will consider several key dimensions of the project.<br/>The maximum weight of each dimension and the aspects that may be considered by the jury include:</p> <ol style="list-style-type: none"> <li>1) Team purpose and project vision (maximum 5 points) <ul style="list-style-type: none"> <li>- Clarity of the team's purpose and motivation.</li> <li>- Clarity and coherence of the project objectives.</li> </ul> </li> <li>2) Team structure and organization (maximum 20 points) <ul style="list-style-type: none"> <li>- Definition and clarity of the team structure.</li> <li>- Coherence between team composition and project needs.</li> <li>- Explanation of the decision-making process.</li> <li>- Definition of responsibilities for each organizational unit.</li> <li>- Overall management and coordination of the team.</li> </ul> </li> <li>3) Definition and quality of project goals (maximum 25 points) <ul style="list-style-type: none"> <li>- Precision and clarity of the defined goals.</li> <li>- Presence of measurable indicators to evaluate progress.</li> <li>- Alignment between goals and team capabilities.</li> <li>- Feasibility of the goals and supporting reasoning.</li> <li>- Definition of timeframes for achieving the goals.</li> <li>- Clear assignment of responsibilities.</li> </ul> </li> <li>4) Project planning, milestones and timeline feasibility (maximum 25 points) <ul style="list-style-type: none"> <li>- Existence of a structured project timeline (e.g., Gantt chart).</li> <li>- Integration of MotoStudent official milestones.</li> <li>- Definition of internal project milestones.</li> <li>- Definition of prototype development stages.</li> <li>- Planning of the different deliveries throughout the competition.</li> </ul> </li> </ol> |
| F.3.5.5 | These values represent the relative weight of each dimension in the evaluation but do not imply that the milestone will be assessed strictly by sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| F.3.5.6 | The jury will assign a global score up to the maximum value of the milestone based on the overall quality and coherence of the project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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## ARTICLE 4: MS1 2<sup>nd</sup> MILESTONE, PRODUCT DEFINITION

### F.4.1 Milestone preamble

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|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.4.1.1 | The objective of this milestone is to define the conceptual and engineering foundations of the motorcycle prototype that will be developed for the MotoStudent competition.                                                                       |
| F.4.1.2 | The prototype to be developed shall be a racing motorcycle powered either by the standardized combustion engine or by the electric powertrain provided by the Competition Organization, depending on the category in which the team participates. |
| F.4.1.3 | Since the vehicle will be operated by a rider, the design must consider appropriate structural loads, ergonomics, safety margins and correct rider integration.                                                                                   |
| F.4.1.4 | The milestone must demonstrate that the team has defined a technically feasible design concept supported by engineering analysis and technical reasoning.                                                                                         |

### F.4.2 Section 1: Executive abstract

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|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.4.2.1 | The Executive Abstract must provide a clear and concise overview of the proposed prototype concept and the main engineering decisions taken by the team.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| F.4.2.2 | The objective of this section is to allow the jury to quickly understand the design philosophy of the prototype and the rationale behind the most relevant technical choices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| F.4.2.3 | <p>The Executive Abstract must synthesize the key engineering aspects of the milestone and provide a clear overview of the proposed prototype definition. In particular, it should summarize:</p> <ul style="list-style-type: none"><li>- The general concept of the motorcycle prototype and the main functional architecture of the vehicle</li><li>- The main design alternatives considered and the justification of the selected technical solutions</li><li>- The most relevant risk, safety and reliability considerations identified during the design phase (e.g. main conclusions from the FMEA)</li><li>- The overall make/buy strategy and manufacturing approach adopted by the team</li><li>- The proposed prototype layout and the integration of the rider within the vehicle design</li><li>- The main project requirements, design criteria and main load assumptions considered for the design of the prototype and the most relevant structural or performance results obtained during the engineering analysis</li><li>- The main engineering results and conclusions supporting the proposed prototype definition</li></ul> |

- F.4.2.4 | The Executive Abstract should demonstrate that the team has developed a coherent prototype concept supported by sound engineering reasoning and consistent design assumptions.
- F.4.2.5 | As part of the Executive Abstract submission, each team must include a Prototype Characteristics Sheet summarizing the main technical parameters of the proposed prototype. Its purpose is to provide with a clear and structured overview of the main technical characteristics of the prototype. At a minimum, the Prototype Characteristics Sheet must include the following information:
- Category of participation (MotoStudent Electric or MotoStudent eFuel)
  - University and team name
  - Prototype wheelbase
  - Estimated total vehicle weight
  - Powertrain type and main characteristics
  - Battery characteristics (Electric category) or fuel system configuration (eFuel category)
  - Front and rear suspension type
  - Front and rear brake system type
  - Wheel and tyre dimensions
  - Main frame concept and material
  - Estimated maximum power or performance target
  - Rider position concept and general prototype layout
- Teams may include additional technical parameters if they consider them relevant to better describe the prototype.
- The information provided in this sheet must be consistent with the rest of the documentation submitted in the milestone.
- F.4.2.6 | The Executive Abstract must remain concise while clearly summarizing the most relevant aspects of the proposed design and its main engineering conclusions.
- As a general guideline, the Executive Abstract should normally range between three and four pages and must not exceed four pages.
- F.4.3      Section 2: Functional systems trade-off**
- F.4.3.1 | The objective of this section is to explain the main design concepts selected for the different functional systems of the prototype.
- F.4.3.2 | Teams must compare possible design alternatives and justify the selected solution.

- F.4.3.3 | The analysis may include references to existing motorcycles, motorsport applications, previous MotoStudent prototypes or other relevant engineering references.
- F.4.3.4 | The advantages, disadvantages and expected impact of the selected solutions should be clearly discussed.
- F.4.4      Section 3: Failure Modes and Effect Analysis (FMEA)**
- F.4.4.1 | The objective of this section is to identify possible failure modes of the main systems of the prototype and evaluate their potential consequences.
- F.4.4.2 | The analysis should identify possible failure modes, their causes, their effects and possible mitigation actions.
- F.4.4.3 | The FMEA should demonstrate that the team has considered reliability, safety and risk mitigation aspects during the design phase.
- F.4.5      Section 4: Make / Buy analysis**
- F.4.5.1 | This section must describe which elements or systems of the prototype are designed and manufactured by the team and which components are sourced externally.
- F.4.5.2 | Components classified as "make" are those whose design and/or manufacturing has been carried out by the team. Minor adaptations or simple assembly of commercial components shall not be considered as "make".
- F.4.5.3 | Components classified as "buy" are those obtained from suppliers or commercial products without significant modification of their original design.
- F.4.5.4 | Teams must clearly identify the main systems of the prototype (e.g. frame, swingarm, suspension elements, braking system, electronics, fairing, etc.) and specify for each of them whether they are classified as make or buy.
- F.4.5.5 | The team should briefly justify the make/buy decision, considering aspects such as technical capability, manufacturing resources, reliability or project strategy.
- F.4.5.6 | If a component from a previous MotoStudent edition is reused, it must be clearly identified. The reuse and evolution of components between editions is considered a natural part of the engineering development process and is therefore allowed and encouraged.
- When a component is reused, its make/buy classification must remain consistent with its original nature. Minor adaptations or improvements to a previously purchased

component shall not change its classification to "make", and components originally designed and manufactured by the team shall remain classified as "make".

#### **F.4.6 Section 5: Proposed design**

F.4.6.1 The objective of this section is to provide a clear visual and geometric representation of the proposed prototype design.

F.4.6.2 The prototype architecture and layout must be presented using both 2D and 3D representations.

F.4.6.3 The design must include at least the following views of the prototype without fairings:

- front view
- side view
- top view

F.4.6.4 The rider must be represented in the design in order to demonstrate correct ergonomic integration with the vehicle. The main rider dimensions used in the design (such as total height, arm reach or leg dimensions) should be indicated.

F.4.6.5 The design must show the main geometric and dimensional constraints of the prototype and should include key parameters such as main dimensions, estimated centre of gravity (CoG) and expected minimum prototype weight.

F.4.6.6 The inclusion of the same views with the fairing installed is highly recommended to help understand the final appearance and packaging of the prototype.

#### **F.4.7 Section 6: Loads definitions**

F.4.7.1 The objective of this section is to define the main load cases that the prototype is expected to withstand during the MotoStudent Final Event. These load cases will form the basis for the structural design and validation of the vehicle.

F.4.7.2 Teams should justify the selected loads using engineering estimations, simplified analytical calculations or numerical simulations where appropriate. If available, empirical data or measurements obtained during testing activities (for example at MotorLand Aragón or similar circuits) may also be used to support the assumptions.

F.4.7.3 The following load cases must be considered in the design of the prototype:

- Static verification loads
- Maximum braking loads
- Maximum acceleration loads
- Cornering loads
- Turning loads (e.g. rapid left-right transitions such as chicanes)



- Aerodynamic loads

F.4.7.4 | The defined loads must include the rider mass in the calculations and must represent realistic operating conditions of the prototype during riding.

#### **F.4.8      Section 7: Structural calculations**

F.4.8.1 | The objective of this section is to evaluate the structural behaviour of the main components of the prototype under the load cases defined in the previous section.

F.4.8.2 | The analysis must assess the stresses and deformations of the structural parts of the vehicle in order to verify that the design can safely withstand the expected operating loads.

F.4.8.3 | Finite element simulations are highly recommended for complex components. For simpler elements, analytical calculations may be used where appropriate.

F.4.8.4 | At minimum, the following structural components must be evaluated:

- frame
- swingarm
- subframe

F.4.8.5 | If a purchased component is structurally critical within the vehicle architecture, its structural behaviour must also be justified by calculation or engineering analysis.

F.4.8.6 | Safety margins must be clearly defined and justified in order to demonstrate the structural integrity of the prototype.

#### **F.4.9      Evaluation and scoring**

F.4.9.1 | The maximum score for the 2<sup>nd</sup> milestone is **100** points.

F.4.9.2 | The milestone will be evaluated as a whole, considering the overall technical coherence, engineering reasoning and quality of the proposed prototype definition.

F.4.9.3 | Although teams are free to structure their submission as they consider appropriate, the evaluation will consider several key dimensions of the project.

F.4.9.4 | The evaluation will consider several key engineering dimensions of the project. The maximum weight of each dimension and the aspects that may be considered by the jury include:

- 1) Engineering concept and system design justification (maximum 10 points)

- Identification of the different technical solutions considered for the main systems of the prototype.
- Analysis of the advantages and disadvantages of the selected solutions.
- Comparison with reference solutions from the market, motorsport or other engineering applications where relevant.
- Technical justification of the selected system architecture.
- Consideration of environmental impact when evaluating alternative solutions.

#### 2) Failure analysis and risk awareness – FMEA (maximum 15 points)

- Completeness of the Failure Modes and Effects Analysis (FMEA).
- Accuracy in identifying possible failure modes and their consequences.
- Correct evaluation of occurrence and detection rankings.
- Definition and consistency of risk priority numbers (RPN).
- Identification of the systems or components requiring special attention.
- Definition and effectiveness of proposed risk reduction actions.

#### 3) Make / Buy strategy and manufacturing approach (maximum 10 points)

- Completeness and clarity of the list of main prototype components.
- Correct classification of elements as designed/manufactured by the team or sourced externally.
- Engineering reasoning behind the make/buy decisions.
- Coherence between the manufacturing strategy and the capabilities of the team.

#### 4) Prototype architecture and design definition (maximum 20 points)

- Clarity and completeness of the proposed prototype architecture.
- Quality and coherence of the 2D and 3D design representations.
- Definition of geometric constraints and main vehicle dimensions.
- Integration and ergonomic positioning of the rider.

#### 5) Definition of load cases and engineering assumptions (maximum 25 points)

- Definition of relevant static and dynamic load cases.
- Consideration of braking, acceleration, cornering and aerodynamic loads.
- Quality of the engineering assumptions used in the calculations.
- Inclusion of the rider mass in the load definition.
- Coherence of the load estimations with realistic vehicle behaviour.

#### 6) Structural analysis and validation (maximum 20 points)

- Structural analysis of the main components of the prototype.
- Evaluation of stresses and deformations under the defined load cases.
- Strength and stiffness calculations or simulations.
- Definition and justification of appropriate safety margins.

- F.4.9.5 | These values represent the approximate weight of each dimension but do not imply that the milestone will be evaluated strictly by sections.
- F.4.9.6 | The jury will assign a global score based on the overall technical quality, engineering reasoning and feasibility of the proposed design.

## ARTICLE 5: MS1 3<sup>rd</sup> MILESTONE, INNOVATION

### F.5.1 Milestone preamble

- F.5.1.1 | Innovation is understood as *the development or implementation of a new or improved solution that provides a measurable technical or process-related benefit compared to an existing approach.*
- F.5.1.2 | The objective of this milestone is to encourage teams to identify a relevant technical or process-related challenge and to develop an innovative solution that improves the prototype, the development process or the understanding of the system.
- F.5.1.3 | The innovation may involve a new technical solution, an improvement of an existing system, a new design methodology, a manufacturing process improvement or a new approach to solving an engineering problem.
- F.5.1.4 | The innovation must be supported by a structured development process including problem identification, concept exploration, development phases, testing and evaluation of results.
- F.5.1.5 | The innovation presented in this milestone must be implemented in the prototype or in its development process and must be available to be demonstrated during the Final Event.  
  
Concepts that remain only at proposal, rendering, simulation or theoretical intention level, without real implementation, shall not be considered sufficient for the purposes of this milestone.
- F.5.1.6 | The expected impact of the innovation should be supported by quantitative or qualitative evidence such as simulations, experimental tests, comparisons with baseline solutions or engineering analysis.
- F.5.1.7 | Innovation in motorsport engineering is often the result of questioning established solutions and exploring new approaches. Teams are therefore encouraged to approach this milestone with creativity, technical reasoning and a clear focus on measurable improvements.

- F.5.1.8 | The jury will evaluate the innovation primarily based on the engineering process followed, the technical justification of the proposed solution and the evidence supporting its impact, rather than solely on the novelty of the idea itself.

## **F.5.2      Section 1: Executive Abstract**

- F.5.2.1 | The Executive Abstract must provide a clear and concise overview of the innovation project developed by the team.

- F.5.2.3 | The objective of this section is to allow the jury to quickly understand the problem addressed, the proposed innovative solution, the development methodology followed and the expected impact on the prototype.

- F.5.2.3 | The Executive Abstract must summarize the key aspects of the milestone, including:
- The problem or limitation identified by the team
  - The objective of the innovation
  - The proposed concept or solution and main alternative solutions considered and the justification of the selected concept
  - The development approach followed
  - The implemented Proof of Concept and the main results obtained, including the comparison between expected and achieved results
  - The main resources required
  - The expected improvement and its engineering justification

- F.5.2.4 | The Executive Abstract should demonstrate that the innovation is supported by sound engineering reasoning and a structured development process.

- F.5.2.5 | The Executive Abstract should be proportional to the complexity of the innovation project and sufficiently detailed to clearly summarize the most relevant aspects of the work carried out. Teams are encouraged to keep this section concise and focused on the key engineering aspects of the innovation. The Executive Abstract must not exceed 4 pages.

## **F.5.3      Section 2: Starting point and innovation objective**

- F.5.3.1 | The objective of this section is to explain the origin of the innovation and the problem that the team intends to solve.

- F.5.3.2 | Teams must clearly describe:
- the initial situation or limitation identified
  - the engineering problem to be addressed
  - the expected improvement or benefit
  - the alignment of the innovation with the team's project goals

#### **F.5.4      Section 3: Solution exploration and alternatives**

- F.5.4.1      The objective of this section is to analyse the different possible solutions considered by the team before selecting the final innovation concept.
- F.5.4.2      The analysis may include:
- existing solutions available in the market or motorsport
  - alternative technical approaches
  - advantages and disadvantages of each option
  - justification of the selected concept
- F.5.4.3      If no direct market solution exists, the comparison should include the closest available technological alternatives.
- F.5.4.4      The proposed innovation must comply with the MotoStudent Technical Regulations. Solutions that affect regulated safety systems or critical vehicle components must demonstrate full compliance with the applicable rules.

#### **F.5.5      Section 4: Development process**

- F.5.5.1      This section must explain the development process followed by the team from the initial concept to the final implementation.
- F.5.5.2      The development process should include:
- concept generation and first sketches
  - design iterations and decision points
  - prototype development phases
  - testing activities performed
- F.5.5.3      It is highly recommended to include a timeline describing the main development milestones of the innovation project (e.g., concept release, design freeze, prototype manufacturing, testing stages).
- Teams are also encouraged to perform experimental tests or validation activities in order to compare the obtained results with the expected improvement and to demonstrate the effectiveness of the proposed solution.

#### **F.5.6      Section 5: Proof of Concept (PoC)**

- F.5.6.1      The objective of this section is to demonstrate that the proposed innovation works as intended.
- F.5.6.2      The Proof of Concept must present the implemented solution and the tests or evaluations used to validate it.

|              |                                                                                                                                                                                                                                                                                                    |
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| F.5.6.3      | <p>The PoC should include:</p> <ul style="list-style-type: none"><li>- description of the implemented concept</li><li>- visual evidence such as photographs or diagrams</li><li>- tests, experiments or validation activities</li><li>- comparison between expected and obtained results</li></ul> |
| <b>F.5.7</b> | <b>Section 6: Bill of Materials (BOM) and development resources</b>                                                                                                                                                                                                                                |
| F.5.7.1      | <p>The objective of this section is to describe the resources required to develop and implement the innovation.</p>                                                                                                                                                                                |
| F.5.7.2      | <p>Teams must include a Bill of Materials (BOM) listing the main components, materials and consumables used during the development of the innovation.</p>                                                                                                                                          |
| F.5.7.3      | <p>The budget estimation should consider:</p> <ul style="list-style-type: none"><li>- cost of the components included in the BOM</li><li>- manufacturing or prototyping costs</li><li>- development resources used during the project</li></ul>                                                    |
| F.5.7.4      | <p>The analysis should demonstrate the relationship between the resources invested and the expected benefit obtained.</p>                                                                                                                                                                          |
| <b>F.5.8</b> | <b>Section 7: Conclusions and future potential</b>                                                                                                                                                                                                                                                 |
| F.5.8.1      | <p>This section must summarize the results of the innovation project.</p>                                                                                                                                                                                                                          |
| F.5.8.2      | <p>Teams should evaluate:</p> <ul style="list-style-type: none"><li>- the final improvement obtained</li><li>- the real impact of the innovation on the prototype</li><li>- possible future developments or improvements</li></ul>                                                                 |
| F.5.8.3      | <p>The potential applicability of the innovation beyond the current prototype may also be discussed.</p>                                                                                                                                                                                           |
| <b>F.5.9</b> | <b>Evaluation and scoring</b>                                                                                                                                                                                                                                                                      |
| F.5.9.1      | <p>The maximum score for the 3<sup>rd</sup> milestone is <b>150</b> points.</p>                                                                                                                                                                                                                    |
| F.5.9.2      | <p>The milestone will be evaluated as a whole, considering the overall coherence of the innovation project, the quality of the engineering reasoning, the development methodology and the demonstrated impact of the proposed innovation.</p>                                                      |

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| F.5.9.3 | <p>Although teams are free to structure their submission as they consider appropriate, the evaluation will consider several key dimensions of the project. The maximum weight of each dimension and the aspects that may be considered by the jury include:</p> <ol style="list-style-type: none"> <li>1) Innovation motivation and problem definition (maximum 10 points) <ul style="list-style-type: none"> <li>- Clarity of the identified problem or improvement opportunity.</li> <li>- Justification of the need for the innovation.</li> <li>- Alignment between the innovation and the team's project goals.</li> </ul> </li> <li>2) Analysis of alternatives and solution selection (maximum 20 points) <ul style="list-style-type: none"> <li>- Identification of existing solutions or technological alternatives.</li> <li>- Comparison between possible approaches.</li> <li>- Justification of the selected concept.</li> <li>- Compliance of the proposed solution with the MotoStudent technical regulations.</li> </ul> </li> <li>3) Development process and engineering methodology (maximum 55 points) <ul style="list-style-type: none"> <li>- Clarity of the development phases.</li> <li>- Definition of development milestones and project evolution.</li> <li>- Engineering reasoning behind design decisions.</li> <li>- Quality of testing activities and validation methods.</li> <li>- Evidence of iterative development and learning during the project.</li> </ul> </li> <li>4) Proof of Concept and validation of results (maximum 35 points) <ul style="list-style-type: none"> <li>- Quality, clarity and level of real implementation of the Proof of Concept.</li> <li>- Demonstration that the concept works as intended.</li> <li>- Comparison between expected and obtained results.</li> <li>- Quality of graphical and technical documentation.</li> </ul> </li> <li>5) Resources, feasibility and development efficiency (maximum 15 points) <ul style="list-style-type: none"> <li>- Completeness and clarity of the Bill of Materials (BOM).</li> <li>- Evaluation of the resources required for the development.</li> <li>- Balance between development effort and expected benefit.</li> </ul> </li> <li>6) Impact and future potential of the innovation (maximum 15 points) <ul style="list-style-type: none"> <li>- Demonstrated or expected impact on the prototype.</li> <li>- Potential benefits in terms of performance, efficiency, reliability or sustainability.</li> <li>- Possibilities for future evolution or further development of the concept.</li> </ul> </li> </ol> |
| F.5.9.4 | <p>These values represent the approximate weight of each dimension but do not imply that the milestone will be evaluated strictly by sections.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

- F.5.8.5 | The jury will assign a global score based on the overall quality of the innovation project, the clarity of the development process, the robustness of the validation carried out and the demonstrated or expected impact of the innovation on the prototype.

#### **ARTICLE 6: MS1 4<sup>th</sup> DELIVERY, PROTOTYPE CAD AND DRAWINGS**

##### **F.6.1 Milestone preamble**

- F.6.1.1 | The objective of this milestone is to demonstrate that the prototype design has reached a sufficient level of technical maturity to allow its correct manufacturing and integration.
- F.6.1.2 | In modern engineering environments, the CAD model represents the digital definition of the product and is the central reference for design, manufacturing and assembly.
- F.6.1.3 | This milestone evaluates the quality, structure and clarity of the CAD model as well as the ability of the team to generate professional manufacturing documentation.
- F.6.1.4 | The documentation provided must demonstrate that the prototype can be manufactured, assembled and understood by a third party using the information delivered by the team.

##### **F.6.2 Section 1: Executive abstract**

- F.6.2.1 | The Executive Abstract must provide a clear overview of the CAD definition of the prototype and the design maturity achieved by the team at this stage of the project.
- F.6.2.2 | The objective of this section is to allow the jury to quickly understand the architecture of the CAD model, the design status of the prototype and the approach followed by the team to prepare the manufacturing documentation.
- F.6.2.3 | The Executive Abstract must synthesize the key aspects of the milestone, including:
- The general architecture of the prototype CAD model
  - The design maturity of the main structural components
  - The modelling strategy used by the team
  - The structure of the assemblies and subsystems
  - The documentation strategy followed to support manufacturing and assembly
  - The main conclusions regarding the completeness, consistency and technical usefulness of the CAD and drawings package
- F.6.2.4 | The Executive Abstract should demonstrate that the team has developed a structured and coherent digital definition of the prototype.



F.6.2.5 | The length of the Executive Abstract should be proportional to the complexity of the prototype design and sufficient to clearly summarize the most relevant aspects of the CAD model and documentation.

F.6.2.6 | The Executive Abstract must not exceed 3 pages.

### **F.6.3      Section 2: Prototype CAD model**

F.6.3.1 | The objective of this section is to deliver the latest CAD definition of the structural elements of the prototype.

F.6.3.2 | This section must be delivered as a single 3D file in. STEP format.

F.6.3.3 | The CAD submission must include a 3D CAD assembly of the motorcycle containing, as a minimum, the main structural components designed by the team.

The assembly shall include at least the following components in their design position:

- frame
- subframe
- swingarm

These components must be modelled with a level of detail sufficient to identify their geometry, interfaces, and structural concept.

F.6.3.4 | The CAD model should represent the current design status of the prototype and demonstrate an appropriate level of engineering definition.

F.6.3.5 | The CAD model must be clean, coherent and free of unnecessary elements that could complicate its interpretation.

### **F.6.4      Section 3: Model tree and CAD structure**

F.6.4.1 | The objective of this section is to present the internal structure of the CAD model and the whole organization of the assemblies, subassemblies and parts.

The model tree must clearly show the hierarchical structure of the prototype design, including the relationships between systems, assemblies and individual parts.

Each assembly and part must include an alphanumeric reference and a clear name.

A part or subassembly may appear more than once if it is used in different assemblies.

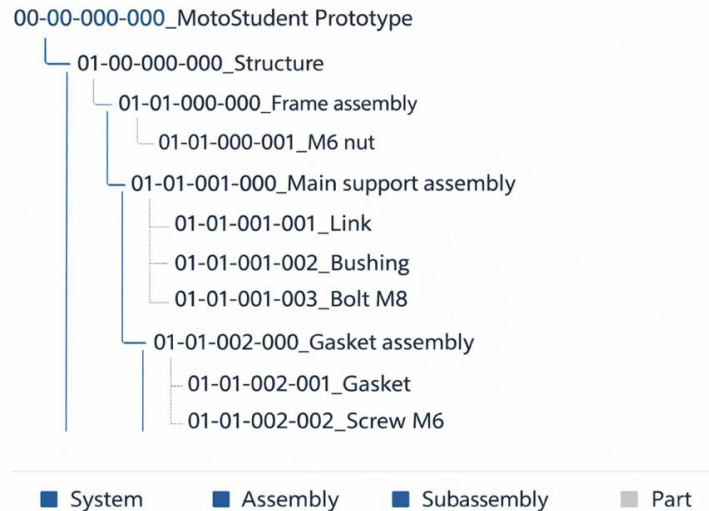


Figure 43 - Example of a model tree

F.6.4.2 The nomenclature for parts and assemblies must follow a consistent alphanumeric reference system defined by the team.

Each element of the prototype must be uniquely identified by an alphanumeric code following the general structure:

***SS-AA-SSS-PPP***

Where:

- SS = System
- AA = Assembly
- SSS = Subassembly
- PPP = Part

This reference must be used consistently across the CAD model, model tree and drawings.

- **System:** The whole prototype must be broken down into the following main 7 systems:
  - 01 – Structure
  - 02 – Powertrain
  - 03 – Suspensions
  - 04 – Wheels
  - 05 – Brakes
  - 06 – Aerodynamics
  - 07 – Others

Teams may further divide these systems into assemblies and subassemblies according to their design structure.

The hierarchical structure of the product must follow the following levels:

- **System**
- **Assembly:** Level set immediately below Systems.
- **Subassembly:** Level set immediately below Assemblies.
- **Part:** The number of the part pertaining the subassembly. Numbering must be ascending. For subassemblies, the part number must be 000.

Each level must clearly represent the functional and structural organization of the prototype.

F.6.4.3 The numbering of parts must be sequential within each subassembly.

When an element corresponds to a subassembly rather than a single part, the part number must be defined as "000" (e.g. 01-01-002-000 → Subassembly; 01-01-002-001 → Part inside that subassembly).

Code:

|               |                 |                    |             |
|---------------|-----------------|--------------------|-------------|
| <u>System</u> | <u>Assembly</u> | <u>Subassembly</u> | <u>Part</u> |
| 00            | AA-             | 111-               | 222         |

*Picture 44 - Drawings nomenclature*

## F.6.5 Section 4: Manufacturing drawings

F.6.5.1 The objective of this section is to demonstrate that the team is capable of producing professional manufacturing drawings derived from the CAD model.

F.6.5.2 Teams are not required to provide drawings for every component of the prototype.

F.6.5.3 Instead, the team must provide representative examples of manufacturing documentation that demonstrate the correct application of engineering drawing standards.

F.6.5.4 At least, the following drawings must be delivered:

- Complete prototype drawing (front, side, top and isometric views)
- Rear suspension assembly drawing
- Top triple clamp part drawing

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.6.5.5      | Assembly drawings must include a Bill of Materials (BOM) identifying the parts included in the assembly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| F.6.5.6      | Parts and subassemblies must be referenced using balloons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| F.6.5.7      | Commercial components must also be identified in the Bill of Materials.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| F.6.5.8      | Drawings should include the necessary information to support manufacturing and assembly, such as dimensions, materials, tolerances and relevant notes when applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>F.6.6</b> | <b>Evaluation and scoring</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| F.6.6.1      | The maximum score for this milestone is <b>60</b> points.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| F.6.6.2      | The milestone will be evaluated as a whole, considering the overall quality, structure and engineering maturity of the CAD definition and the manufacturing documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| F.6.6.3      | <p>Although teams are free to structure their submission as they consider appropriate, the evaluation will consider several key dimensions of the project. The maximum weight of each dimension and the aspects that may be considered by the jury include:</p> <ol style="list-style-type: none"> <li>1) CAD model definition and engineering maturity (maximum 10 points) <ul style="list-style-type: none"> <li>- Correct delivery format of the CAD model.</li> <li>- Level of definition of the structural components.</li> <li>- Coherence and cleanliness of the model.</li> </ul> </li> <li>2) Model tree organization and digital structure (maximum 20 points) <ul style="list-style-type: none"> <li>- Correct application of the nomenclature system.</li> <li>- Completeness of the assemblies and subassemblies.</li> <li>- Inclusion of the relevant parts within the model structure.</li> <li>- Clarity of the relationships between systems and assemblies.</li> </ul> </li> <li>3) Manufacturing drawings and documentation quality (maximum 30 points) <ul style="list-style-type: none"> <li>- Correct application of engineering drawing standards.</li> <li>- Inclusion of relevant dimensions.</li> <li>- Definition of materials and relevant technical information.</li> <li>- Correct use of Bill of Materials and balloons.</li> <li>- Overall clarity and usability of the documentation for manufacturing.</li> </ul> </li> </ol> |
| F.6.6.4      | These values represent the approximate weight of each dimension but do not imply that the milestone will be evaluated strictly by sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

- F.6.6.5 | The jury will assign a global score based on the overall quality of the CAD definition, the clarity of the model structure and the professional level of the manufacturing documentation.

## ARTICLE 7: MS1 5<sup>TH</sup> MILESTONE, ECONOMICAL PLAN

### F.7.1 Milestone preamble

- F.7.1.1 | Engineering projects are not only defined by their technical performance but also by their economic feasibility. Every engineering decision has an associated economic impact that must be understood and justified.
- F.7.1.2 | The objective of this milestone is to introduce teams to the economic dimension of engineering projects, encouraging them to analyse costs, resources and financial sustainability as part of the project development process.
- F.7.1.3 | Teams are expected to demonstrate a structured and transparent approach to the economic management of the project, including cost tracking, budgeting and financial planning.
- F.7.1.4 | The purpose of this milestone is not to reward teams that achieve the lowest costs or the highest financial results, but to evaluate the reasoning process, justification and understanding of the economic implications of their engineering decisions.
- F.7.1.5 | This milestone also encourages multidisciplinary collaboration within the team, involving members with backgrounds in engineering, economics, management or related fields.
- F.7.1.6 | Teams participating in MotoStudent may operate under different economic conditions and levels of available resources.
- For this reason, this milestone does not aim to compare the absolute financial capacity of teams, but rather to evaluate their ability to understand, manage and justify the economic implications of their engineering decisions.
- The evaluation will therefore focus on the transparency of the economic analysis, the identification of the main cost drivers and the coherence between technical decisions and their economic impact on the project.
- F.7.1.7 | All economic values presented in this milestone must be expressed in euros (€).
- If the original transaction was performed in another currency, the value must be converted to euros using a reasonable exchange rate, which must be indicated by the team.

The same exchange rate must be used consistently throughout the entire submission.

## **F.7.2 Section 1: Executive abstract**

F.7.2.1 The Executive Abstract must provide a clear overview of the economic situation of the project and the methodology used by the team to analyse and manage the project's financial resources.

F.7.2.2 The objective of this section is to allow the jury to quickly understand the economic logic of the project and the key conclusions derived from the financial analysis.

F.7.2.3 The Executive Abstract must summarise the most relevant aspects of the milestone, including:

- The overall economic approach of the team
- The current financial situation of the project
- The estimated value of the prototype and the main criteria used for its calculation
- The main cost drivers of the project
- The expected future financial situation of the team
- The reasoning behind the economic conclusions reached
- The main conclusions regarding the economic feasibility and management of the Project

F.7.2.4 The Executive Abstract should demonstrate that the team understands the relationship between engineering decisions, project management and economic impact.

F.7.2.5 The length of the Executive Abstract should be sufficient to clearly explain the economic logic of the project and should not exceed 3 pages.

## **F.7.3 Section 2: Total costs**

F.7.3.1 The objective of this section is to record all expenses incurred by the team from the team enrolment date to the milestone delivery date.

F.7.3.2 This section must include **all expenses related to the development of the project and prototype.**

F.7.3.3 Expenses must be categorized at least into the following groups:

- **Production related costs** (e.g., materials, parts acquisition, prototype manufacturing).
- **Marketing related costs** (e.g., team clothing, promotional materials, events).

|              |                                                                                                                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <ul style="list-style-type: none"> <li>- <b>Operational costs</b> (e.g., rent, utilities, workspace costs).</li> </ul>                                                                                                                |
| F.7.3.4      | The appropriate taxes must be included when applicable.                                                                                                                                                                               |
| F.7.3.5      | <p>Each cost entry must include at least the following fields:</p> <ul style="list-style-type: none"> <li>- Date of the transaction</li> <li>- Concept of the expense</li> <li>- Cost without tax</li> <li>- Cost with tax</li> </ul> |
| F.7.3.6      | At the end of the section, the total expenses (with tax included) must be calculated.                                                                                                                                                 |
| F.7.3.7      | Donations in kind (e.g., parts or manufacturing services provided by sponsors) must be accounted as expenses using their estimated market value.                                                                                      |
| <b>F.7.4</b> | <b>Section 3: Total income</b>                                                                                                                                                                                                        |
| F.7.4.1      | The objective of this section is to record all sources of income received by the team from the team enrolment date to the milestone delivery date.                                                                                    |
| F.7.4.2      | Income sources may include sponsorship contributions, institutional funding, university support or other financial contributions.                                                                                                     |
| F.7.4.3      | Both monetary and in-kind contributions from sponsors must be accounted for.                                                                                                                                                          |
| F.7.4.4      | <p>Each income entry must include at least the following fields:</p> <ul style="list-style-type: none"> <li>- Date when the income was received</li> <li>- Concept of the income</li> <li>- Amount received</li> </ul>                |
| F.7.4.5      | At the end of the section, the total income must be calculated.                                                                                                                                                                       |
| <b>F.7.5</b> | <b>Section 4: Prototype value</b>                                                                                                                                                                                                     |
| F.7.5.1      | The objective of this section is to estimate the theoretical cost of reproducing the prototype without external sponsorship support.                                                                                                  |
| F.7.5.2      | All parts and assemblies must follow the nomenclature defined in the CAD structure of the prototype.                                                                                                                                  |
| F.7.5.3      | The prototype value must be presented in table format.                                                                                                                                                                                |
| F.7.5.4      | For each system, assembly, subassembly or part, the following information must be included:                                                                                                                                           |

|              |                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <ul style="list-style-type: none"> <li>- Alphanumeric reference</li> <li>- Name or part reference</li> <li>- Manufacturing hours (if applicable)</li> <li>- Total estimated cost</li> </ul>                                                                      |
| F.7.5.5      | <p>For manufactured parts, the total cost must include:</p> <p><i>Total cost of the part = (30 €/h * manufacturing hours) + material costs</i></p>                                                                                                               |
| F.7.5.6      | <p>If a component has been purchased from a supplier, manufacturing hours must be considered zero, as the cost of the component already includes the associated manufacturing effort.</p>                                                                        |
| F.7.5.7      | <p>If a component has been manufactured by the team, the hours required to manufacture that part must be estimated and included.</p>                                                                                                                             |
| F.7.5.8      | <p>If a component has been donated by a sponsor, it must be accounted as a purchased component using its estimated market value.</p>                                                                                                                             |
| F.7.5.9      | <p>Engineering activities such as design time or prototype testing must not be included in the prototype manufacturing value.</p> <p>However, tools, moulds or manufacturing resources required to build the prototype must be included when applicable.</p>     |
| <b>F.7.6</b> | <b>Section 5: Team liquidity</b>                                                                                                                                                                                                                                 |
| F.7.6.1      | <p>The objective of this section is to determine the current financial position of the team.</p>                                                                                                                                                                 |
| F.7.6.2      | <p>The value must correspond to the <b>difference between the total income (Section 3) and the total costs (Section 2)</b>.</p>                                                                                                                                  |
| F.7.6.3      | <p>The resulting value must represent the team's current cash balance.</p>                                                                                                                                                                                       |
| F.7.6.4      | <p>If the team started the project with an initial financial balance, this amount must be clearly explained and included in the calculation.</p>                                                                                                                 |
| <b>F.7.7</b> | <b>Section 6: Future budget</b>                                                                                                                                                                                                                                  |
| F.7.7.1      | <p>The objective of this section is to estimate the financial resources required from the milestone submission date until the end of the competition (including also the costs that may arise after the competition, e.g., traveling back to your hometown).</p> |
| F.7.7.2      | <p>This section must estimate both expected income and expected expenses.</p>                                                                                                                                                                                    |



|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.7.7.3      | <p>At minimum, the following categories must be considered:</p> <ul style="list-style-type: none"> <li>- <b>Expected income</b> (e.g., confirmed sponsorship agreements not yet received)</li> <li>- <b>Expected production costs</b> (e.g., missing prototype components, spare parts, testing activities)</li> <li>- <b>Marketing related costs</b></li> <li>- <b>Travel and logistics costs</b> (e.g., transportation of the prototype, accommodation)</li> <li>- <b>Operational costs</b></li> </ul> |
| F.7.7.4      | Taxes must be included when applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F.7.7.5      | The final balance may be positive, negative or zero. No penalties will be applied for a negative balance.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>F.7.8</b> | <b>Section 7: Economic viability of the project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| F.7.8.1      | The objective of this section is to analyse the economic viability of the project based on the current liquidity and the estimated future budget.                                                                                                                                                                                                                                                                                                                                                        |
| F.7.8.2      | This section must explain the team's economic results and the reasoning behind them.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| F.7.8.3      | Teams should explain the causes of the result obtained and the actions planned to manage their financial situation.                                                                                                                                                                                                                                                                                                                                                                                      |
| F.7.8.4      | In case of a negative balance, teams should explain how they plan to manage or mitigate the financial situation.                                                                                                                                                                                                                                                                                                                                                                                         |
| F.7.8.5      | In case of a positive balance, teams should explain how the available resources will be used.                                                                                                                                                                                                                                                                                                                                                                                                            |
| F.7.8.6      | The evaluation will focus on the reasoning process and economic understanding rather than the final financial outcome.                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>F.7.9</b> | <b>Evaluation and scoring</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| F.7.9.1      | The maximum score for the 5 <sup>th</sup> milestone is <b>75</b> points.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| F.7.9.2      | The milestone will be evaluated as a whole, considering the clarity, consistency and reasoning of the economic analysis.                                                                                                                                                                                                                                                                                                                                                                                 |
| F.7.9.3      | The evaluation will consider several key dimensions of the project. The maximum weight of each dimension and the aspects that may be considered by the jury include:                                                                                                                                                                                                                                                                                                                                     |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>1) Cost tracking and expense documentation (maximum 10 points)</p> <ul style="list-style-type: none"> <li>- Completeness of the expense records</li> <li>- Correct classification of expenses</li> <li>- Transparency of the cost tracking process</li> </ul> <p>2) Income structure and funding sources (maximum 10 points)</p> <ul style="list-style-type: none"> <li>- Completeness of the income records</li> <li>- Clarity of sponsorship contributions</li> <li>- Transparency in financial reporting</li> </ul> <p>3) Prototype value estimation (maximum 25 points)</p> <ul style="list-style-type: none"> <li>- Completeness of the prototype cost breakdown</li> <li>- Correct differentiation between purchased and manufactured components</li> <li>- Reasonable estimation of manufacturing hours</li> <li>- Consistency with the CAD structure of the prototype</li> </ul> <p>4) Financial planning and budget forecasting (maximum 15 points)</p> <ul style="list-style-type: none"> <li>- Completeness of the future budget estimation</li> <li>- Realistic estimation of expected costs and income</li> <li>- Identification of financial risks</li> </ul> <p>5) Economic reasoning and project viability (maximum 15 points)</p> <ul style="list-style-type: none"> <li>- Clarity of the financial conclusions</li> <li>- Understanding of the economic implications of the project</li> <li>- Coherence between technical decisions and economic reasoning</li> </ul> |
| F.7.9.4 | These values represent the approximate weight of each dimension but do not imply that the milestone will be evaluated strictly by sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| F.7.9.5 | The jury will assign a global score based on the overall quality, transparency and reasoning of the economic analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## ARTICLE 8: MS1 6<sup>th</sup> MILESTONE, PITCH PRESENTATION

### F.8.1 Milestone preamble

|         |                                                                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.8.1.1 | Engineering projects must not only be technically sound, but also clearly communicated. The ability to explain a project, present its value and convince others of its potential is a fundamental skill for engineers, entrepreneurs and innovators. |
| F.8.1.2 | The objective of this milestone is for teams to present their project in a clear, engaging and persuasive way, summarising the most relevant aspects of the work carried out during the MotoStudent competition.                                     |

- F.8.1.3 | Teams must demonstrate their ability to communicate the essence of their project, including the team structure, the prototype design, the innovation developed and the overall vision behind the project.
- F.8.1.4 | The presentation should follow the spirit of an entrepreneurial pitch, where the team explains what they have built, how they developed it and why their project is relevant.
- F.8.1.5 | This milestone aims to evaluate communication skills, storytelling ability and the capacity to present complex engineering work in a clear and engaging way.
- F.8.1.6 | The pitch should not be understood as a high-budget video production. The focus of the evaluation will be on clarity of the message, structure of the presentation and the ability of the team to communicate their project effectively.
- F.8.1.7 | MotoStudent may use the submitted videos for communication, promotion and dissemination purposes, including public projection during the Final Event or publication through MotoStudent channels.
- F.8.2 | Pitch video requirements**
- F.8.2.1 | Teams must submit a pitch video presenting their MotoStudent project.
- F.8.2.2 | The video must clearly explain the main aspects of the project, including:
- the team and its organisation
  - the prototype concept and design
  - the innovation developed
  - the development process followed
  - the overall vision of the project
- F.8.2.3 | The pitch should aim to communicate the project in a clear, engaging and dynamic way, similar to an entrepreneurial pitch presentation.
- F.8.2.4 | The recommended duration of the video is between 4 and 6 minutes. The video must not exceed 6 minutes.
- F.8.2.5 | The video must be presented in vertical format.
- F.8.2.6 | The video must be recorded in English.
- F.8.2.7 | The first seconds of the video must include the MotoStudent logo and the team logo.
- F.8.2.8 | The final seconds of the video must include the team logo and the logos of MEF and TechnoPark with the text *"Powered by"* placed above them.

- F.8.2.9 | Teams may include MotoStudent branding elements during the video if desired. Official graphic resources will be available through the MotoStudent platform.
- F.8.2.10 | Teams are encouraged to support their explanations using visual material such as prototype components, CAD images, development footage or test results.
- F.8.2.11 | The video must be published on the team's Instagram account before the milestone deadline and the link must be submitted through the MotoStudent platform.
- F.8.2.12 | Teams may also publish the video on other social media platforms.
- F.8.3      Evaluation and scoring**
- F.8.3.1 | The maximum score for this milestone is **80** points.
- F.8.3.2 | The milestone will be evaluated as a whole, considering the overall quality of the communication, the clarity of the message and the ability of the team to present their project in an engaging way.
- F.8.3.3 | The evaluation will consider several key dimensions of the presentation. The maximum weight of each dimension and the aspects that may be considered by the jury include:
- 1) Communication and storytelling (maximum 25 points)
    - Clarity of the message
    - Structure of the presentation
    - Ability to communicate complex ideas clearly
    - Engagement and persuasiveness
  - 2) Project content and technical explanation (maximum 25 points)
    - Quality of the explanation of the prototype
    - Inclusion of the key project milestones
    - Integration of engineering and innovation aspects
    - Coherence between the different elements of the project
  - 3) Creativity and originality of the presentation (maximum 15 points)
    - Creativity of the format
    - Originality of the narrative
    - Ability to capture attention
  - 4) Visual and production quality (maximum 15 points)
    - Video clarity and editing
    - Audio quality
    - Effective use of visual material

- F.8.3.4 | These values represent the approximate weight of each dimension but do not imply that the milestone will be evaluated strictly by sections.
- F.8.3.5 | The jury will assign a global score based on the overall quality, clarity and impact of the pitch presentation.

## **ARTICLE 9: MS1 7<sup>th</sup> MILESTONE, FINAL EVENT PRESENTATION**

### **F.9.1 Milestone preamble**

- F.9.1.1 | This milestone consists of a formal presentation delivered during the Final Event in front of an international jury composed of professionals from the motorsport, mobility and engineering industries.
- F.9.1.2 | The Final Presentation represents the closing moment of the MS1 phase and provides the team with the opportunity to present the overall development of the project and the final prototype built for the competition.
- F.9.1.3 | The objective of this milestone is to allow the jury to understand the engineering process followed by the team, the main technical decisions taken during the project and the results achieved.
- F.9.1.4 | Special attention should be given to the evolution of the project, including the differences between the initially proposed concept and the final prototype presented at the event. Teams are encouraged to explain the reasons behind any relevant design, manufacturing or economic deviations.
- F.9.1.5 | This presentation should demonstrate the team's capacity to clearly communicate complex engineering work to a professional audience.
- F.9.1.6 | The Final Presentation may also serve as an opportunity for teams to interact with industry professionals and potential future employers present at the event.

### **F.9.2 Presentation format**

- F.9.2.1 | The presentation will take place during the Final Event according to the schedule defined by the Organization.
- F.9.2.2 | The maximum duration of the presentation will be 30 minutes.
- F.9.2.3 | The jury may ask questions during the presentation.
- After the presentation, the jury may continue the discussion with the team in the team's box in the presence of the prototype.

- F.9.2.4 | A maximum of three students may deliver the presentation.
- Other team members, the Team Leader and the Team Tutors may attend the presentation but may not intervene during the exposition.
- F.9.2.5 | A digital screen with HDMI connection will be provided by the Organization. Presenting students must bring their own laptop and the required adapters or chargers.
- Internet connection will not be guaranteed during the Final Event. Teams are responsible for ensuring that their presentation materials can run offline.
- F.9.3      Section1: Product definition and project evolution**
- F.9.3.1 | The objective of this section is to explain the definition of the prototype and the development process followed by the team during the competition.
- F.9.3.2 | The presentation should highlight the main engineering decisions taken during the design and manufacturing of the prototype.
- Teams should explain how the initial concept evolved during the project and describe any relevant deviations from the original design presented in the MS1 milestones.
- These deviations may include design modifications, manufacturing challenges, changes in materials or adjustments required to achieve a functional prototype.
- F.9.3.3 | The objective of this section is not only to present the final prototype but also to demonstrate the engineering reasoning and problem-solving capabilities of the team.
- F.9.4      Section 2: Innovation project**
- F.9.4.1 | The objective of this section is to present the innovation project developed by the team during the competition.
- F.9.4.2 | Teams must explain the motivation behind the selected innovation and how it contributes to improving the prototype.
- The presentation should describe the development process of the innovation, the decisions taken during the project and the validation methods used.
- If the innovation has been integrated into the final prototype, the team should clearly demonstrate how it operates and what impact it has on the vehicle performance or project objectives.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.9.4.3      | The jury may ask additional questions regarding the technical feasibility, development process and potential impact of the innovation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>F.9.5</b> | <b>Evaluation and scoring</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| F.9.5.1      | The maximum score for this milestone is <b>60</b> points.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| F.9.5.2      | <p>The milestone will be evaluated as a whole, considering the clarity of the presentation, the engineering reasoning shown by the team and the ability to communicate the project to a professional audience.</p> <p>Although teams are free to structure their presentation as they consider appropriate, the evaluation will consider several key dimensions of the project.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| F.9.5.3      | <p>The maximum weight of each dimension and the aspects that may be considered by the jury include:</p> <ol style="list-style-type: none"> <li>1) Product definition and project evolution (maximum 25 points) <ul style="list-style-type: none"> <li>- Clarity of the explanation of the initial project concept.</li> <li>- Realism and coherence of the objectives defined at the start of the project.</li> <li>- Explanation of the main engineering decisions taken during design and manufacturing.</li> <li>- Identification and justification of relevant changes or deviations during the development process.</li> <li>- Overall clarity and quality of the explanation.</li> </ul> </li> <li>2) Innovation project (maximum 35 points) <ul style="list-style-type: none"> <li>- Justification of the selected system or component to be improved.</li> <li>- Clarity in explaining the expected improvement and the objectives of the innovation.</li> <li>- Explanation of the development process and decisions taken during the project.</li> <li>- Demonstration of the innovation implemented in the prototype.</li> <li>- Technical understanding and mastery of the subject by the presenting students.</li> </ul> </li> </ol> |
| F.9.5.4      | These values represent the approximate weight of each dimension but do not imply that the presentation will be evaluated strictly by sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| F.9.5.5      | The jury will assign a global score based on the overall quality of the presentation, the technical understanding demonstrated by the team and the clarity of communication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## ARTICLE 10: MOTOSTUDENT ELECTRIC DELIVERIES

### F.10.1 MSE 1<sup>st</sup> Delivery: Electric and battery pack scheme

F.10.1.1 | The objective of this milestone is to review the electrical architecture and the battery pack concept of the prototypes participating in the MotoStudent Electric category.

This delivery must allow the Organization to understand the general design of the electrical system and verify its compliance with the regulation before the Final Event.

F.10.1.2 | The submission must include at least the following elements:

- A general unifilar type schematic in order to clearly represent the electrical architecture and connections of the prototype.
- The complete wiring scheme of the prototype.
- The complete wiring harness schematics
- A general description of the battery pack architecture.
- A 3D representation of the battery pack containers or housing.
- The type of cells used and their main technical characteristics.
- The mechanical protection systems implemented for the battery pack and cells.
- A list of the main electrical components of the battery pack and the vehicle, including their name and location within the prototype. Datasheets are not required unless specifically requested by the Organization.

F.10.1.3 | The Organization reserves the right to request additional information or propose modifications if the submitted design does not clearly demonstrate compliance with the regulation or presents potential safety concerns.

F.10.1.4 | The Organization will provide feedback to the teams within 90 calendar days after the submission deadline.

### F.10.2 MSE 2<sup>nd</sup> Delivery: Electrical precheck

F.10.2.1 | The objective of this milestone is to perform a preliminary verification of the electrical compliance of the prototype before the Final Event and the official scrutineering process.

F.10.2.2 | The submission must review the relevant requirements described in **Section D** of this regulation and demonstrate their compliance.

F.10.2.3 | Teams must include graphical evidence of the implemented solutions whenever possible (for example photographs, diagrams or screenshots).

F.10.2.4 | The Organization reserves the right to request additional information or corrections if the provided documentation is insufficient to verify compliance.



The Organization will provide feedback to the teams within 90 calendar days after the submission deadline.

## **ARTICLE 11: RIDER ELIGIBILITY DECLARATION**

### **F.11.1 Rider Eligibility Declaration**

F.11.1.1 | Each team must submit the rider declaration available on the MotoStudent Platform.

F.11.1.2 | Each declared rider must include their riding experience and background.

F.11.1.3 | The rider information must comply with the requirements defined in Article G.2.

If a team selects or works with a rider prior to the formal declaration period, it shall be the team's responsibility to verify in advance that such rider is expected to comply with the eligibility requirements defined in these Rules & Regulations.

F.11.1.4 | If the documentation is incomplete or inconsistent, the Organization may request additional information.

## **ARTICLE 12: MS2 PHASE**

### **F.12.1 Objectives and procedure**

F.12.1.1 | The MS2 dynamic tests consist of a series of exercises designed to demonstrate and evaluate the behaviour, performance and reliability of the prototypes developed during the MotoStudent International Competition.

F.12.1.2 | Each category will have its own independent classification and will compete in parallel during the Final Event.

F.12.1.3 | All dynamic tests must be carried out by the rider declared by the team during the Administrative Milestone.

F.12.1.4 | Minor adjustments to dimensions, schedules or operational aspects of the tests may be introduced by the Organization during the Final Event if required for safety or operational reasons.

F.12.1.5 | The dynamic tests are designed to evaluate the overall engineering performance of the prototype. The objective is not only to reward the fastest vehicle, but the most balanced and well-developed prototype across multiple performance dimensions such as braking capability, acceleration, agility, consistency and race performance.

Teams are therefore encouraged to develop a complete and reliable prototype rather than optimizing the vehicle exclusively for a single test.

F.12.1.6 | All measurements used for scoring purposes will be those recorded by the Organization's official timing and measurement systems. Data recorded by teams through their own devices (such as GPS loggers, telemetry systems or onboard instrumentation) may differ due to measurement methodology and will not be considered valid for the official competition results.

F.12.1.7 | Tests 1, 2 and 3 may have two attempts. In such case, the best result obtained by the team will be considered as the valid result for scoring purposes.

F.12.1.8 | The Competition Organization will publish the official results of each dynamic test as soon as reasonably possible, taking into account the operational needs of the Final Event and any verification or incident review that may be required before validation of the results.

## **F.12.2 Test 1: Brake**

F.12.2.1 | The brake test evaluates the distance required for the prototype to come to a complete stop from a minimum speed of 80 km/h.

F.12.2.2 | The start will be given by a signal from a sport marshal at the starting point.

F.12.2.3 | The rider will have a maximum time of 10 seconds from the signal to start the test.

F.12.2.4 | The total length of the acceleration straight is 200 m.

F.12.2.5 | The speed reached at the end of the acceleration straight will be measured by the Competition Organization.

F.12.2.6 | The braking distance will be measured from the speed trap to the foremost part of the prototype once the motorcycle has come to a complete stop.

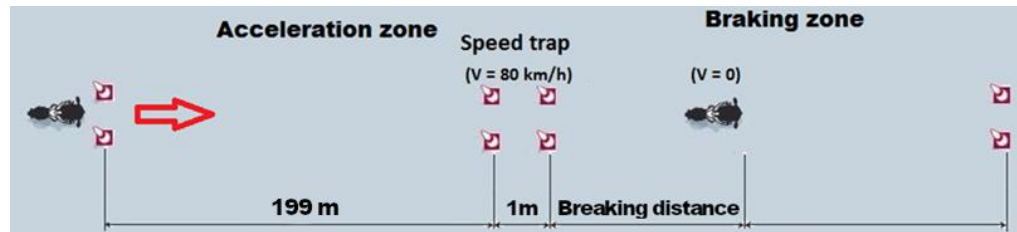
F.12.2.7 | If the speed measured by the Organization is lower than the required minimum speed, the following penalties will be applied to the measured braking distance:

- < 80.00 km/h and  $\geq$  79.00 km/h: +4 m
- < 79.00 km/h and  $\geq$  78.00 km/h: +6 m
- < 78.00 km/h and  $\geq$  77.00 km/h: +8 m
- < 77.00 km/h and  $\geq$  76.00 km/h: +10 m
- < 76.00 km/h and  $\geq$  75.00 km/h: +12 m
- < 75.00 km/h and  $\geq$  70.00 km/h: +20 m

F.12.2.8 | If the speed measured is lower than 70.00 km/h the result of the test will be considered as NC (Not Compliant).

F.12.2.9 | Two photocells will be placed before the braking zone, separated by 1 meter, in order to record the speed during the test.

F.12.2.10 | Graphic description of the test (for illustrative purposes only):



Picture 45 - Graphic description of MS1 Test 1: Brake

F.12.2.11 | The score of the test will be assigned according to the ranking of the braking distances obtained (including any applicable penalties), from the lowest distance to the highest. The points will be awarded according to the following table.

| Rank             | Score | Rank             | Score | Rank             | Score |
|------------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup>  | 90    | 18 <sup>th</sup> | 43    | 35 <sup>th</sup> | 26    |
| 2 <sup>nd</sup>  | 85    | 19 <sup>th</sup> | 42    | 36 <sup>th</sup> | 25    |
| 3 <sup>rd</sup>  | 80    | 20 <sup>th</sup> | 41    | 37 <sup>th</sup> | 24    |
| 4 <sup>th</sup>  | 76    | 21 <sup>st</sup> | 40    | 38 <sup>th</sup> | 23    |
| 5 <sup>th</sup>  | 72    | 22 <sup>nd</sup> | 39    | 39 <sup>th</sup> | 22    |
| 6 <sup>th</sup>  | 68    | 23 <sup>rd</sup> | 38    | 40 <sup>th</sup> | 21    |
| 7 <sup>th</sup>  | 65    | 24 <sup>th</sup> | 37    | 41 <sup>st</sup> | 20    |
| 8 <sup>th</sup>  | 62    | 25 <sup>th</sup> | 36    | 42 <sup>nd</sup> | 19    |
| 9 <sup>th</sup>  | 59    | 26 <sup>th</sup> | 35    | 43 <sup>rd</sup> | 18    |
| 10 <sup>th</sup> | 56    | 27 <sup>th</sup> | 34    | 44 <sup>th</sup> | 17    |
| 11 <sup>th</sup> | 54    | 28 <sup>th</sup> | 33    | 45 <sup>th</sup> | 16    |
| 12 <sup>th</sup> | 52    | 29 <sup>th</sup> | 32    | 46 <sup>th</sup> | 15    |
| 13 <sup>th</sup> | 50    | 30 <sup>th</sup> | 31    | 47 <sup>th</sup> | 14    |
| 14 <sup>th</sup> | 48    | 31 <sup>st</sup> | 30    | 48 <sup>th</sup> | 13    |
| 15 <sup>th</sup> | 46    | 32 <sup>nd</sup> | 29    | 49 <sup>th</sup> | 12    |
| 16 <sup>th</sup> | 45    | 33 <sup>rd</sup> | 28    | 50 <sup>th</sup> | 11    |
| 17 <sup>th</sup> | 44    | 34 <sup>th</sup> | 27    | ...              | 10    |

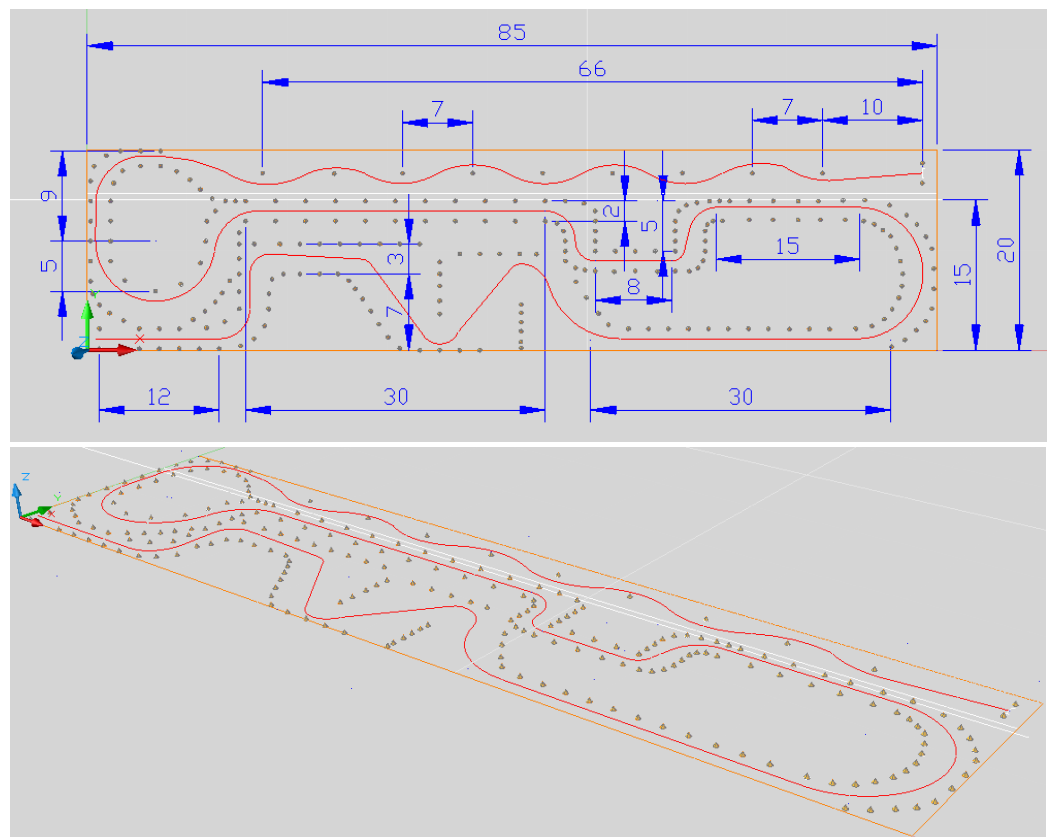
Table 2 - MS2 Test 1: Brake scores

### F.12.3 Test 2: Gymkhana

F.12.3.1 | The second test consists of a short circuit designed to evaluate the agility, manoeuvrability and acceleration capability of the prototype.

F.12.3.2 | The start will be given by a signal from a sport marshal.

- F.12.3.3 | Two photocells will be placed at the start and finish line in order to measure the total time required to complete the course.
- F.12.3.4 | The rider will have a maximum time of 10 seconds from the signal to start the test.
- F.12.3.5 | If a cone is moved or knocked down by the rider or the prototype, a penalty of 5 seconds will be applied for each cone. If more than 5 cones are moved, the test will result as NC.
- F.12.3.6 | If a cone is passed on the wrong side, a penalty of 10 seconds will be applied. If more than 3 cones are incorrectly passed, the test will result as NC.
- F.12.3.7 | Graphic description of the test (for illustrative purposes only) (TBC):



Picture 46 - Graphical description of MS2 Test 2: Gymkhana. Measures indicated in meters (m)

- F.12.3.8 | The score of the test will be assigned according to the ranking of the final times obtained, including any applicable penalties, from the lowest time to the highest. Points will be awarded according to the following table.

| Rank            | Score | Rank             | Score | Rank             | Score |
|-----------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup> | 70    | 18 <sup>th</sup> | 37    | 35 <sup>th</sup> | 20    |
| 2 <sup>nd</sup> | 66    | 19 <sup>th</sup> | 36    | 36 <sup>th</sup> | 19    |
| 3 <sup>rd</sup> | 62    | 20 <sup>th</sup> | 35    | 37 <sup>th</sup> | 18    |

|                  |    |                  |    |                  |    |
|------------------|----|------------------|----|------------------|----|
| 4 <sup>th</sup>  | 59 | 21 <sup>st</sup> | 34 | 38 <sup>th</sup> | 17 |
| 5 <sup>th</sup>  | 56 | 22 <sup>nd</sup> | 33 | 39 <sup>th</sup> | 16 |
| 6 <sup>th</sup>  | 53 | 23 <sup>rd</sup> | 32 | 40 <sup>th</sup> | 15 |
| 7 <sup>th</sup>  | 51 | 24 <sup>th</sup> | 31 | 41 <sup>st</sup> | 14 |
| 8 <sup>th</sup>  | 49 | 25 <sup>th</sup> | 30 | 42 <sup>nd</sup> | 13 |
| 9 <sup>th</sup>  | 47 | 26 <sup>th</sup> | 29 | 43 <sup>rd</sup> | 12 |
| 10 <sup>th</sup> | 45 | 27 <sup>th</sup> | 28 | 44 <sup>th</sup> | 11 |
| 11 <sup>th</sup> | 44 | 28 <sup>th</sup> | 27 | 45 <sup>th</sup> | 10 |
| 12 <sup>th</sup> | 43 | 29 <sup>th</sup> | 26 | 46 <sup>th</sup> | 9  |
| 13 <sup>th</sup> | 42 | 30 <sup>th</sup> | 25 | 47 <sup>th</sup> | 8  |
| 14 <sup>th</sup> | 41 | 31 <sup>st</sup> | 24 | 48 <sup>th</sup> | 7  |
| 15 <sup>th</sup> | 40 | 32 <sup>nd</sup> | 23 | 49 <sup>th</sup> | 6  |
| 16 <sup>th</sup> | 39 | 33 <sup>rd</sup> | 22 | 50 <sup>th</sup> | 5  |
| 17 <sup>th</sup> | 38 | 34 <sup>th</sup> | 21 | ...              | 5  |

Table 3 - MS2 Test 2: Gymkhana scores

#### F.12.4 Test 3: Acceleration

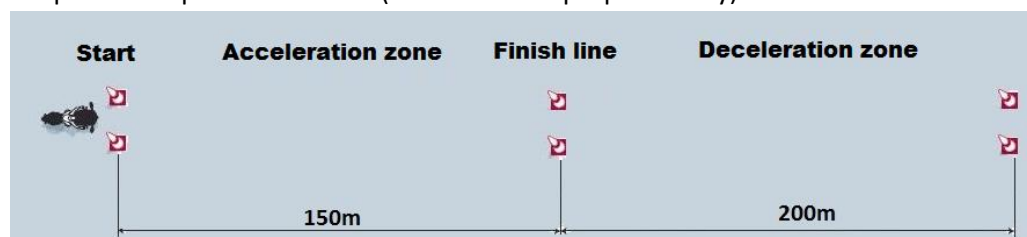
F.12.4.1 | The third test consists of a straight-line acceleration run over a distance of 150 m in order to evaluate the acceleration performance of the prototype.

F.12.4.2 | The start will be given by a signal from a sport marshal at the starting point.

F.12.4.3 | Two photocells will be placed at the start and at the end of the 150 m straight in order to record the start and end time for each prototype. The difference between both measurements will determine the total time used to evaluate the acceleration performance.

F.12.4.4 | The rider will have a maximum time of 10 seconds from the signal to start the test.

F.12.4.5 | Graphic description of the test (for illustrative purposes only):



Picture 47 - Graphical description of MS2 Test 3: Acceleration

F.12.4.6 | The score of the test will be assigned according to the ranking of the times obtained during the test, from the lowest time to the highest. Points will be awarded according to the following table.

| Rank            | Score | Rank             | Score | Rank             | Score |
|-----------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup> | 90    | 18 <sup>th</sup> | 43    | 35 <sup>th</sup> | 26    |

|                  |    |                  |    |                  |    |
|------------------|----|------------------|----|------------------|----|
| 2 <sup>nd</sup>  | 85 | 19 <sup>th</sup> | 42 | 36 <sup>th</sup> | 25 |
| 3 <sup>rd</sup>  | 80 | 20 <sup>th</sup> | 41 | 37 <sup>th</sup> | 24 |
| 4 <sup>th</sup>  | 76 | 21 <sup>st</sup> | 40 | 38 <sup>th</sup> | 23 |
| 5 <sup>th</sup>  | 72 | 22 <sup>nd</sup> | 39 | 39 <sup>th</sup> | 22 |
| 6 <sup>th</sup>  | 68 | 23 <sup>rd</sup> | 38 | 40 <sup>th</sup> | 21 |
| 7 <sup>th</sup>  | 65 | 24 <sup>th</sup> | 37 | 41 <sup>st</sup> | 20 |
| 8 <sup>th</sup>  | 62 | 25 <sup>th</sup> | 36 | 42 <sup>nd</sup> | 19 |
| 9 <sup>th</sup>  | 59 | 26 <sup>th</sup> | 35 | 43 <sup>rd</sup> | 18 |
| 10 <sup>th</sup> | 56 | 27 <sup>th</sup> | 34 | 44 <sup>th</sup> | 17 |
| 11 <sup>th</sup> | 54 | 28 <sup>th</sup> | 33 | 45 <sup>th</sup> | 16 |
| 12 <sup>th</sup> | 52 | 29 <sup>th</sup> | 32 | 46 <sup>th</sup> | 15 |
| 13 <sup>th</sup> | 50 | 30 <sup>th</sup> | 31 | 47 <sup>th</sup> | 14 |
| 14 <sup>th</sup> | 48 | 31 <sup>st</sup> | 30 | 48 <sup>th</sup> | 13 |
| 15 <sup>th</sup> | 46 | 32 <sup>nd</sup> | 29 | 49 <sup>th</sup> | 12 |
| 16 <sup>th</sup> | 45 | 33 <sup>rd</sup> | 28 | 50 <sup>th</sup> | 11 |
| 17 <sup>th</sup> | 44 | 34 <sup>th</sup> | 27 | ...              | 10 |

Table 4 - MS2 Test 3: Acceleration test score

#### F.12.5 Test 4: $V_{\max}$

- F.12.5.1 | This test consists of recording the maximum speed reached at the official speed trap during the Free Practice 2 session.
- F.12.5.2 | Any valid passage through the speed trap (including laps without passing through the finish line), respecting track limits, will be considered for the test.
- F.12.5.3 | The highest recorded speed for each team will be used to determine the classification.
- F.12.5.4 | The score of the test will be assigned according to the ranking of the maximum speeds recorded for each team, from the highest speed to the lowest. Points will be awarded according to the following table.

| Rank             | Score | Rank             | Score | Rank             | Score |
|------------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup>  | 30    | 18 <sup>th</sup> | 10    | 35 <sup>th</sup> | 7     |
| 2 <sup>nd</sup>  | 27    | 19 <sup>th</sup> | 10    | 36 <sup>th</sup> | 6     |
| 3 <sup>rd</sup>  | 25    | 20 <sup>th</sup> | 10    | 37 <sup>th</sup> | 6     |
| 4 <sup>th</sup>  | 23    | 21 <sup>st</sup> | 9     | 38 <sup>th</sup> | 6     |
| 5 <sup>th</sup>  | 22    | 22 <sup>nd</sup> | 9     | 39 <sup>th</sup> | 6     |
| 6 <sup>th</sup>  | 21    | 23 <sup>rd</sup> | 9     | 40 <sup>th</sup> | 6     |
| 7 <sup>th</sup>  | 20    | 24 <sup>th</sup> | 9     | 41 <sup>st</sup> | 5     |
| 8 <sup>th</sup>  | 19    | 25 <sup>th</sup> | 9     | 42 <sup>nd</sup> | 5     |
| 9 <sup>th</sup>  | 18    | 26 <sup>th</sup> | 8     | 43 <sup>rd</sup> | 5     |
| 10 <sup>th</sup> | 17    | 27 <sup>th</sup> | 8     | 44 <sup>th</sup> | 5     |
| 11 <sup>th</sup> | 16    | 28 <sup>th</sup> | 8     | 45 <sup>th</sup> | 5     |
| 12 <sup>th</sup> | 15    | 29 <sup>th</sup> | 8     | 46 <sup>th</sup> | 4     |
| 13 <sup>th</sup> | 14    | 30 <sup>th</sup> | 8     | 47 <sup>th</sup> | 4     |
| 14 <sup>th</sup> | 13    | 31 <sup>st</sup> | 7     | 48 <sup>th</sup> | 4     |

|                  |    |                  |   |                  |   |
|------------------|----|------------------|---|------------------|---|
| 15 <sup>th</sup> | 12 | 32 <sup>nd</sup> | 7 | 49 <sup>th</sup> | 4 |
| 16 <sup>th</sup> | 11 | 33 <sup>rd</sup> | 7 | 50 <sup>th</sup> | 4 |
| 17 <sup>th</sup> | 10 | 34 <sup>th</sup> | 7 | ...              | 3 |

Table 5 - MS2 Test 4:  $V_{max}$  scores

#### F.12.6 Test 5: Regularity

F.12.6.1 The regularity test evaluates the consistency of the prototype and rider combination. The objective of this test is to determine the consistency of the lap times achieved in Sector 2 of the Speed Circuit during the Warm-up session.

F.12.6.2 All Sector 2 times recorded during the test will be considered for the evaluation, including laps that do not complete the full lap, provided that track limits are respected.

F.12.6.3 For each team, the regularity value will be calculated using the Sector 2 times recorded during the session. The recorded sector times will be sorted in ascending order and the minimum difference between the  $n^{\text{th}}$  and the  $(n+2)^{\text{th}}$  sector times will be determined, where  $n$  ranges from 1 to  $m-2$  and  $m$  represents the total number of recorded sector times.

$$\min (|x_n - x_{n+2}|)$$

F.12.6.4 The score of the test will be assigned according to the ranking of the regularity values obtained by each team, from the lowest time difference to the highest. Points will be awarded according to the following table.

| Rank             | Score | Rank             | Score | Rank             | Score |
|------------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup>  | 50    | 18 <sup>th</sup> | 23    | 35 <sup>th</sup> | 9     |
| 2 <sup>nd</sup>  | 46    | 19 <sup>th</sup> | 22    | 36 <sup>th</sup> | 8     |
| 3 <sup>rd</sup>  | 43    | 20 <sup>th</sup> | 21    | 37 <sup>th</sup> | 8     |
| 4 <sup>th</sup>  | 40    | 21 <sup>st</sup> | 20    | 38 <sup>th</sup> | 8     |
| 5 <sup>th</sup>  | 38    | 22 <sup>nd</sup> | 19    | 39 <sup>th</sup> | 8     |
| 6 <sup>th</sup>  | 36    | 23 <sup>rd</sup> | 18    | 40 <sup>th</sup> | 8     |
| 7 <sup>th</sup>  | 34    | 24 <sup>th</sup> | 17    | 41 <sup>st</sup> | 7     |
| 8 <sup>th</sup>  | 33    | 25 <sup>th</sup> | 16    | 42 <sup>nd</sup> | 7     |
| 9 <sup>th</sup>  | 32    | 26 <sup>th</sup> | 15    | 43 <sup>rd</sup> | 7     |
| 10 <sup>th</sup> | 31    | 27 <sup>th</sup> | 14    | 44 <sup>th</sup> | 7     |
| 11 <sup>th</sup> | 30    | 28 <sup>th</sup> | 13    | 45 <sup>th</sup> | 7     |
| 12 <sup>th</sup> | 29    | 29 <sup>th</sup> | 12    | 46 <sup>th</sup> | 6     |
| 13 <sup>th</sup> | 28    | 30 <sup>th</sup> | 11    | 47 <sup>th</sup> | 6     |
| 14 <sup>th</sup> | 27    | 31 <sup>st</sup> | 10    | 48 <sup>th</sup> | 6     |
| 15 <sup>th</sup> | 26    | 32 <sup>nd</sup> | 9     | 49 <sup>th</sup> | 6     |
| 16 <sup>th</sup> | 25    | 33 <sup>rd</sup> | 9     | 50 <sup>th</sup> | 6     |
| 17 <sup>th</sup> | 24    | 34 <sup>th</sup> | 9     | ...              | 5     |

Table 6 - MS2 Test 5: Regularity scores

## F.12.7 Test 6: Pole position

- F.12.7.1 | This test corresponds to the fastest lap time obtained during the official Qualifying session.
- F.12.7.2 | The score of the test will be assigned according to the ranking of the lap times recorded during the Qualifying session, from the lowest time to the highest. Points will be awarded according to the following table.

| Rank             | Score | Rank             | Score | Rank             | Score |
|------------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup>  | 30    | 18 <sup>th</sup> | 10    | 35 <sup>th</sup> | 7     |
| 2 <sup>nd</sup>  | 27    | 19 <sup>th</sup> | 10    | 36 <sup>th</sup> | 6     |
| 3 <sup>rd</sup>  | 25    | 20 <sup>th</sup> | 10    | 37 <sup>th</sup> | 6     |
| 4 <sup>th</sup>  | 23    | 21 <sup>st</sup> | 9     | 38 <sup>th</sup> | 6     |
| 5 <sup>th</sup>  | 22    | 22 <sup>nd</sup> | 9     | 39 <sup>th</sup> | 6     |
| 6 <sup>th</sup>  | 21    | 23 <sup>rd</sup> | 9     | 40 <sup>th</sup> | 6     |
| 7 <sup>th</sup>  | 20    | 24 <sup>th</sup> | 9     | 41 <sup>st</sup> | 5     |
| 8 <sup>th</sup>  | 19    | 25 <sup>th</sup> | 9     | 42 <sup>nd</sup> | 5     |
| 9 <sup>th</sup>  | 18    | 26 <sup>th</sup> | 8     | 43 <sup>rd</sup> | 5     |
| 10 <sup>th</sup> | 17    | 27 <sup>th</sup> | 8     | 44 <sup>th</sup> | 5     |
| 11 <sup>th</sup> | 16    | 28 <sup>th</sup> | 8     | 45 <sup>th</sup> | 5     |
| 12 <sup>th</sup> | 15    | 29 <sup>th</sup> | 8     | 46 <sup>th</sup> | 4     |
| 13 <sup>th</sup> | 14    | 30 <sup>th</sup> | 8     | 47 <sup>th</sup> | 4     |
| 14 <sup>th</sup> | 13    | 31 <sup>st</sup> | 7     | 48 <sup>th</sup> | 4     |
| 15 <sup>th</sup> | 12    | 32 <sup>nd</sup> | 7     | 49 <sup>th</sup> | 4     |
| 16 <sup>th</sup> | 11    | 33 <sup>rd</sup> | 7     | 50 <sup>th</sup> | 4     |
| 17 <sup>th</sup> | 10    | 34 <sup>th</sup> | 7     | ...              | 3     |

Table 7 - MS2 Test 6: Pole position scores

## F.12.8 Test 7: Best Race Lap

- F.12.8.1 | This test corresponds to the fastest lap time achieved during the Play-Off race and / or the Final race.
- F.12.8.2 | For teams participating in both races, only the fastest lap obtained in the Final race will be considered.
- F.12.8.3 | The score of the test will be assigned according to the ranking of the best lap times recorded during the race sessions, from the lowest time to the highest. Points will be awarded according to the following table.

| Rank            | Score | Rank             | Score | Rank             | Score |
|-----------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup> | 30    | 16 <sup>th</sup> | 11    | 31 <sup>st</sup> | 7     |
| 2 <sup>nd</sup> | 27    | 17 <sup>th</sup> | 10    | 32 <sup>nd</sup> | 7     |
| 3 <sup>rd</sup> | 25    | 18 <sup>th</sup> | 10    | 33 <sup>rd</sup> | 7     |
| 4 <sup>th</sup> | 23    | 19 <sup>th</sup> | 10    | 34 <sup>th</sup> | 7     |



|                  |    |                  |    |                  |   |
|------------------|----|------------------|----|------------------|---|
| 5 <sup>th</sup>  | 22 | 20 <sup>th</sup> | 10 | 35 <sup>th</sup> | 7 |
| 6 <sup>th</sup>  | 21 | 21 <sup>st</sup> | 9  | 36 <sup>th</sup> | 6 |
| 7 <sup>th</sup>  | 20 | 22 <sup>nd</sup> | 9  | 37 <sup>th</sup> | 6 |
| 8 <sup>th</sup>  | 19 | 23 <sup>rd</sup> | 9  | 38 <sup>th</sup> | 6 |
| 9 <sup>th</sup>  | 18 | 24 <sup>th</sup> | 9  | 39 <sup>th</sup> | 6 |
| 10 <sup>th</sup> | 17 | 25 <sup>th</sup> | 9  | 40 <sup>th</sup> | 6 |
| 11 <sup>th</sup> | 16 | 26 <sup>th</sup> | 8  | 41 <sup>st</sup> | 5 |
| 12 <sup>th</sup> | 15 | 27 <sup>th</sup> | 8  | 42 <sup>nd</sup> | 5 |
| 13 <sup>th</sup> | 14 | 28 <sup>th</sup> | 8  | 43 <sup>rd</sup> | 5 |
| 14 <sup>th</sup> | 13 | 29 <sup>th</sup> | 8  | 44 <sup>th</sup> | 5 |
| 15 <sup>th</sup> | 12 | 30 <sup>th</sup> | 8  | ...              | 5 |

Table 8 - MS2 Test 7: Best Race Lap scores

## F.12.9 Test 8: Race result

F.12.9.1 The result of Test 8: Race Result will be determined according to the official classification of the Final Race and the Play-Off Race. For the purposes of Test 8 scoring, teams classified in the Final Race shall be ranked first, followed by teams classified in the Play-Off Race.

Teams not classified in either race, including DNFs, shall be ranked after all classified teams and ordered according to the distance completed in their respective race.

F.12.9.2 The score of the test will be assigned according to the ranking defined in Art. F.12.9.1, from the highest finishing position to the lowest. Points will be awarded according to the following table.

| Rank             | Score | Rank             | Score | Rank             | Score |
|------------------|-------|------------------|-------|------------------|-------|
| 1 <sup>st</sup>  | 150   | 16 <sup>th</sup> | 75    | 31 <sup>st</sup> | 43    |
| 2 <sup>nd</sup>  | 143   | 17 <sup>th</sup> | 72    | 32 <sup>nd</sup> | 42    |
| 3 <sup>rd</sup>  | 136   | 18 <sup>th</sup> | 69    | 33 <sup>rd</sup> | 41    |
| 4 <sup>th</sup>  | 130   | 19 <sup>th</sup> | 66    | 34 <sup>th</sup> | 40    |
| 5 <sup>th</sup>  | 124   | 20 <sup>th</sup> | 63    | 35 <sup>th</sup> | 39    |
| 6 <sup>th</sup>  | 118   | 21 <sup>st</sup> | 60    | 36 <sup>th</sup> | 38    |
| 7 <sup>th</sup>  | 113   | 22 <sup>nd</sup> | 58    | 37 <sup>th</sup> | 37    |
| 8 <sup>th</sup>  | 108   | 23 <sup>rd</sup> | 56    | 38 <sup>th</sup> | 36    |
| 9 <sup>th</sup>  | 103   | 24 <sup>th</sup> | 54    | 39 <sup>th</sup> | 35    |
| 10 <sup>th</sup> | 98    | 25 <sup>th</sup> | 52    | 40 <sup>th</sup> | 34    |
| 11 <sup>th</sup> | 94    | 26 <sup>th</sup> | 50    | 41 <sup>st</sup> | 33    |
| 12 <sup>th</sup> | 90    | 27 <sup>th</sup> | 48    | 42 <sup>nd</sup> | 32    |
| 13 <sup>th</sup> | 86    | 28 <sup>th</sup> | 46    | 43 <sup>rd</sup> | 31    |
| 14 <sup>th</sup> | 82    | 29 <sup>th</sup> | 45    | 44 <sup>th</sup> | 30    |
| 15 <sup>th</sup> | 78    | 30 <sup>th</sup> | 44    | ...              | 0     |

Table 9 - MS2 Test 8: Race result scores

## ARTICLE 13: MOTOSTUDENT AWARDS

### F.13.1 General Principles

- F.13.1.1 | The MotoStudent Awards recognise the teams that demonstrate excellence in project development, technical performance and sporting results throughout the MotoStudent International Competition.

### F.13.2 MS1 Awards

- F.13.2.1 | The **Best MS1 Award** is granted to the team from each category that achieves the highest MS1 score (maximum of **600** points), calculated as the sum of the following milestones minus any applicable penalties:
- MS1 1<sup>st</sup> milestone: Team plan (maximum of **75** points)
  - MS1 2<sup>nd</sup> milestone: Product definition (maximum of **100** points)
  - MS1 3<sup>rd</sup> milestone: Innovation (maximum of **150** points)
  - MS1 4<sup>th</sup> milestone: Prototype CAD and Drawings (maximum of **60** points)
  - MS1 5<sup>th</sup> milestone: Economical Plan (maximum of **75** points)
  - MS1 6<sup>th</sup> milestone: Pitch Presentation (maximum of **80** points)
  - MS1 7<sup>th</sup> milestone: Final Event Presentation (maximum of **60** points)
- F.13.2.2 | The **Best Design Award** recognises the team that achieves the highest score (maximum of **220** points) considering the following milestones minus any applicable penalties:
- MS1 2<sup>nd</sup> milestone: Product Definition (maximum of **100** points)
  - MS1 4<sup>th</sup> milestone: Prototype CAD and Drawings (maximum of **60** points)
  - MS1 7<sup>th</sup> milestone: Final Event Presentation (maximum of **60** points)
- F.13.2.3 | The **Best Innovation Award** recognises the team that achieves the highest score (maximum of **210** points) considering the following milestones minus any applicable penalties:
- MS1 3<sup>rd</sup> milestone: Innovation (maximum of **150** points)
  - MS1 7<sup>th</sup> milestone: Final Event presentation (maximum of **60** points)
- F.13.2.4 | In case of a tie in any MS1 award, the Organization will apply a justified tie-breaker methodology.
- F.13.2.5 | Tie-breakers apply only to determine the awarded team and do not modify the official classification.
- F.13.2.6 | For clarity and reference, the following table summarizes the MS1 awards and the milestones contributing to each award:

| MS1 Awards                      |                                     | Milestones |     |     |     |     |     |     |
|---------------------------------|-------------------------------------|------------|-----|-----|-----|-----|-----|-----|
|                                 |                                     | 1st        | 2nd | 3rd | 4th | 5th | 6th | 7th |
| Best MS1<br>(out of 600 points) | Best Design (out of 220 points)     |            | 100 |     | 60  |     |     | 60  |
|                                 | Best Innovation (out of 210 points) |            |     | 150 |     |     |     |     |
|                                 |                                     | 75         |     |     |     | 75  | 80  |     |

Table 10 - MS1 Awards

### F.13.3 MS2 Awards

F.13.3.1 The total MS2 score (maximum of **540** points) is obtained by adding the scores achieved in the following dynamic tests:

- Test 1: Brake (maximum of **90** points)
- Test 2: Gymkhana (maximum of **70** points)
- Test 3: Acceleration (maximum of **90** points)
- Test 4:  $V_{\max}$  (maximum of **30** points)
- Test 5: Regularity (maximum of **50** points)
- Test 6: Pole Position (maximum of **30** points)
- Test 7: Best Race Lap (maximum of **30** points)
- Test 8: Race Result (maximum of **150** points)

F.13.3.2 The **Best MS2 Award** is granted to the teams finishing in 1<sup>st</sup>, 2<sup>nd</sup> and 3<sup>rd</sup> position in the MS2 classification for each category.

F.13.3.3 If, for reasons beyond the Organization's control, one or more scoring tests must be cancelled, the Competition Organization reserves the right to adjust the allocation of points for the affected tests, with prior notification to the participating teams.

F.13.3.4 In case of a tie in the MS2 classification, the following criterion will be applied until the tie is resolved:

- **Best MS1 Award** score

F.13.3.5 Tie-breakers apply only to determine the awarded teams and do not modify the official classification.

### F.13.4 Overall Category Awards

F.13.4.1 The **Best MotoStudent Award** is granted to the team from each category that achieves the highest overall score, obtained by summing the following scores and applying any applicable penalties:

- **Best MS1 Award** score.
- **Best MS2 Award** score.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.13.4.2      | <p>The <b>Best Rookie Award</b> is granted to the rookie team (representing a campus not previously registered in that MotoStudent category) that achieves the highest overall score, calculated as the sum of the following scores minus any applicable penalties:</p> <ul style="list-style-type: none"> <li>- <b>Best MS1 Award</b> score.</li> <li>- <b>Best MS2 Award</b> score.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| F.13.4.3      | <p>In case of a tie for the <b>Best MotoStudent</b> or <b>Best Rookie awards</b>, the deciding criterion will be:</p> <ul style="list-style-type: none"> <li>- <b>Best MS1 Award</b> score</li> </ul> <p>If the tie persists, the jury will determine the awarded team. Tie-breakers apply only to determine the awarded team and do not modify the official classification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>F.13.5</b> | <b>Special Nominations</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F.13.5.1      | <p>The <b>MotoStudent Fair Play nomination</b> highlights the team that best represents the values of respect, integrity, teamwork, and professional conduct throughout all phases of the competition.</p> <p>This recognition has a special character, as it will be voted on by the participating teams themselves, who will be able to select the team they believe has demonstrated exemplary behaviour both on and off the track.</p> <p>The winning team will be the one that receives the highest number of nominations through a quiz available on the MotoStudent platform, which will be open from the first to the last day of the Final Event. In this way, the teams themselves will recognize those participants who best represent the spirit of the competition.</p> <p>This recognition does not affect the scoring or the official ranking of the competition. In the event of a tie, the Organization will apply the tie-breaking criteria it considers appropriate.</p> |
| F.13.5.2      | <p>The <b>MotoStudent Media Hero nomination</b> aims to recognize the team that stands out the most in communicating, promoting and increasing the visibility of its project throughout the Competition.</p> <p>As MotoStudent teams are multidisciplinary, this nomination also seeks to highlight the work carried out in areas such as communication, marketing, and project management visibility, which play a key role in enhancing the team's outreach and visibility.</p> <p>This is a recognition-based nomination, and therefore it does not impact the official scoring or final ranking of the Competition. It shall be awarded to the team that stands</p>                                                                                                                                                                                                                                                                                                                     |

out the most in terms of visibility and communication, based on an overall evaluation carried out by a designated jury.

The evaluation is based on the team's overall performance throughout the entire competition. In this regard, the following aspects will be considered:

- Consistency and level of activity across communication channels, digital platforms, and social media (especially Instagram).
- The quality, creativity, and relevance of the content produced.
- The ability to communicate clearly and effectively both the project's development and the team's identity.
- The coherence, planning, and evolution of the communication strategy throughout the competition.
- Proactivity in promoting the project through the organization or participation in fairs, events, collaborations, promotional actions, or activities within the university environment.
- The development of initiatives aimed at increasing the visibility of the team and its project, regardless of their scale, format, or reach.
- Participation in initiatives promoted by the MotoStudent organization, such as the Poster Contest, campaigns, prototype presentations or similar activities.

Teams are encouraged to create and publish audiovisual content throughout the Competition that reflects the evolution of their project, such as team presentation videos, prototype development updates, innovation-related content or other materials of similar nature.

Although such audiovisual content is not a mandatory deliverable, its development and publication may be positively considered in the evaluation of this nomination, as it constitutes one of the most effective ways to communicate the evolution, identity and visibility of the project over time.

The absence of audiovisual content does not automatically exclude a team from this nomination, but it may significantly reduce the team's ability to demonstrate sustained communication effort and project visibility throughout the Competition.

Furthermore, this content may serve as a foundation for the preparation of the MS1 6<sup>th</sup> Milestone, Pitch Presentation, helping teams deliver a more complete and structured presentation of their project.

Through this nomination, MotoStudent aims to encourage teams to actively share their work, connect with the public, and strengthen the visibility of their project beyond the technical field.

#### **F.13.6 General Provisions**

- |          |                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| F.13.6.1 | The economic or material value of each award will be communicated by the Organization to all registered teams.                          |
| F.13.6.2 | Awards cannot be shared or transferred to another team.                                                                                 |
| F.13.6.3 | The Organization reserves the right to introduce additional awards during the development of the MotoStudent International Competition. |
| F.13.6.4 | If a test is declared null or cancelled, the Organization reserves the right to define a new scoring distribution for that test.        |

#### ARTICLE 14: DELIVERABLE SUBMISSIONS AND PENALTIES

##### **F.14.1      General submission requirements**

- |          |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.14.1.1 | All deliverables must be submitted in digital format via the MotoStudent Platform, except for the Final Event Presentation.                                                                                                                                                                                                                                                                                                                |
| F.14.1.2 | <p>The main document of each deliverable must be submitted in PDF format.</p> <p>All relevant information required for the evaluation must be included in the main PDF document. Supporting files are considered complementary and may not be reviewed by the jury.</p>                                                                                                                                                                    |
| F.14.1.3 | Additional supporting files (e.g. CAD models, datasets or technical drawings) may be submitted only when they are relevant to the deliverable and must be provided in commonly accepted engineering formats.                                                                                                                                                                                                                               |
| F.14.1.4 | <p>All submitted files must be properly optimized and compressed to ensure efficient upload, storage and review. Teams are responsible for minimizing file sizes without compromising the clarity and quality of the information.</p> <p>The Organization reserves the right not to accept or not to consider files that are excessively large, not properly optimized, or not clearly required for the evaluation of the deliverable.</p> |
| F.14.1.5 | Each deliverable must be prepared and submitted by registered Team Members. If a deliverable is submitted by a person not registered as an official Team Member, the delivery score will result in NC.                                                                                                                                                                                                                                     |
| F.14.1.6 | Deliverables must be submitted in one of the official competition languages. Otherwise, the delivery score will result in NC.                                                                                                                                                                                                                                                                                                              |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F.14.1.7      | Once a deliverable has been submitted, it cannot be modified, replaced or resubmitted unless expressly authorized by the Organization.                                                                                                                                                                                                                                                                                                                                                                                                          |
| F.14.1.8      | Teams are responsible for ensuring that the submission is correctly completed. Technical issues, upload errors or incorrect submissions will not justify resubmission unless expressly accepted by the Organization.                                                                                                                                                                                                                                                                                                                            |
| <b>F.14.2</b> | <b>Deliverable format requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| F.14.2.1      | Each deliverable must include a cover and an index page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| F.14.2.2      | All pages must be numbered, except the cover and index pages, which may not display the page number but shall count towards the total page limit.                                                                                                                                                                                                                                                                                                                                                                                               |
| F.14.2.3      | The maximum length of each deliverable is 50 pages, including cover, index, figures, tables and references.<br><br>Any content exceeding the page limit will not be considered during evaluation.                                                                                                                                                                                                                                                                                                                                               |
| F.14.2.4      | Only DIN A4 page format is allowed, except for tables and graphs, which may be presented in DIN A3 format. Each DIN A3 page may only contain a single table or a single graph and its corresponding title.                                                                                                                                                                                                                                                                                                                                      |
| F.14.2.5      | The format must comply with the following: <ul style="list-style-type: none"> <li>- Margins: <ul style="list-style-type: none"> <li>- Left: 3 cm</li> <li>- Right: 3 cm</li> <li>- Top: 2.5 cm</li> <li>- Bottom: 2.5 cm</li> </ul> </li> <li>- Line spacing: 1.5</li> <li>- Paragraph spacing: <ul style="list-style-type: none"> <li>- Before: 0 pt</li> <li>- After: 6 pt</li> </ul> </li> <li>- Font: <ul style="list-style-type: none"> <li>- Type: Arial</li> <li>- Size: 11 (except titles, subtitles and tables)</li> </ul> </li> </ul> |
| F.14.2.6      | No annexes or additional documentation may be included unless explicitly required by these Rules & Regulations.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>F.14.3</b> | <b>Penalties applicable to MS1 deliverables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| F.14.3.1      | Failure to comply with any applicable technical regulation defined in Sections B, C or D will result in a NC score.                                                                                                                                                                                                                                                                                                                                                                                                                             |

- F.14.3.2 | If a deliverable does not include a cover or index page, or does not comply with the required format, the delivery score will be penalized with a MP.
- F.14.3.3 | Submission of a deliverable within 10 days after the deadline (240 hours) will result in a MP in that delivery score.
- F.14.3.4 | Deliverables submitted more than 10 days after the deadline (240 hours) may not be accepted by the Organization and will result in a NP.
- F.14.3.5 | The minimum possible score of a deliverable is 0 points.
- F.14.4      MotoStudent Electric deliverables specific penalties**
- F.14.4.1 | Submission of any documentation within 10 days after its deadline (240 hours later) will result in a Minor Penalty (MP) applied to the total MS1 score.
- F.14.4.2 | Failure to submit or submission later than 10 days after the deadline (240 hours later) will result in a Severe Penalty (SP) applied to the total MS1 score. In this case, the team must inform the Organization via ticket in order to review the delivery.
- F.14.4.3 | The Organization reserves the right not to accept any delivery submitted more than 10 days after the deadline (240 hours later), resulting in a Not Presented (NP).
- F.14.4.4 | In case of a NP score in any of the MSE deliveries before the Final Event, the team shall contact the Organization. The Organization reserves the right not to scrutineer any bike that has a NP score in any MSE delivery.



# SECTION G

## SPORTING REGULATIONS FOR MS2 PHASE

## ARTICLE 1: GOALS AND CONDITIONS

### G.1.1 MS2 definition

- G.1.1.1 | The MS2 Phase consists of a series of dynamic tests aimed at evaluating the performance and behaviour of the manufactured prototype.

### G.1.2 Requirements

- G.1.2.1 | In order to participate in the MS2 Phase, teams must comply with the requirements defined in this section.
- G.1.2.2 | Only prototypes that have successfully passed the Scrutineering process (Section E) are allowed to participate in the MS2 Phase. Prototypes must be presented in safe and suitable conditions to take part in the dynamic tests. The Organization Technical Staff may inspect the condition of the prototype at any time during the Final Event.

### G.1.3 MS2 Sports Committee

- G.1.3.1 | The MS2 Phase will be supervised and managed by the MS2 Sports Committee, which will include at least the following officials:
- Competition General Manager
  - Competition Sport Director
  - Race Director
  - Race Director Assistant
  - Sporting Jury (Officials' coordinators)
  - Timekeeping Director
  - Competition Technical Director
  - Chief Scrutineer

## ARTICLE 2: RIDER

### G.2.1 Rider Definition

- G.2.1.1 | Each team must nominate one rider responsible for riding the prototype during the MS2 Phase tests. As MotoStudent aims to evaluate the engineering work and the prototype performance rather than the rider ability, specific eligibility restrictions are applied to ensure fair conditions among teams.

### G.2.2 Rider Eligibility

- G.2.2.1 | Once approved by the Competition Organization, the rider shall be officially registered with the team exclusively for the purpose of riding the prototype during

the MS2 Phase and the Final Event. The rider shall be registered through the submission of the ***Rider Eligibility Declaration***. Riders are not required to be registered as Team Members through the MotoStudent Platform.

The rider may access the areas and activities strictly necessary to perform the riding duties during the Final Event and must comply with all rules, safety instructions and obligations applicable to participants.

The rider is not considered a Team Member and therefore shall not take part in the design, development, modification, repair or technical work on the prototype, unless the rider is simultaneously registered as a Team Member in accordance with the applicable requirements.

Riders are also exempt from the academic eligibility requirements applicable to Team Members.

G.2.2.2 | The rider is not required to pay any entry fee to participate in the Competition.

G.2.2.3 | The rider must be at least 18 years old on the first day of the Final Event.

G.2.2.4 | The participation of riders who have competed within the last 10 years prior to the Final Event in Circuit Racing events organized under the authority of FIM, FIM Continental Unions (Europe, Asia, North America, Latin America, Africa, Oceania), or in the highest-level National Circuit Racing Championship officially organized or recognized by the corresponding national motorcycle federation is not allowed.

This restriction also applies to riders who have participated in such competitions as Wild Card or replacement riders in one or more events.

For the purpose of this regulation, the highest-level National Circuit Racing Championships includes the highest-level circuit racing championships organized and designated by national motorcycle federations. In case of doubt regarding the eligibility of a rider or the status of a specific championship, the Organization reserves the right to determine whether the competition is considered a high-level National Championship for the purposes of this regulation.

Participation in previous editions of MotoStudent is exempt from this restriction.

G.2.2.5 | A student may hold a federation license and participate as a rider. However, if the rider is not registered as an official Team Member, he/she will only be allowed to ride the prototype.

G.2.2.6 | Each rider must be registered with a single team and may only participate with that team in the corresponding category. A rider cannot participate with multiple

teams, even if they belong to the same university or compete in different categories.

G.2.2.7 | The rider must attend all mandatory briefing sessions organized prior to the MS2 Phase tests and any additional meetings requested by the Organization.

G.2.2.8 | If the rider withdraws during the Final Event due to medical reasons or force majeure, no replacement rider will be allowed.

### G.2.3 Rider Federative requirements

G.2.3.1 | The rider must hold a valid federative license allowing participation in an international FIM competition such as MotoStudent.

Track day insurance policies or federation licenses that do not cover participation in FIM international competitions will not be accepted.

G.2.3.2 | Riders participating in MotoStudent must hold one of the following licenses \*:

- RFME:
  - RFME homologated license
  - Or
  - RFME One Event homologated license
- FIM
  - FIM International Racing Meetings Annual license
  - Or
  - FIM International Racing Meetings - One Event license
  - Or
  - FIM Superbike/Supersport World Championship license
  - Or
  - FIM World Endurance Championship license
- Others:
  - **\*\*For FIM Europe Countries only\*\***: National license + Start Permission.

\*\*\* These federative requirements may vary depending on the conditions imposed by the motorcycle federations involved (FIM, RFME and FARAM) for the corresponding competition season.

G.2.3.3 | Teams may process temporary licenses for their riders through:

- Aragonese Motorcycling Federation ([www.faram.es](http://www.faram.es)).
- Royal Spanish Motorcycling Federation ([www.rfme.com](http://www.rfme.com)).

### G.2.4 Communication with the rider

- G.2.4.1 | Radio communication or any other electronic communication between the team and the rider during the MS2 Phase tests is strictly forbidden.
- G.2.4.2 | Communication between the team and the rider is only allowed through a pit board displayed from the Pit Wall during Free Practice, Warm Up, Qualifying and Race sessions.
- G.2.4.3 | The pit board must be lightweight and must not exceed 1 m in width and 1.5 m in height.
- G.2.4.4 | The pit board must be handled by a single Team Member, who must hold it securely to avoid any risk of falling onto the track.
- G.2.4.5 | The pit board may only be shown when the rider passes through the Finish Straight and must immediately be removed afterwards. The board must never be placed on the track surface.

## **G.2.5 Rider replacement**

- G.2.5.1 | Any rider replacement must be authorized by the Organization. Typical reasons for a replacement may include:
  - Injury of the main rider
  - Unavailability of the main rider due to schedule changes affecting the Final Event
- G.2.5.2 | Rider replacements requested up to one week before the beginning of the Final Event (previous Wednesday not included) will be subject to an administrative fee.
- G.2.5.3 | Rider replacements requested during the week prior to the Final Event (from the previous Wednesday until the administrative checks) will be subject to a **double administrative fee**.
- G.2.5.4 | Once the rider administrative checks have been completed during the Final Event, rider replacement will not be permitted.

## **G.2.6 Rider Medical Fitness**

- G.2.6.1 | After any crash, incident or situation that may affect the physical condition of the rider, the Organization may require a medical evaluation before allowing the rider to continue participating in the session or in the Event.
- G.2.6.2 | The Official Medical Service has the authority to declare a rider FIT or UNFIT to continue participating in the Event.

Race Direction may prevent a rider from continuing to participate if the rider is considered a safety risk for himself/herself or for other participants.

G.2.6.3 | The decision of the Official Medical Service regarding the medical fitness of a rider is final and cannot be appealed.

### ARTICLE 3: SPORT BEHAVIOUR

#### G.3.1 Weather conditions

G.3.1.1 | In case of rain, the Organization may declare two different rain conditions:

- Moderate rain
- Intense rain

G.3.1.2 | If moderate rain is declared, both MotoStudent eFuel and MotoStudent Electric teams may continue the MS2 Phase tests. The use of wet weather tires may be advisable.

G.3.1.3 | If intense rain is declared, all MS2 Phase tests will be suspended and any prototype outside its Box must return immediately.

G.3.1.4 | In case of intense rain, the Organization will inform the teams whether the affected test is cancelled or postponed.

G.3.1.5 | For any other weather conditions or unforeseen circumstances, the MS2 Sports Committee will decide whether the tests are postponed, modified or cancelled.

G.3.1.6 | The Organization reserves the right to modify the schedule or evaluation procedures due to weather conditions or other unforeseen events.

#### G.3.2 Access to the track

G.3.2.1 | Prototypes must enter the track only through the designated access points established by the Organization for each test.

#### G.3.3 Behaviour on the track:

G.3.3.1 | Riding in the opposite direction of the track is strictly forbidden.

G.3.3.2 | In the event of an accident or mechanical failure, the rider must immediately move the prototype off-track and off the racing line to a safe area where it does not endanger other riders.

G.3.3.3 | No person may approach a stopped prototype on the track except the rider, Race Direction officials, marshals or authorized Technical Staff.

G.3.3.4 | In the event of a crash, the rider must move to a safe area as soon as possible. Track marshals will assess the situation and determine whether the rider may safely continue the session. If the prototype is considered unsafe or has suffered significant damage, the marshals may require the prototype to be removed from the track and inspected by the Technical Scrutineers before it is allowed to rejoin any session.

#### **G.3.4 Behaviour at the Pit Lane**

G.3.4.1 | The maximum speed limit in the Pit Lane is 60 km/h.

G.3.4.2 | Only properly accredited personnel are allowed in the Pit Lane. The Organization may remove any person whose presence is not justified.

G.3.4.3 | If oil or any other liquid is spilled in the Pit Lane working area, the prototype may not start until both the prototype and the affected area have been completely cleaned.

G.3.4.4 | For the MotoStudent eFuel category, if the engine automatic starting system does not work, up to two people may assist the rider by pushing the prototype to the Pit Lane exit. External batteries or auxiliary starting devices are not allowed.

G.3.4.5 | If a rider experiences a problem before reaching the Pit Lane exit, the prototype may return to the Box with the engine / motor switched off through the work zone and accompanied by a Pit Lane marshal.

G.3.4.6 | During pit stops only registered members can manipulate the prototype in pit lane. Failure to comply with this rule may result in the exclusion of the team.

G.3.4.7 | During pit stops, only registered Team Members may manipulate the prototype. Failure to comply with this rule may result in the exclusion of the team from the session. A maximum of five (5) people may be in contact with or manipulating the prototype, including the rider. Any physical contact with the prototype will be considered as manipulation for the purpose of this limit, even if the rider remains on the motorcycle without actively intervening.

G.3.4.8 | If work is carried out inside the Box, the number of Team Members working on the prototype is not limited (the rider is not considered as Team Member for this effect). **During race sessions, introducing the prototype into the Box may result in exclusion from that session** (see Articles G.6.2.1(17) and G.6.2.2(17)).

- G.3.4.9 | Oil or liquid changes and cleaning operations must be performed inside the Box using appropriate containers. Teams must ensure the area remains clean after these operations.
- G.3.4.10 | Stopping the prototype in the fast lane of the Pit Lane (between the red line and the Pit Wall) is strictly forbidden.
- G.3.4.11 | Smoking, eating, drinking, or using bicycles, electric scooters or any other personal vehicles is forbidden in the Pit Lane.
- G.3.4.12 | The use of chairs or tables in the Pit Lane is not allowed.

### G.3.5 Flag Signals

- G.3.5.1 | The following flag signals will be used to alert riders during the MS2 Phase tests:
- Red flag: waved at all marshal posts to indicate that the session is stopped.
  - Black flag: shown together with the rider's number. The rider must return to the pits at the end of the current lap.
  - Black flag with orange disc: shown together with the rider's number to indicate a mechanical problem that may endanger the rider or others. The rider must leave the track **immediately**.
  - Yellow flag: indicates danger on or near the track. Riders must slow down and overtaking is forbidden.
    - One (1) yellow flag: hazard beside the track.
    - Two (2) yellow flags: hazard on the track.
  - White flag with red diagonal cross: rain affecting that section of the track.
  - Yellow and red striped flag: reduced adhesion on the track surface due to substances other than rain.
  - Green flag: track is clear. End of hazard zone and end of overtaking prohibition.
  - Blue flag: shown to indicate that a faster rider is approaching and the slower rider must allow safe overtaking.
  - Chequered black/white flag: indicates the end of a practice session or race.
  - Chequered black/white flag together with blue flag: used when a rider closely follows the leader on the final lap (see Article G.6.4.1).
- G.3.5.2 | Any failure to respect flag signals may be penalized.

### G.3.6 Regulations priority order

- G.3.6.1 | Any situation not covered by the MotoStudent Rules & Regulations will be governed by the current RFME ESBK Sporting Regulations. If the situation is not covered there, the current RFME General Sporting Regulations for Speed Competitions will apply.



The current RFME ESBK Sporting Regulations can be consulted at:

<https://rfme.com/campeonatos/campeonato-de-espana-de-superbike/>

The RFME General Sporting Regulations can be consulted at:

<https://rfme.com/rfme/estatutos-y-codigos-rfme/>

#### ARTICLE 4: PRACTICE SESSIONS (INCLUDING WARM-UP)

##### G.4.1 Practice session track

- G.4.1.1 Practice sessions will take place on the MotorLand Aragón FIM Grand Prix Circuit layout, with a total length of 5,077.65 m. The circuit layout is shown in Annex 2 of Section J of these Rules & Regulations.

##### G.4.2 Development of the sessions

- G.4.2.1 Riders may enter the track from the pit lane when the green light is displayed at the pit exit.

- G.4.2.2 The practice session time will start when the pit exit green light is switched on. A visible board or countdown display in the pit lane will indicate the remaining session time.

- G.4.2.3 The end of the session will be indicated by the chequered flag waved at the finish line, after which the pit exit will be closed. Lap times will continue to be recorded until each rider crosses the finish line after the allotted session time has elapsed. Riders may then complete the current lap and return to the pit lane.

- G.4.2.4 If a practice session is interrupted due to an incident or any other reason, a red flag will be shown at the start line and at all marshal posts. All riders must slow down and return safely to the pit lane.

When the session is restarted, the remaining time will be the time displayed on the official timekeeping monitors and the pit lane countdown at the moment the red flag was shown.

- G.4.2.5 If the number of authorized prototypes exceeds the maximum number allowed on track simultaneously, the Organization may divide the session into two groups.

##### G.4.3 Number and duration of Qualifying/Practices sessions for each Category

- G.4.3.1 Two **Free Practice** session of **20 minutes** for each Category. (TBC)

G.4.3.2 | One **Qualifying** session of **20 minutes** for each Category. (TBC)

G.4.3.3 | One **Warm-Up** session of **15 minutes** for each Category. (TBC)

#### **G.4.4 Lap Time**

G.4.4.1 | All laps will be timed.

G.4.4.2 | Lap time is measured as the time elapsed between two consecutive crossings of the finish line timing plane.

#### **G.4.5 Qualifying session results**

G.4.5.1 | The qualifying classification will be determined by the fastest lap time achieved by each rider during the Qualifying session.

G.4.5.2 | If the Qualifying session is cancelled, the classification will be determined using the fastest lap time recorded during the Free Practice sessions.

G.4.5.3 | In the event of a tie, the riders' second-best lap times, and if necessary subsequent lap times, will be taken into account.

G.4.5.4 | The Qualifying results will become final 30 minutes after publication of the provisional results. After this time, the results will not be subject to protest or appeal and will not be modified based on any infringements reported after this deadline.

The Play-Off Race grid will be established according to these results.

#### **G.4.6 Qualification for the Final Race**

G.4.6.1 | Direct qualification:

- Based on the Qualifying session results, the fastest half of the riders authorized to participate in the Qualifying session (rounded up to the nearest whole number) will qualify directly for the Final Race.
- Riders included in the authorized list who have not recorded a lap time during the Qualifying session will also be considered in this calculation.

G.4.6.2 | Promotion to the Final Race from the Play-Off Race:

- All riders authorized to participate in the Qualifying session who do not qualify directly for the Final Race will participate in the Play-Off Race.
- The 1<sup>st</sup>, 2<sup>nd</sup> and 3<sup>rd</sup> classified riders in the Play-Off Race will also qualify for the Final Race.

- If a team withdraws between the publication of the Qualifying results and the publication of the Final Race grid, the next classified rider from the Play-Off Race will be promoted to the Final Race.

G.4.6.3 | If the Play-Off Race is cancelled, qualification for the Final Race will require a lap time equal to or faster than 112% of the fastest lap recorded during the Qualifying session.

G.4.6.4 | The Organization reserves the right to exclude any prototype considered to represent a safety risk for the race, based on factors such as lap times, track behaviour or top speed.

#### **G.4.7 MS2 tests carried out during sessions**

G.4.7.1 | Test 4 (Top Speed) will take place during the second Free Practice session.

G.4.7.2 | Test 6 (Pole Position) will take place during the Qualifying session.

G.4.7.3 | Test 5 (Regularity) will take place during the Warm-Up session.

G.4.7.4 | Test 7 (Best Race Lap) and Test 8 (Race Result) will take place during the race sessions.

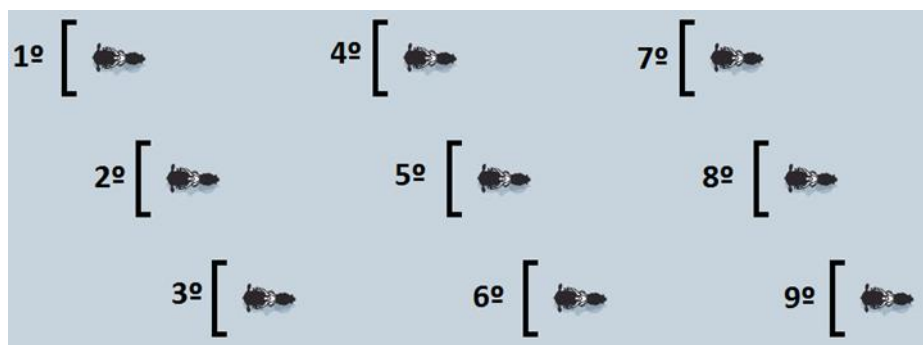
### **ARTICLE 5: GRID POSITIONS**

#### **G.5.1 Grid configuration**

G.5.1.1 | The starting grid will be arranged in a 3–3–3 configuration in staggered formation.

G.5.1.2 | Each row will be offset.

G.5.1.3 | The distance between rows will be 9 metres.



*Picture 48 - Starting grid positions*

## **G.5.2 Grid Allocation**

### **G.5.2.1 Play-Off Race**

Grid positions will be determined by the fastest lap time recorded during the Qualifying session.

Teams not directly qualified for the Final Race will be placed on the grid in order of their Qualifying lap times, starting with the fastest.

### **G.5.2.2 Final Race**

Grid positions will be determined based on the Qualifying session results and the results of the Play-Off Race.

Riders directly qualified for the Final Race will occupy the first grid positions according to their Qualifying times.

The riders promoted from the Play-Off Race will then be placed on the grid according to their finishing order in the Play-Off Race.

**G.5.2.3** | If a team is unable to participate in the Qualifying session due to a technical problem or crash, Race Direction may authorize the team to participate in the Play-Off Race upon request, provided that the team has completed at least one full lap during the Free Practice sessions. In such case, the team will start from the last grid position.

If multiple teams are in this situation, their order on the grid will be determined according to their best lap time achieved during their last Free Practice sessions.

## **G.5.3 MS1 Handicaps**

The MS1 Phase results may influence the starting grid allocation through the application of time handicaps.

**G.5.3.1** | If a team has achieved less than 250 points in the MS1 Phase, a time handicap will be applied to its grid allocation time (Play-Off Race or Final Race grid allocation), but not to the promotion to the Final Race.

**G.5.3.2** | If a team is not directly qualified for the Final Race, the handicap will only be applied for the Play-Off Race grid allocation. If the rider subsequently qualifies for the Final Race, the handicap will not be applied again.

**G.5.3.3** | The handicap time applied according to the MS1 score is:

| Scoring intervals     | Handicap time |
|-----------------------|---------------|
| 250 > MS1 Score ≥ 200 | 3"            |
| 200 > MS1 Score ≥ 150 | 7"            |
| 150 > MS1 Score ≥ 100 | 15"           |
| 100 > MS1 Score       | 30"           |

Table 11 - Handicap time

## ARTICLE 6: RACES

### G.6.1 Number of races

- G.6.1.1 During the Final Event each Category will run two (2) race sessions:
- Play-Off Race: Qualification session for the Final Race. The first three classified riders will qualify for the Final Race.  
Race distance:  
Electric Category: **6 laps**  
eFuel Category: **6 laps**
  - Final Race: The final positions for the MS2 tests developed during the races will be determined in this race.  
Race distance:  
Electric Category: **7 laps**  
eFuel Category: **9 laps**
- G.6.1.2 The Play-Off Race may be cancelled if the MS2 Sports Committee considers it necessary.
- G.6.1.3 Electric and eFuel categories will run independent races.

### G.6.2 Start Procedures

- G.6.2.1 Standard Start Procedure - eFuel Category
1. Approximately 40 minutes before the Start of the Race (except in the case of a restarted or rescheduled race), Race Direction will inform the teams via the official timing system if the race is declared "wet". The race may also be declared "dry", although this is not mandatory. If no information is displayed, the race will automatically be considered "dry".
  2. Only riders who have completed at least one sighting lap will be permitted to start the race from their published grid position. Under no circumstances may a rider push the motorcycle onto the grid from the pit lane.
  3. Approximately 20 minutes before the Start of the Race (except in the case of a restarted or rescheduled race):
    - Pit lane exit opens for sighting laps.
    - Green lights are displayed at the pit lane exit.

- Count-down boards are shown at the pit exit.

Riders may complete more than one sighting lap by passing through the pit lane, where adjustments may be made, tyres may be changed or refuelling may be carried out.

4. Approximately 15 minutes before the Start of the Race.

-Pit lane exit closes.

-Red lights are displayed at the pit lane exit.

5. Riders who do not proceed to the grid may start the warm-up lap from the pit lane under the instructions of the marshal positioned at the pit lane exit. Riders starting the warm-up lap from the pit lane must start the race from the back of the grid.

6. When riders arrive on the grid after the sighting lap(s), they must stop at the rear of the grid and switch off the engine. The motorcycle will then be pushed at walking pace by a Team Member to the assigned grid position. The rider may remain seated on the motorcycle or dismount while it is pushed.

All persons present on the grid must wear a valid grid access pass.

Officials will display panels indicating the grid rows to assist riders in locating their grid positions.

7. At this stage the Race Director will declare the race as “wet” or “dry” by displaying a board on the grid and in the pit lane. If no board is displayed the race will automatically be considered “dry”.

8. Riders on the grid may make adjustments to the motorcycle or change tyres according to track conditions.

Tyre warmers may be used on the grid.

One generator per prototype may be used exclusively to power tyre warmers. The generator must be a compact portable unit with a maximum output of 2 kW. **Generators must be installed on a trolley with wheels and equipped with a closed tray capable of containing any potential fluid leakage.** Any generator considered unsafe by the Technical Scrutineers will be removed immediately from the grid. The decision of the Technical Scrutineers is final.

Generator trolleys must be positioned behind the prototypes.

The size and type of grid equipment must allow rapid clearing of the grid and are subject to approval by the Organization.

All adjustments must be completed before the display of the 3 Minutes board.

After the display of the 3 Minutes board, any rider wishing to continue working on the motorcycle must push the motorcycle to the pit lane. The rider and motorcycle must be clear of the grid before the 1 Minute board is displayed. Work may then continue in the pit lane.

Such riders will start the warm-up lap from the pit lane and will start the race from the back of the grid.

9. Refuelling or changing the fuel tank on the grid is strictly forbidden.

10. 5 Minutes before the Start of the Warm-Up Lap – the 5 Minutes board is displayed on the grid.
11. 3 Minutes before the Start of the Warm-Up Lap – the 3 Minutes board is displayed.  
Generators must be disconnected and all trolleys removed from the grid as quickly as possible.  
At this point only, the rider and three Team Members per team may remain on the grid. All other team personnel must leave the grid immediately.  
Riders must put their helmets on.  
No additional persons may access the grid after this point.
12. 1 Minute before the Start of the Warm-Up Lap – the 1 Minute board is displayed.  
Tyre warmers must be removed immediately.  
Engines must be started.  
Only the rider may remain on the grid. All Team Members must immediately leave the grid.
13. 30 Seconds before the Start of the Warm-Up Lap – the 30 Seconds board is displayed.  
Any rider whose engine is not running will be removed to the pit lane under the control of the grid marshals. Such riders may start the warm-up lap from the pit lane and will start the race from the back of the grid.
14. Approximately 5 minutes before the Start of the Race – the green flag is waved to start the warm-up lap.  
Riders will complete one lap at unrestricted speed followed by a Safety Car. The Safety Car may overtake slow riders.  
Any rider who stalls the engine on the grid or encounters a problem must signal by raising an arm. Under the supervision of an official the rider and motorcycle will be moved to the pit lane where mechanics may provide assistance.  
Such riders may start the warm-up lap from the pit lane and will start the race from the back of the grid, provided they exit before the pit lane closes and reach the grid before the Safety Car.  
If they do not exit before the pit lane closes they will start the race from the pit lane.  
When the riders who started from the grid pass the pit lane exit, the pit exit light will turn green to allow riders waiting in the pit lane to join the warm-up lap. Thirty seconds later the light will turn red closing the pit exit.  
If two or more riders must start from the back of the grid, they will line up in the order of their original grid positions.  
An official will stand at the front of the grid holding a stationary red flag.  
Any rider arriving at pit lane entry after the Safety Car must enter the pit lane and start the race from the pit lane exit.  
The reference lap time of the Safety Car will be 4'30".

Any rider encountering a problem during the warm-up lap may return to the pit lane for repairs.

As each row of the grid is completed, the officials will lower the panels indicating that the row is correctly formed. When all panels have been lowered and the Safety Car has taken its position, an official at the rear of the grid will wave a green flag.

The Starter will then instruct the official at the front of the grid, holding the red flag, to move to the side of the track.

15. Race Start.

A red light will be displayed for between 2 and 5 seconds. When the red light goes out the race will start.

A Safety Car will follow the motorcycles during the entire first lap and may overtake slow riders.

Any rider who anticipates the start will be penalized by Race Direction. The standard penalty is two Long Lap Penalties.

In case of minor movement Race Direction will be the sole judge of whether an advantage has been gained.

16. If a rider stalls the motorcycle after the start, **officials** will assist in pushing the motorcycle into the pit lane where the team may provide assistance.

17. After the riders pass the pit lane exit point, the green light will be switched on to allow riders still in the pit lane to start the race.

A pit lane start will be made from a stationary position as directed by the officials. However, a rider who is delayed and arrives after the pit lane exit is open (and before it gets closed) will not be required to stop at pit exit before joining the race.

If more than six riders must start the race from the pit lane, the start will be delayed and a Quick Start Procedure will be initiated.

A rider starting from the pit lane may join the race until the race leader crosses the finish line to complete the first lap.

18. If a motorcycle that has been active in the race enters the pit box, it is considered retired and may not rejoin the race.

#### G.6.2.2 Standard Start Procedure - Electric Category

1. Approximately 40 minutes before the Start of the Race (except in the case of a restarted or rescheduled race), Race Direction will inform the teams via the official timing system if the race is declared "wet". The race may also be declared "dry", although this is not mandatory. If no information is displayed, the race will automatically be considered "dry".

2. Approximately 25 minutes before the Start of the Race.

Teams must be ready in the "Slow Lane" of the Pit Lane in order to push their prototypes onto the grid through the pit wall access door.

3. Approximately 20 minutes before the Start of the Race (except in the case of a restarted or rescheduled race).



- The pit wall access to the grid will open for Teams and prototypes.
4. Approximately 15 minutes before the Start of the Race.  
Pit wall access to the grid will close.
  5. Riders who do not go onto the grid may start the warm-up lap from the pit lane under the instructions of the marshal positioned at the pit lane exit.  
Riders starting the warm-up lap from the pit lane must start the race from the back of the grid.
  6. When riders reach the grid through the pit wall, one mechanic must push the prototype at walking pace to the assigned grid position.  
The use of the electric motor to reach the grid is not permitted. All persons present on the grid must wear a valid grid access pass.  
Officials will display panels indicating the grid rows to assist riders in locating their grid positions.
  7. At this stage the Race Director will declare the race as “wet” or “dry” by displaying a board on the grid and in the pit lane. If no board is displayed the race will automatically be considered “dry”.
  8. Riders on the grid may make adjustments to the prototype or change tyres according to track conditions.  
Tyre warmers may be used on the grid.  
One generator per prototype may be used exclusively to power tyre warmers on the grid. **The generator must not be powered by any kind of fuel** and must consist of a sealed Pb battery (gel / AGM / sealed battery) together with a commercial sinusoidal wave inverter with a maximum output of 2 kW.  
**All generators must be installed on a wheeled trolley and equipped with a closed tray capable of containing any possible fluid leakage.**  
Any generator considered unsafe by the Technical Scrutineers will be removed immediately from the grid. The decision of the Technical Scrutineers will be final.  
Generator trolleys must be positioned behind the prototypes.  
To ensure rapid clearing of the grid, the size and type of grid equipment (including generator trolleys) are subject to approval by the Organization.  
All adjustments must be completed before the display of the 3 Minutes board.  
After the display of the 3 Minutes board, any rider wishing to continue adjustments must push the prototype to the pit lane.  
The rider and prototype must be clear of the grid before the 1 Minute board is displayed. Work may then continue in the pit lane.  
Such riders will start the warm-up lap from the pit lane and will start the race from the back of the grid.
  9. Charging the prototype battery on the grid is strictly forbidden at all times.
  10. 5 Minutes before the Start of the Warm-Up Lap – the 5 Minutes board will be displayed on the grid.

11. 3 Minutes before the Start of the Warm-Up Lap – the 3 Minutes board will be displayed.  
Generators must be disconnected and all trolleys must be removed from the grid as quickly as possible.  
At this point only the rider and three Team Members per team may remain on the grid. All other team personnel must leave the grid immediately.  
Riders must put their helmets on.  
No additional persons may access the grid after this moment.
12. 1 Minute before the Start of the Warm-Up Lap – the 1 Minute board will be displayed.  
Tyre warmers must be removed immediately.  
Prototypes must be powered on and ready to race.  
Only the rider may remain on the grid. All Team Members must immediately vacate the grid.
13. 30 Seconds before the Start of the Warm-Up Lap – the 30 Seconds board will be displayed.  
Any rider whose prototype is not powered on will be removed to the pit lane under the control of the grid marshals. Such riders may start the warm-up lap from the pit lane and will start the race from the back of the grid.
14. Approximately 5 minutes before the Start of the Race – the green flag will be waved to start the warm-up lap.  
Any rider who stalls the powertrain on the grid or encounters another problem must signal by raising an arm. Attempting to restart the prototype on the grid is not permitted. Under the supervision of an official, the rider and prototype will be moved to the pit lane where mechanics may provide assistance.  
Such riders may start the warm-up lap from the pit lane and will start the race from the back of the grid, provided they exit before the pit lane is closed and reach the grid before the Safety Car.  
If they do not exit before the pit lane closes, they will start the race from the pit lane.  
Riders will complete one lap at unrestricted speed followed by a Safety Car. The Safety Car may overtake slow riders.  
When the riders who started from the grid pass the pit lane exit, the pit lane exit light will turn green to allow riders waiting in the pit lane to join the warm-up lap. Thirty seconds later the light will turn red closing the pit exit. If two or more riders must start from the back of the grid, they will line up in the order of their original grid positions.  
An official will stand at the front of the grid holding a stationary red flag.  
Any rider arriving at the pit lane entry point after the Safety Car must enter the pit lane and start the race from the pit lane exit.  
The reference lap time of the Safety Car will be 4'30".  
Any rider encountering a problem during the warm-up lap may return to the pit lane to make repairs.

As each row of the grid is completed, officials will lower the panels indicating that the row is correctly formed. When all panels have been lowered and the Safety Car has taken its position, an official at the rear of the grid will wave a green flag.

The Starter will then instruct the official at the front of the grid, holding the red flag, to move to the side of the track.

15. Race Start.

A red light will be displayed for between 2 and 5 seconds. When the red light goes out the race will start.

A Safety Car will follow the motorcycles for the entire first lap and may overtake slow riders.

Any rider who anticipates the start will be penalized by Race Direction. The standard penalty is two Long Lap Penalties.

In case of minor movement, Race Direction will be the sole judge of whether an advantage has been gained.

16. If a rider stalls the prototype after the start, **officials** will assist in pushing the motorcycle into the pit lane where the Team may provide assistance.

17. After the riders pass the pit lane exit point, the green light at the pit lane exit will be switched on to allow riders still in the pit lane to start the race. A pit lane race start will be made from a stationary position as directed by the officials.

However, a rider who is delayed and arrives after the pit lane exit is open (and before it gets closed) will not be required to stop at pit exit before joining the race.

If more than six riders must start the race from the pit lane, the start will be delayed and a Quick Start Procedure will take place.

Any rider starting the race from the pit lane may join the race until the race leader crosses the finish line to complete the first lap.

18. If a prototype that has been active in the race enters the pit box, the prototype will be considered retired and may not re-enter the race.

### G.6.2.3 Start Delayed

If any problem occurs during the start procedure that may represent a safety risk, the Starter will invoke the Start Delayed procedure as follows:

- A red flag will be waved from the Starter's rostrum and the red start light will remain on.
- The Starter will display the "Start Delayed" board and marshals will wave yellow flags at each row of the starting grid.
- Riders must remain in their grid positions with helmets on. Engines may be switched off.
- The prototype that caused the Start Delayed will be removed to the pit lane, regardless of whether the problem can be solved before the new start. If the problem is resolved, the rider may start the warm-up lap from the pit lane and will start the race from the back of the grid.

- After the Start Delayed board is displayed, only two Team Members per prototype are allowed to remain on the grid. Only tyre warmers and stands are permitted. Generators and trolleys are not allowed on the grid.
- As soon as possible the start procedure will be resumed from the 3 Minutes board.
- After the 30 Seconds board is displayed, riders will complete one additional warm-up lap and the race distance will automatically be reduced by one lap. If Race Direction considers it necessary, the race distance may be reduced by more than one lap.

#### G.6.2.4 Quick Start Procedure

When a race is stopped, riders must return to the pit lane unless otherwise instructed by officials. If the race is to be restarted, minor repairs may be carried out. The following procedure will apply:

- Upon arrival in the pit lane, Teams may make adjustments to their machine. Refuelling is permitted in pit boxes for the eFuel Category and Recharging is permitted in the pit lane for the Electric Category.
- When all riders have entered the pit lane, the Race Director will announce the time remaining until the re-opening of the pit lane.
  - a) The time between the red flag and the opening of the pit lane exit will be a minimum of 10 minutes.
  - b) The re-opening time will be published on the timing screens and may also be communicated to Teams through any other official communication channel. Count-down boards may be displayed at the pit lane exit.
- When this time period has elapsed:
  - a) (For eFuel Category) The pit lane will be opened for SIXTY SECONDS only. Riders will complete one lap at unrestricted speed to the starting grid, followed by a Safety Car. Any rider delaying the progress of the sighting lap may be overtaken by the Safety Car. Any rider arriving behind the Safety Car must enter the pit lane. Such riders must start the warm-up lap from the pit lane and will start the race from the back of the grid.
  - b) (For Electric Category) The pit wall access will be opened for TWO MINUTES only. Prototypes will be placed in their grid positions.
- Any riders remaining in the pit lane after the pit lane exit or pit wall access has been closed must start the warm-up lap from the pit lane exit and will start the race from the back of the grid.
- After the closure of the pit lane exit or pit wall access, tyre warmers must be removed from all machines remaining in the pit lane.
- Only one Team Member per prototype is allowed on the grid, without tools, solely to indicate the rider's grid position.
- Once the rider reaches their grid position, the mechanic must **immediately** leave the grid.

- For the eFuel Category, engines must remain running on the grid. For the Electric Category, prototypes must be powered on as soon as they reach their grid positions.
- As soon as the Safety Car arrives at the rear of the grid, a 30 Seconds board will be displayed. These 30 seconds may be reduced at the discretion of Race Direction if the grid is cleared and conditions are considered safe.
- After the 30 seconds have elapsed, a green flag will be shown to start the warm-up lap.
- The warm-up lap will be completed at unrestricted speed, followed by a Safety Car. At the end of the warm-up lap, the normal start procedure will apply and the race will start in the standard manner.

G.6.2.5 | If a race in the Electric Category is declared “wet” while all previous sessions have been run in “dry” conditions, the start procedure of the eFuel Category will apply and the race distance will be reduced by one lap.

G.6.2.6 | If a race in the eFuel Category is declared “wet” while all previous sessions have been run in “dry” conditions, the standard start procedure for that category will apply.

### G.6.3 Finish of a Race and Race Results

G.6.3.1 | When the leading rider has completed the designated race distance, he/she will be shown the Chequered Flag by an official positioned at the finish line behind the first protection line. The Chequered Flag will continue to be displayed to the subsequent riders.

Once the Chequered Flag has been shown to the leading rider, no other rider will be permitted to enter the track from the pit lane.

Immediately after the Chequered Flag is shown to the leading rider, the red light will be switched on at the pit lane exit and a marshal displaying a red flag will take position at the pit lane exit.

If a rider closely precedes the leader during the final lap before the finish line, the official will show the Chequered Flag together with the Blue Flag simultaneously to both the leader and the rider(s) concerned. This indicates that the race has finished for the leader, while the rider(s) concerned must complete their lap and then take the Chequered Flag.

G.6.3.2 | To be classified as a finisher in the race and be included in the official results, a rider must:

- a) Complete at least 75% of the race distance, expressed in laps and rounded up to the nearest whole number of laps.

- b) Cross the finish line on the race track (not in the pit lane) within five minutes of the race winner.

If the rider is not in contact with the prototype when crossing the finish line, the finishing time will be determined by the last part of either the rider or the prototype to cross the finish line.

- G.6.3.3 | The race results will be determined according to the order in which riders cross the finish line and the number of laps completed.

- G.6.3.4 | In the event of a photo-finish between two or more riders, the result will be decided in favour of the rider whose front wheel leading edge crosses the plane of the finish line first.

If the rider is not in contact with the prototype when crossing the finish line, the finishing time will be determined by the last part of either the rider or the prototype to cross the finish line.

- G.6.3.5 | At the end of the race, all prototypes that have taken the start must proceed to Parc Fermé under the instructions of the officials. This requirement applies whether the prototype has finished the race, retired, entered the pit box, or been removed from the track due to an incident. No work or intervention may be carried out on the prototype after the race unless authorized by the Technical Scrutineers or Race Direction.

Failure to comply with this requirement may result in disqualification from the race.

- G.6.3.6 | For the Electric Category only, the victory lap and return to Parc Fermé will be carried out using the “National Track” layout (2,379.12 m).

#### **G.6.4 Interruption of a Race**

- G.6.4.1 | If the Race Director decides to interrupt a race at any point from the start of the warm-up lap onwards, red flags will be displayed at the finish line and at all marshals’ posts, and red lights around the circuit will be switched on.

- G.6.4.2 | The race results will be established taking into consideration the last point at which the leader had completed a full lap without the red flag being displayed. The classification will be established as follows:

- a) For riders who crossed the finish line on the same lap as the leader before the red flag was displayed, a partial classification will be established at the end of that lap.

- b) For riders who had not crossed the finish line on the same lap as the leader before the red flag was displayed, a partial classification will be established at the end of the previous lap.
  - c) The complete classification will be established by combining both partial classifications according to the lap/time procedure.
  - d) Riders who have entered the pit lane which is then closed due to the red flag may be classified according to the number of laps completed and the time of crossing the virtual finish line in the pit lane as follows:
    - If the rider has crossed the virtual finish line in the pit lane, the rider's position will be determined based on that crossing.
    - If the rider has not crossed the virtual finish line in the pit lane, the rider's last crossing of the track finish line will be used.
  
- G.6.4.3 | At the moment the red flag is displayed, riders who are not actively competing in the race will not be classified.
  
- G.6.4.4 | Within five minutes after the red flag has been displayed, riders who have not entered the pit lane, crossing the designated pit lane entry timing point together with their prototype, will not be classified.  
  
 An exemption may be granted in exceptional circumstances (e.g. post-race celebration in an interrupted race deemed to have been completed).
  
- G.6.4.5 | If more than two thirds of the original race distance have been completed by the leader of the race, rounded down to the nearest whole number of laps, the race will be deemed completed and full points will be awarded.
  
- G.6.4.6 | If less than two thirds of the original race distance have been completed by the leader of the race, rounded down to the nearest whole number of laps, the race will be restarted according to Article G.6.5. If it is not possible to restart the race, the results will be awarded as follows:
  - Less than 50%\* of the original race distance completed = Null Race, no points awarded.
  - More than 50%\* of the original race distance completed = Full points awarded.

\*50% rounded down to the nearest whole number of laps.
  
- G.6.4.7 | For the purposes of these Rules & Regulations, "active" and "actively competing" are defined as a rider riding on track, attempting to repair or restart the engine / motor, or attempting to rejoin the track or return to the pit lane with the prototype in a condition fit to rejoin the race. Race Direction will be the sole judge of whether a rider is actively competing, including the condition of the prototype. No appeal is possible against the decision of Race Direction.

## **G.6.5 Re-Starting a Race that has been interrupted**

G.6.5.1 Re-started races will, in principle, follow the procedures defined in this Article. However, as local track, climatic conditions, and other circumstances may vary, Race Direction may reschedule a re-started race within the race program if necessary. Race Direction will make the final decision on whether, when, and how many times an interrupted race will be re-started, according to the circumstances.

Note that in all cases of restarted races, the calculation of the race distance required to declare a result (2/3 distance) will be based on the number of laps of the current race, and not on the original race distance.

G.6.5.2 If a race has to be re-started, the time at which the new start procedure will begin will be displayed on the timekeeping screens. Additionally, this information may also be communicated to Teams through any of the official communication channels of the Organization.

G.6.5.3 For any race that is re-started, the start procedure will follow the Quick Start Procedure.

G.6.5.4 The conditions for the re-started race will be as follows:

- a) If less than three laps have been completed:
  - All riders may start, including riders who may not have completed the sighting lap or warm-up lap for the original start.
  - Prototypes may be repaired.
  - Refuelling or recharging is permitted.
  - The race distance will be the number of laps required to complete the original race distance.
  - Grid positions will be the same as for the original race.
- b) If three laps or more than three laps have been completed but less than two thirds of the race distance:
  - Only riders classified in the interrupted race may re-start. To be eligible to re-start, the rider must enter the pit lane, riding or pushing the prototype, within five minutes after the red flag was displayed in the interrupted race (using the homologated track).
  - Prototypes may be repaired.
  - Refuelling or recharging is permitted.
  - The race distance will be the number of laps required to complete the original race distance.
  - Grid positions will be determined according to the classification at the moment of the interruption (Article G.6.5.2).



## ARTICLE 7: PARC FERMÉ

### G.7.1 Parc Fermé (closed park area)

G.7.1.1 | Parc Fermé is a controlled area used during the Final Event to ensure sporting fairness and to allow the Technical Organization to verify that the motorcycles comply with the Technical Regulations following a practice session, qualifying session or race. When a motorcycle is placed in Parc Fermé, no intervention, modification, adjustment or work of any kind may be carried out on the motorcycle unless expressly authorized by the Technical Director or Technical Staff.

Any motorcycle required to enter Parc Fermé must proceed there immediately after the relevant session and remain under the supervision of the Technical Organization until released.

### G.7.2 Practice Sessions

G.7.2.1 | At the end of any practice session or on-track test, the first three classified prototypes of the session, or in any test performed during the session, may be required to proceed at the Technical Control Area in order to perform any checks requested by the Chief Scrutineer.

Additionally, other prototypes may be selected at random and directed to the Technical Control Area by the Technical Stewards for inspection.

### G.7.3 Race Sessions

G.7.3.1 | At the end of a race, or after the final part of a race that has been interrupted, all prototypes must proceed directly to the Parc Fermé area designated by the Competition Organization.

In this area the prototypes shall remain at the disposal of the Technical Stewards until their release is authorized, in order to allow:

- Technical inspections
- Regulatory compliance checks
- Verification in case of protests or technical investigations

It is the responsibility of the Teams and riders to ensure that their prototype is brought to Parc Fermé in accordance with the instructions of the officials.

This requirement applies to all prototypes that have taken the start of the race, whether they have finished the race, retired, entered the pit box, or been removed from the track due to an incident.

Once the prototype has crossed the finish line, no work, intervention or adjustment of any kind may be carried out on the motorcycle unless expressly authorized by the Technical Stewards or Race Direction.

Failure to comply with Parc Fermé procedures may result in disqualification from the race.

#### **G.7.4 Parc Fermé responsibility and selection for inspection**

G.7.4.1 It is the sole responsibility of the Team and the rider to ensure that the prototype proceeds immediately to Parc Fermé when required.

The selection of prototypes for technical inspection may take place without prior notice and forms part of the standard technical control procedures.

The Technical Director, Technical Stewards or Race Direction may require any prototype to proceed to Parc Fermé or to undergo technical inspection at any time during the Competition.

### **ARTICLE 8: ABANDONEMENT DURING THE EVENT**

#### **G.8.1 Notification of Withdrawal**

G.8.1.1 Any Team entered in the Final Event must inform the Organization if, at any time, they decide not to participate in the event or decide to withdraw during the Event.

#### **G.8.2 Abandonment Process**

G.8.2.1 If a Team is unable to continue in the competition due to irreparable technical problems, rider injury, or any other reason, the Team Leader or Faculty Advisor must report to the Paddock General Office in order to formally declare the Team's abandonment.

Once the abandonment has been declared, the engine / motor must be immediately returned to the Technical Inspection Area or to any other location designated by the Organization for its return and subsequent inspection and rebuilding. The prototype shall also remain available for possible technical checks if requested by the Technical Staff.

G.8.2.2 The abandonment declaration is **irreversible**. Once a Team has formally declared its abandonment, it will not be permitted to participate in any remaining sessions or races, even if the original issue has been resolved.

## ARTICLE 9: TEST 1, 2 & 3 BEHAVIOR

### G.9.1 Rider behaviour

- G.9.1.1 | Riders must strictly follow the instructions given by the marshals at all times.
- G.9.1.2 | Riders must circulate at a calm and moderate speed when transiting between tests. This requirement applies to all movements within the test area during Rounds 1 and 2 of Tests 1, 2 and 3.

### G.9.2 Test Rounds Layout

- G.9.2.1 | The entry point for all rounds of Tests 1, 2 and 3 will be located at the pit entry.
- G.9.2.2 | The exit point for all rounds of Tests 1, 2 and 3 will be located at the pit exit.
- G.9.2.3 | During these rounds, circulation inside the Pit Lane will be in the opposite direction to the normal racing direction of the circuit.
- G.9.2.4 | The approximate location of the tests will be as follows:

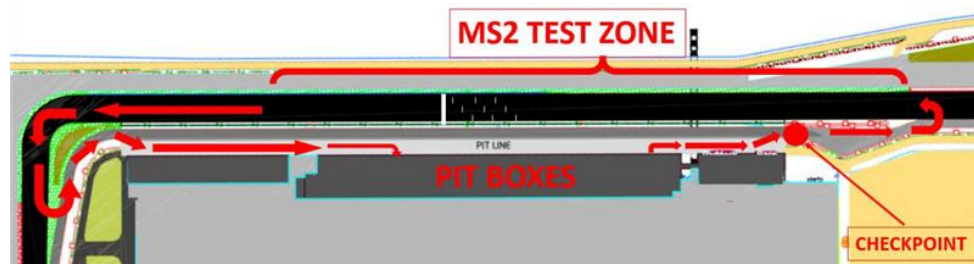


Figure 49 - Rounds 1 and 2 layouts

### G.9.3 Waiting Area Regulations

- G.9.3.1 | A designated waiting area will be established at the entrance of each test round.
- G.9.3.2 | Each prototype will be assigned access to three electrical outlets within the waiting area.
- G.9.3.3 | Only two Team Members and one rider per team are allowed inside the waiting area.
- G.9.3.4 | The only permitted equipment in the waiting area are two tyre warmers (front and rear) and one cooling or heating fan for the thermal management of the prototype.
- G.9.3.5 | Marshals will indicate the starting order of the rounds based on the arrival order of the teams in the waiting area.

G.9.3.6 | If a team refuses to proceed with the test when instructed by the marshals, the round will be recorded as NP (Not Performed) for that team.

G.9.3.7 | Any incident or interference with the correct operation of the test may be penalised by Race Direction depending on the severity of the infringement.

#### **G.9.4 Time Control Procedure**

G.9.4.1 | All rounds of Tests 1, 2 and 3 will be subject to a time control procedure.

G.9.4.2 | The time control will be located in the designated tent within the waiting area.

G.9.4.3 | The rider and the prototype in operational condition must be present in the waiting area at the exact scheduled time published by the Organization.

G.9.4.4 | Teams arriving earlier than their scheduled time must wait in the area indicated by the marshals.

G.9.4.5 | If a rider arrives after the scheduled control time, participation in that round will not be permitted and the round will be recorded as NP.

# **SECTION H**

FINAL EVENT

## ARTICLE 1: SCHEDULE

### H.1.1 Official Schedules during the Final Event

- H.1.1.1 | The Organization will establish an official schedule assigning each team the corresponding time slots to carry out the tests related to Scrutineering, the MS1 Phase and the MS2 Phase. These schedules will be published and communicated to the teams prior to the Final Event.
- H.1.1.2 | Teams must comply with the schedules established by the Organization during the Final Event. Failure to respect the assigned schedule may result in a penalty in the corresponding test or, if operational conditions require it, exclusion from that test.
- H.1.1.3 | In the MotoStudent Electric Category, a minimum interval of 40 minutes between tests will be guaranteed to allow teams to recharge the battery.
- H.1.1.4 | The Organization reserves the right to modify the schedule of the tests if required for operational or safety reasons. The Organization will make every effort to ensure the efficient development of the Final Event and to provide teams with sufficient time to prepare between tests.

## ARTICLE 2: AREAS AND FACILITIES DURING THE FINAL EVENT

### H.2.1 Definition of Areas

- H.2.1.1 | Participating teams undertake to respect and take care of the facilities where the Final Event takes place. Any misuse or damage to the facilities may result in sanctions imposed by the Organization, including expulsion from the Competition.
- H.2.1.2 | Annex 3 provides an indicative distribution of the different areas of the Final Event described below. The final layout will be confirmed in the Event Guide for Teams document.
- H.2.1.3 | Start–Finish Straight  
The finish straight of the FIM Grand Prix International circuit at MotorLand Aragón. In addition to being the starting and finishing point of the races, this straight will host several dynamic tests of the MS2 Phase of the Competition, as well as other parallel activities.
- H.2.1.4 | Pit Lane  
The route connecting the circuit layout with the Team Boxes. Access to this area is restricted to registered students, tutors and riders participating in the Competition. Public access to the Pit Lane is strictly prohibited.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H.2.1.5  | <p><b>Pit Building</b></p> <p>The main building of the circuit, composed of three floors:</p> <ul style="list-style-type: none"><li>- Ground Floor: Team Boxes, bathrooms and showers.</li><li>- First Floor: presentation rooms, media centre, restaurant, services, bathrooms, terraces and briefing room.</li><li>- Second Floor: public terraces.</li></ul> <p>The floors are connected by stairways through Towers A, B, C, D, E and F. Tower E includes an elevator accessible for people with reduced mobility.</p> |
| H.2.1.6  | <p><b>Public Terraces</b></p> <p>Located along the Pit Building, these terraces are open to the public to observe the activities taking place on track.</p>                                                                                                                                                                                                                                                                                                                                                                |
| H.2.1.7  | <p><b>Media Centre</b></p> <p>Dedicated workspace for accredited media professionals and the MotoStudent communication team. The Media Centre provides working facilities for journalists, photographers and broadcasters covering the Final Event. Access to the Media Centre is restricted to media personnel accredited by the Organization. The Organization reserves the right to regulate access and use of the facilities.</p>                                                                                      |
| H.2.1.8  | <p><b>Team Boxes</b></p> <p>Working garages assigned to the teams during the Final Event.</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| H.2.1.9  | <p><b>Technical Control Area</b></p> <p>Scrutineering area where the Organization performs the technical inspections of the participating prototypes (Static Scrutineering).</p>                                                                                                                                                                                                                                                                                                                                           |
| H.2.1.10 | <p><b>MotoStudent General Office</b></p> <p>Main office of the Competition Organization responsible for assisting registered teams during the Final Event.</p>                                                                                                                                                                                                                                                                                                                                                             |
| H.2.1.11 | <p><b>Paddock General Office (PGO)</b></p> <p>Paddock office where teams may be summoned by different organizational committees (MS1, MS2, Technical, Disciplinary, etc.) to address matters related to protests, irregularities, disciplinary procedures or organizational issues.</p>                                                                                                                                                                                                                                    |
| H.2.1.12 | <p><b>MotoStudent Shop</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Official shop of the Organization. During the Final Event, payments may be made in cash (€) or by credit card.



Picture 50 - Accepted credit cards

- H.2.1.13 | Tires Technical Service  
Service point for the supply and mounting of tyres.
- H.2.1.14 | Fuel Area  
Facility designated for the supply of the Official Fuel.
- H.2.1.15 | Dynamic Area  
Paddock area reserved for dynamic activities, including the Dynamic Safety Check and the regulated practice areas.
- H.2.1.16 | Parc Fermé  
Closed control area where prototypes may be retained for technical inspection after sessions or races.
- H.2.1.17 | MotoStudent Arena  
Paddock area where official activities such as the MS1 Awards Ceremony, presentations and other event activities will take place.
- H.2.1.18 | Fan Zone  
Area dedicated to commercial activities, exhibitions and entertainment for visitors.
- H.2.1.19 | Medical Centre  
Medical facility providing healthcare assistance during the Final Event.
- H.2.1.20 | Working Vehicles Parking  
Area reserved for operational vehicles used by teams and Organization, with a maximum vehicle length of 8 meters.

## H.2.2 Team Installations

- H.2.2.1 | The Organization will provide and assign, free of charge, a working area (located inside a Team Box) to each team. All repairs, adjustments and preparation of the prototype must be carried out in this area.



- H.2.2.2 | The Team Boxes are equipped with 230V power sockets with European standard plugs, as well as 230V industrial sockets and three-phase 400V sockets. The boxes also include a TV connection with access to live timing, compressed air supply, water connections and access to bathroom facilities.
- H.2.2.3 | Personal and transport vehicles must be parked only in the areas designated for this purpose. Access to the track and test areas is restricted to vehicles authorized by the Organization.

## ARTICLE 3: PARTICIPANT ACCREDITATION

### H.3.1 Accreditation

- H.3.1.1 | Upon arrival at the Final Event, the Organization will provide an accreditation pass and wristband to each registered Team Member, rider and tutor. This identification will allow the Organization to monitor participation in the different tests and any interventions carried out on the prototype.
- H.3.1.2 | Accreditation passes and wristbands will be issued to registered Team Members upon entry to the Final Event and must be worn visibly at all times during the event.  
The accreditation passes and bracelets will be placed to the registered Team Members at the time of entering the Final Event, and they must wear them until it finishes.
- H.3.1.3 | Accreditation passes and wristbands will only be issued to officially registered members of the team. Any person accompanying the team who is not officially registered will be considered part of the general public and will therefore not have the same access rights or operational privileges as registered participants.

## ARTICLE 4: BEHAVIOUR RULES

### H.4.1 Behaviour of Participants during the Final Event

- H.4.1.1 | To ensure a safe and respectful environment during the Final Event, the following actions are strictly prohibited:
- Cooking inside the Boxes or in any areas not specifically authorized by the Organization.
  - Sleeping inside the Boxes.
  - Installing tents in areas not specifically authorized by the Organization.
  - Bringing pets into the Paddock.
  - Smoking, lighting fires, or storing gas containers inside the Boxes.
  - Parking unauthorized vehicles inside the Paddock.
  - Exchanging personal or vehicle accreditation passes.

- Painting, drilling, disassembling or otherwise damaging circuit facilities.
- Carrying out any commercial activity that may compromise public health or safety.
- Driving in a reckless manner inside the Paddock. Reckless driving includes driving without all tyres in contact with the ground, driving at speeds inappropriate for pedestrian safety, or carrying more passengers than permitted by local regulations and/or circuit rules. The use of helmets (for motorcycles) or safety belts (for cars) is mandatory.
- Walking around the Paddock stripped to the waist or wearing only sandals.

#### H.4.1.2 Alcohol and Drugs

The consumption, possession or influence of alcohol, drugs or any substance that may impair the physical or mental condition of participants is strictly prohibited during the Final Event.

This prohibition applies to all participants including Team Members, riders, tutors and any accredited personnel.

The Organization reserves the right to carry out checks whenever it considers necessary. Any person found under the influence of alcohol or drugs may be immediately removed from the Event and may lead to penalties or disqualification of the Team.

### H.4.2 **Authorized Vehicles in the Paddock**

H.4.2.1 The prototype may only move under its own power during official tests. For any other movement within the Paddock, it must be pushed or moved with the engine or motor switched off.

H.4.2.2 Vehicle circulation inside the Paddock is restricted to Organization vehicles and the following light vehicles:

- Non-motorized vehicles (scooters, bicycles, etc.).
- Personal mobility vehicles (electric scooters, assisted bicycles, etc.).
- Small scooters.

H.4.2.3 The circulation of any vehicle on the Pit Lane (except for prototypes) is strictly prohibited, except for the use of non-motorized vehicles during the time allowed for track inspection.

H.4.2.4 The circulation of the vehicles described in Article H.4.2.2 must comply with the following rules:

- No more than two persons may travel on the vehicle.
- Children under 14 years of age are not allowed to drive any vehicle.
- Reckless driving inside the Paddock is strictly prohibited.

- Vehicle circulation is prohibited during quiet hours (23:00 – 07:00).
- **The use of a helmet is mandatory.**

#### **H.4.3 Prototype Manipulation**

H.4.3.1 | For safety reasons, any manipulation of the prototype must comply with the following rules. Failure to comply may result in penalties or even expulsion from the Competition.

H.4.3.2 | All repairs and manipulations of the prototype during the Final Event must be carried out exclusively inside the Team Box assigned to each team.

H.4.3.3 | The prototype must remain within the circuit facilities during the entire Final Event. The prototype, or any of its components (including batteries, fuel systems or critical parts), may not be removed from the circuit facilities under any circumstance without the explicit authorization of the Organization.

Exceptionally, the Organization may authorize the temporary transfer of the prototype to the designated karting practice area under controlled conditions and only within the MotorLand facilities.

Any unauthorized removal of the prototype or its components from the circuit may result in penalties or exclusion from the Competition.

### **ARTICLE 5: COMMUNICATION**

#### **H.5.1 Contact with the Organization during the Final Event**

H.5.1.1 | Communication between teams and the Organization during the Final Event must be carried out primarily through the MotoStudent General Office.

H.5.1.2 | During the Final Event, the Team Tutor, Team Leader or rider may be summoned by the Organization to report incidents or address operational matters. Teams must attend these calls at the indicated location within a maximum period of 15 minutes from the moment the notification is issued.

#### **H.5.2 Briefings**

H.5.2.1 | During the Final Event, the Organization will hold several briefings to explain the development of the Competition to teams, tutors and riders.

H.5.2.2 | Team Leaders, tutors or riders must attend the briefings when summoned by the Organization. Failure to attend without justified cause may result in penalties.

### **H.5.3 Communication of Schedules and Results**

- H.5.3.1 | The Organization will communicate the schedules of tests and sessions through different official communication channels.
- H.5.3.2 | The Organization will publish the results obtained in each test on the official notice board located in the Paddock.
- H.5.3.3 | The timing results of the “Free Practice”, “Qualifying” and “Race” sessions will be broadcast live through the circuit’s internal TV signal. In addition to screens located in common areas, all Team Boxes and the Pit Wall will have TV connections to access the live timing broadcast.
- H.5.3.4 | The timing results of the “Free Practice”, “Qualifying” and “Race” sessions will also be broadcast live via the internet. The Organization will communicate to the teams the web platform where live timing can be followed.
- H.5.3.5 | The final results obtained in each test may be published through the official communication channels of the Competition once the Competition has concluded.

## **ARTICLE 6: ELECTRICAL SAFETY**

### **H.6.1 Individual Safety Protections**

- H.6.1.1 | In addition to the mandatory elements specified in Article D.5.3, the use of the following protective equipment is strongly recommended:
  - Helmet or protective visor against fire or electric arc hazards.
  - Insulated protective boots.
  - Insulating mat CEI 61111 – Class 0.
  - Emergency hook (CEI 61235 / CEI 60230 / CEI 50508 / NF C18-510).
- H.6.1.2 | It is strongly recommended that students who will work with the electric prototype have completed a High Voltage safety training course related to electrical systems or industrial automation.

### **H.6.2 Quarantine Zone**

- H.6.2.1 | A quarantine zone will be designated by the Organization where prototypes may be isolated if the Technical Director considers it necessary.
- H.6.2.2 | Only one prototype may remain inside the quarantine zone at any given time.
- H.6.2.3 | The prototype may only be removed from the quarantine zone with the explicit authorization of the Technical Director.

- H.6.2.4 | The rider and mechanics from the team may enter the quarantine zone only with the explicit authorization of the Technical Director. They must wear all mandatory personal safety equipment before entering the area.

**H.6.3 Procedure in Case of Electrical Fire in Boxes, Paddock or Pit Lane**

- H.6.3.1 | In the event of an electrical hazard in any of the above-mentioned areas, the PAS principle (Protect, Alert, Support) must be followed.
- H.6.3.2 | After ensuring personal safety, the Organization must be immediately alerted. If no official staff member is nearby, the Paddock General Office must be informed.
- H.6.3.3 | To facilitate the safe removal of the prototype, remove any obstacles that may hinder the intervention of the Organization. Do not move any object that could endanger yourself or others nearby.
- H.6.3.4 | Follow the instructions of the Organization at all times. This is a critical safety situation; any confrontation, obstruction or lack of cooperation may result in penalties.

# **SECTION I**

## **MODIFICATIONS GLOSSARY**

# **SECTION J**

## **ANNEXES**

### ANNEX 1: CLAIMS AND IMPUGNATIONS FORM

Mr/Ms \_\_\_\_\_, holder of ID/Passport number \_\_\_\_\_,  
acting as Team Tutor / Team Leader of Team \_\_\_\_\_,  
representing the University \_\_\_\_\_, with prototype  
number \_\_\_\_\_, participating in the MotoStudent eFuel / Electric (*delete as appropriate*)  
Category, hereby submits (*mark with an X*):

☐ A **CLAIM** addressed to the Organization of the MotoStudent International Competition.

☐ **AN IMPUGNATION** (PROTEST) against Team \_\_\_\_\_  
\_\_\_\_\_, prototype number \_\_\_\_\_.

The claimant considers that the following Article of the Competition Rules & Regulations has  
been violated:

Article \_\_\_\_\_ of the MotoStudent Rules & Regulations, which states:

The following evidence is provided in support of this claim/impugnation and may be considered  
by the Organization during the verification process:

The undersigned declares that the information provided in this document is true and submitted  
in good faith in accordance with the MotoStudent Regulations.

Signed in \_\_\_\_\_, on \_\_\_\_ / \_\_\_\_ / 20\_\_\_\_.

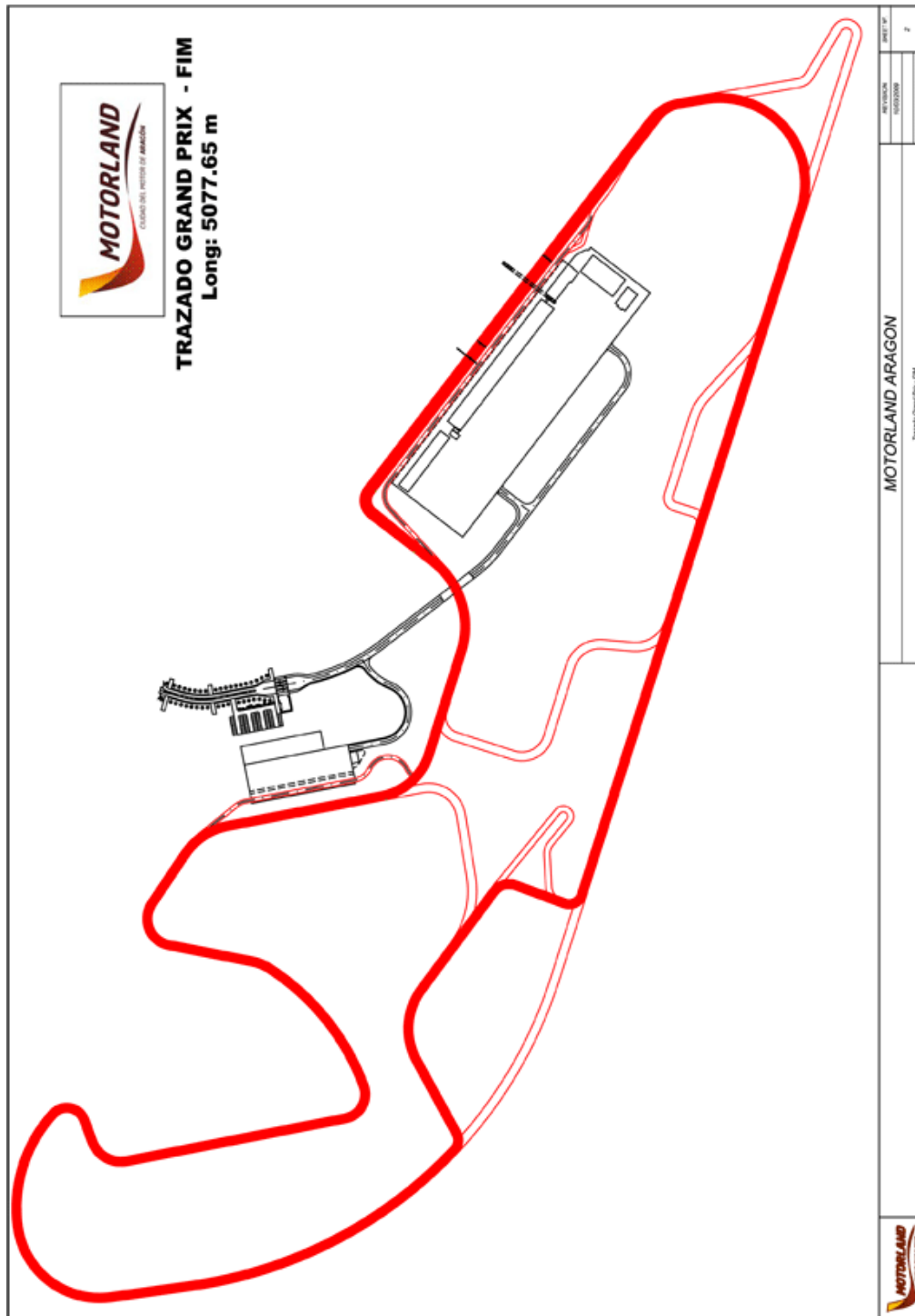
Signature:

Mr/Ms \_\_\_\_\_

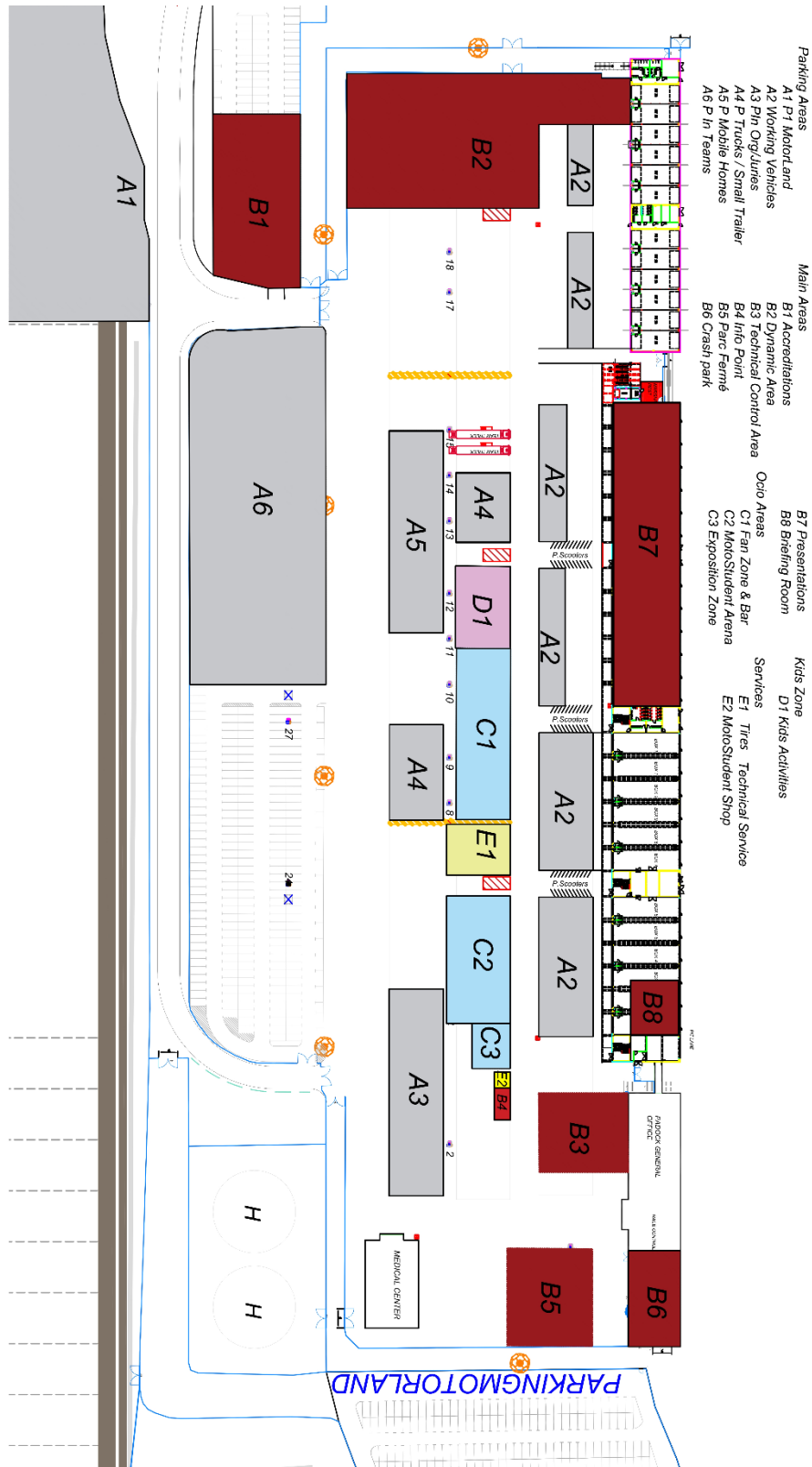
Team Tutor / Team Leader (*delete as appropriate*)



ANNEX 2: MOTORLAND ARAGÓN – FIM GRAND PRIX INTERNATIONAL TRACK LAYOUT



**ANNEX 3: INDICATIVE FINAL EVENT MAP (TO BE CONFIRMED)**



#### ANNEX 4: COLLABORATION AGREEMENT

##### COLLABORATION AGREEMENT BETWEEN THE UNIVERSITY \_\_\_\_\_ AND THE UNIVERSITY \_\_\_\_\_

In \_\_\_\_\_, on the \_\_\_\_ day of \_\_\_\_\_ of the year \_\_\_\_\_.

On the one hand, \_\_\_\_\_, acting in the capacity of \_\_\_\_\_  
of the University \_\_\_\_\_, acting for and on behalf of the aforementioned  
University.

And, on the other hand, \_\_\_\_\_, acting in the capacity of  
\_\_\_\_\_ of the University \_\_\_\_\_, acting for and on behalf of the  
aforementioned University.

##### HEREBY STATE

That the \_\_\_\_\_ team participates in the MotoStudent Competition and carries  
out activities related to the design, development and testing of a competition motorcycle  
prototype.

That the University of the student is interested in allowing its students to collaborate in the  
development and realization process of the electric/eFuel prototype participating in the current  
Edition of MotoStudent.

That, by virtue of the foregoing, the parties mutually recognize each other's legal capacity to be  
bound in the representation they respectively hold and agree to sign the present Agreement,  
which shall be governed by the following

##### CLAUSES

###### FIRST. Purpose

The parties agree to the participation of the students \_\_\_\_\_ (full name  
and ID number) in the \_\_\_\_\_ team for the development of the prototype  
participating in the MotoStudent International Competition.

###### SECOND. Student requirements

In order to facilitate access to the facilities of \_\_\_\_\_ and the development of  
activities related to the MotoStudent Competition, the above-mentioned students must prove  
that they meet the following requirement:

- They must be enrolled in the academic program during the period in which their participation  
in the Competition takes place.

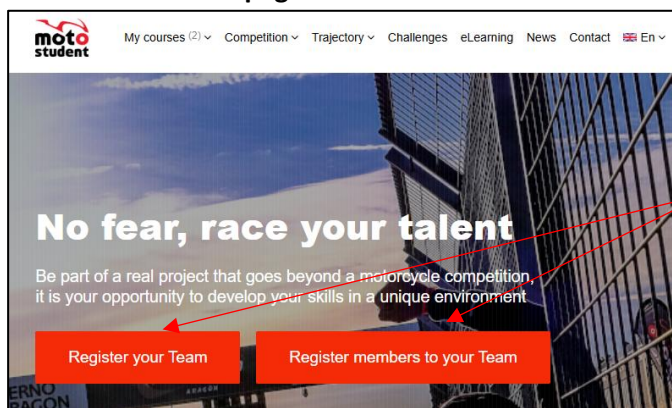
## ANNEX 5: MOTOSTUDENT REGISTRATION PROCESS

Registration will open in accordance with the schedule indicated in the Competition Rules & Regulations. The **deadline** for registration is **30 days before the Final Event**. Please note that places are limited and that teams must complete the different Milestones established in the Competition Regulations.

The Team Registration process is explained below:

- Each team must register their Team through the MotoStudent Platform, where you will find the option “Register your team”.

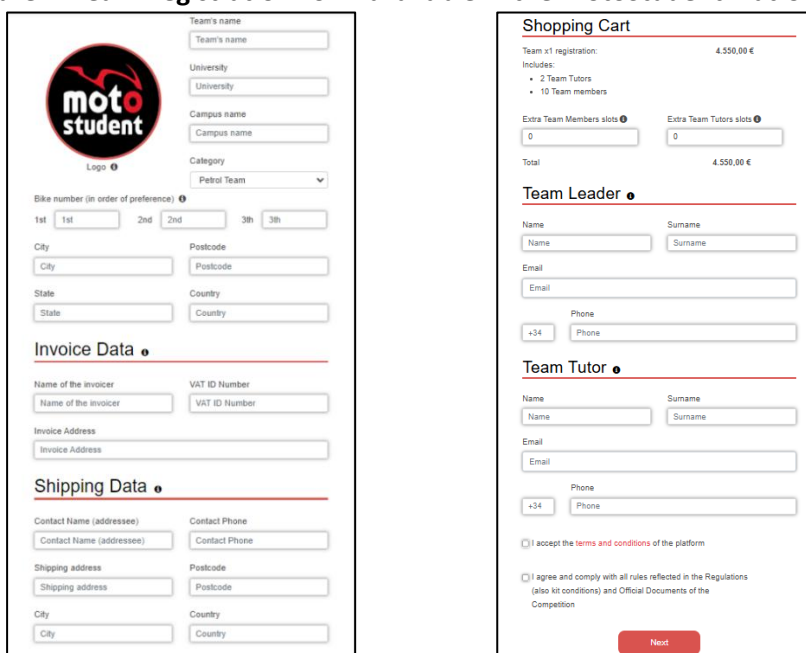
**Picture 1: Main page MotoStudent Platform**



New buttons

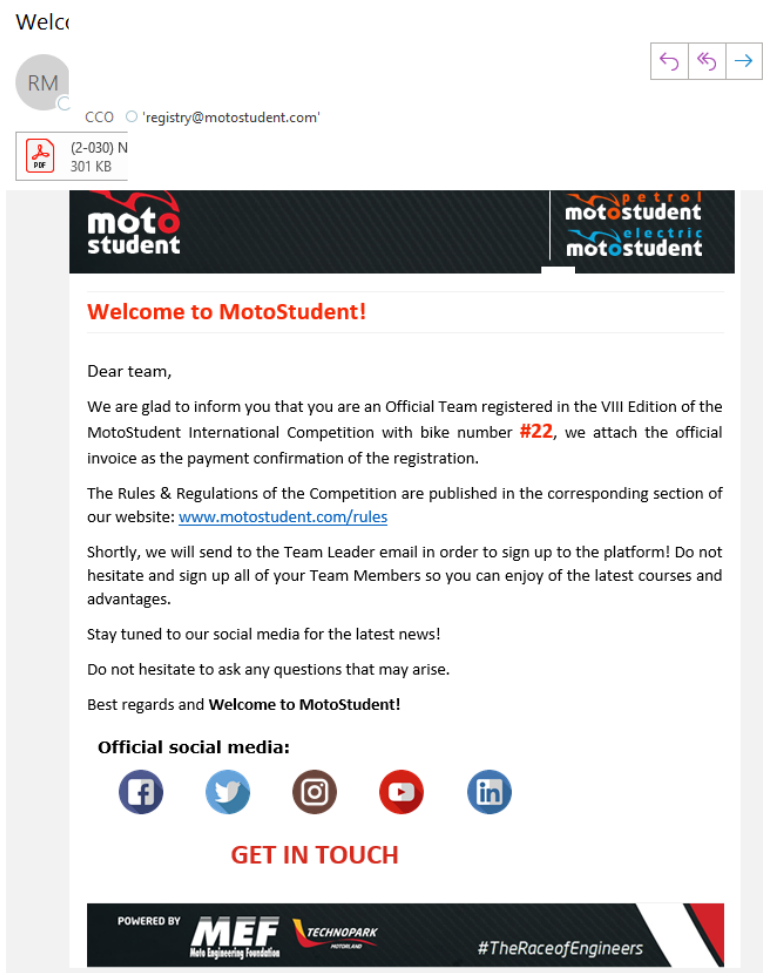
- Once the team clicks on the above button a new window will open and there you will find all the team information to fill in.

**Picture 2: Team Registration Form available in the MotoStudent Platform**



- Once the Team Registration has been submitted, the Administration Department will validate the data, prepare the proforma invoice and send it to the Team Leader and Team Tutor contact indicated in the Registration Form. **The Team will be accepted as an official MotoStudent Team once the proforma invoice has been paid.**
- Once the Administration Department confirms the payment with the bank details, the team will be contacted by email, sending to the team leader and tutor the **WELCOME TO MOTOSTUDENT** email.

**Picture 3: Welcome to MotoStudent email**



### TEAM LEADER/TEAM MEMBERS/TEAM TUTORS REGISTRATION

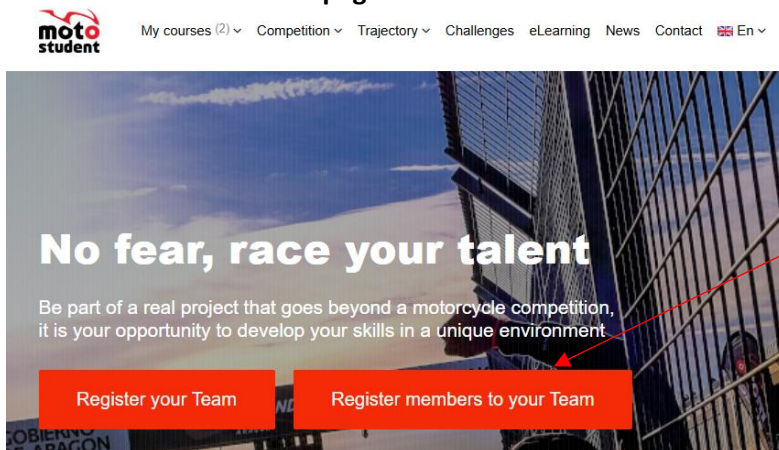
The first member who must be officially registered in the team is the Team Leader. This step is very important, as the Team Leader is the person in charge of accepting the rest of Team

Members/Team Tutors in the team.

Once the Team Leader is registered, team members/team tutors can start registering through the MotoStudent Platform.

To apply for the registration, team leader/team member/team tutor must go to the MotoStudent platform and click on “Register a Member”:

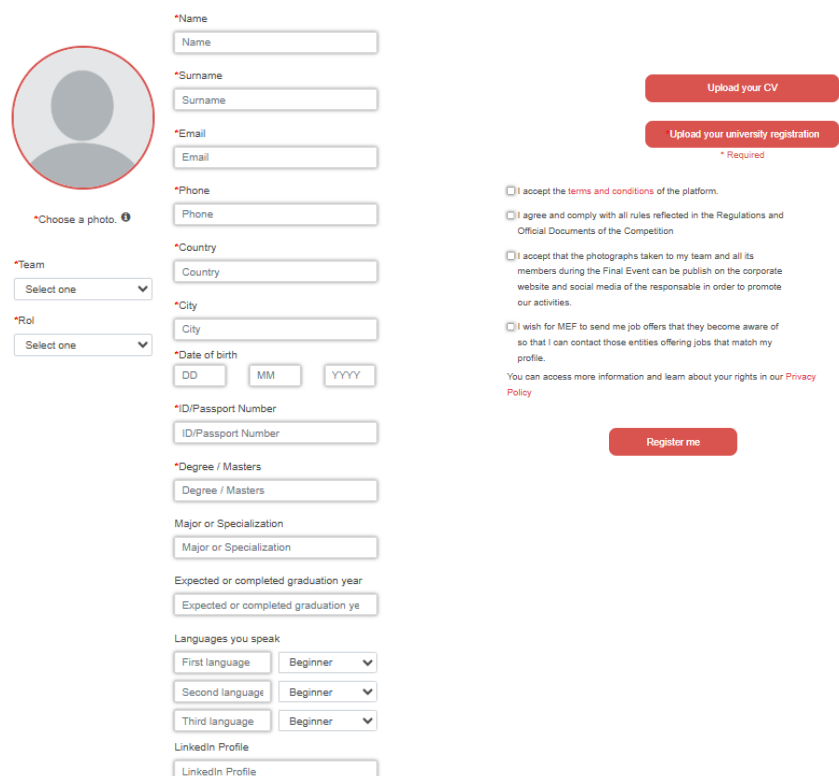
**Picture 4: Main page MotoStudent Platform**



Click here to register a member

- You must fill the following registration form. Here, we explain some of the fields that may generate doubts:

**Picture 5: Members Registration Form available in the MotoStudent Platform**



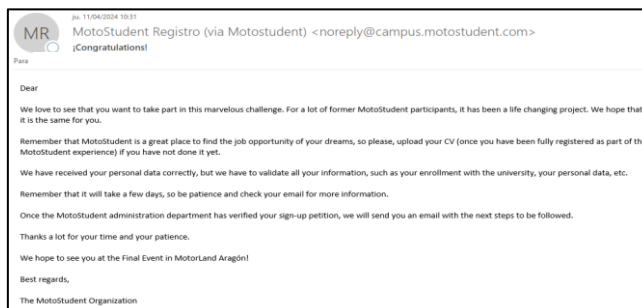
- Add a **personal photo**.
  - **Team**: select your Team Name from the list.
  - **Rol**: select your Team Rol.
  - Complete your **personal data**, information about your **studies** and your **LinkedIn Profile**.
  - **Upload your CV**: uploading your CV will help you to attract the top automotive companies in order to find your dreams job.
  - Upload your **University Registration**: here, you have to upload an official document that justifies your relation with your University:
    - **Team Leader/Team Member**: Official university enrolment document for the period 2025-2026 or 2026-2027.
    - **Team Tutor**: University contract or official certificate from the University justifying the employment contract with the University.
  - Accept the terms and conditions of the platform.
  - Agreement and compliance with the Rules and Official Competition Documents.
  - Authorize the taking and use of photographs and audiovisual material of the team and its members during the Final Event.
  - Accept the terms and conditions of the job board.
- Save the information and **wait for the Organisation to validate the user**. Once the validation process is finished, the team leader/team member/team tutor will receive an **email confirming the official registration** as a team leader.

There is **difference** in the **acceptation** of registration of the Team Leader, from the Team Members/Team tutors.

**1. Team Leader only has an Administration acceptance:**

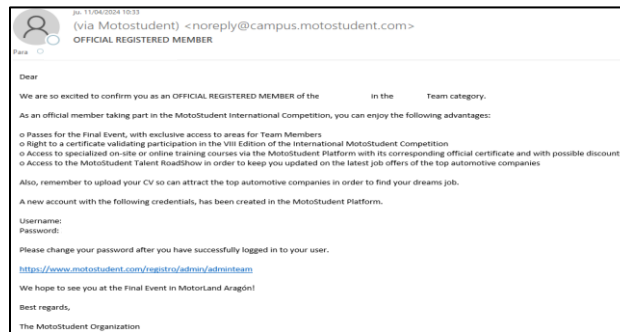
- MotoStudent confirms the reception of the Team Leader data and start to revise it.

**Picture 6: Notification that the Team Leader data has been received correctly and will be reviewed**



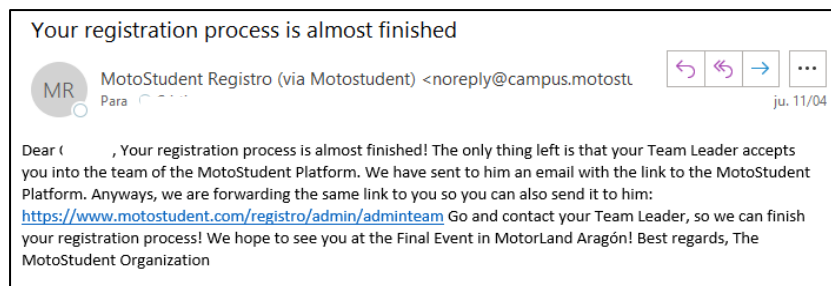
- MotoStudent confirms the Team Leader registration after reviewing all the data

Picture 7: Email confirming that the team leader's user has been validated.



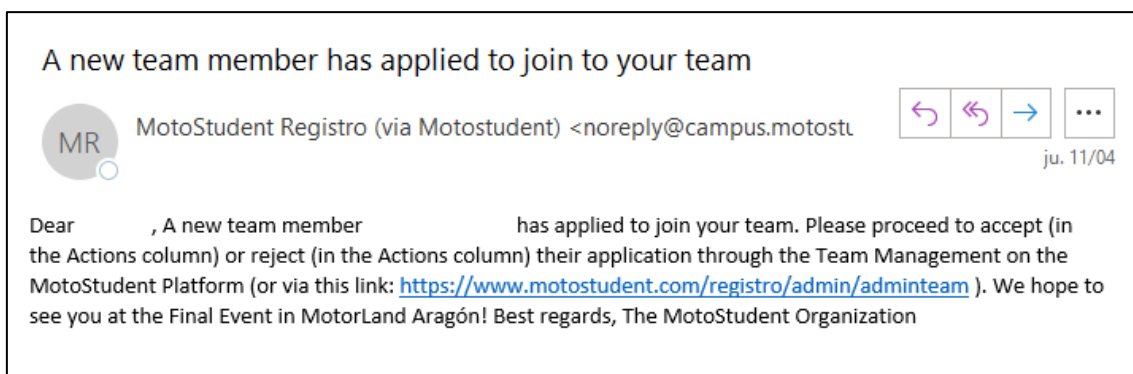
2. **Team Members/Team Tutors** have a **double acceptance** to be registered in the team:
  - In the first instance, **verification** by the **Administration Department**.
  - Secondly and finally the **acceptance** of the **Team Leader** in the Team.
- Once MotoStudent validate Team Member/Team Tutor data, the following email is sent:

Picture 8: Email received by the team member/team tutor who wants to register, when the info has been validated by administration department but has to be validated also by the team leader



- The Team Leader will be notified that a team member/team tutor is applying to join the team.

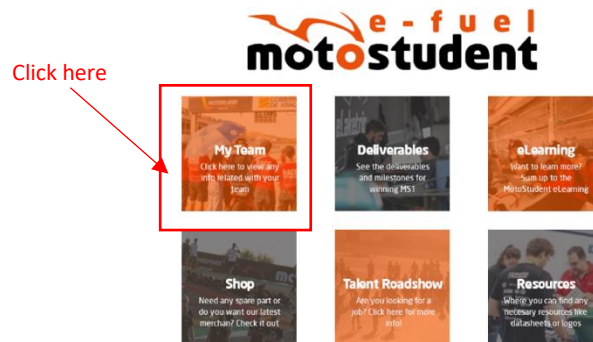
Picture 9: Notification received by the team leader when a team member/team tutor applies to join the team.



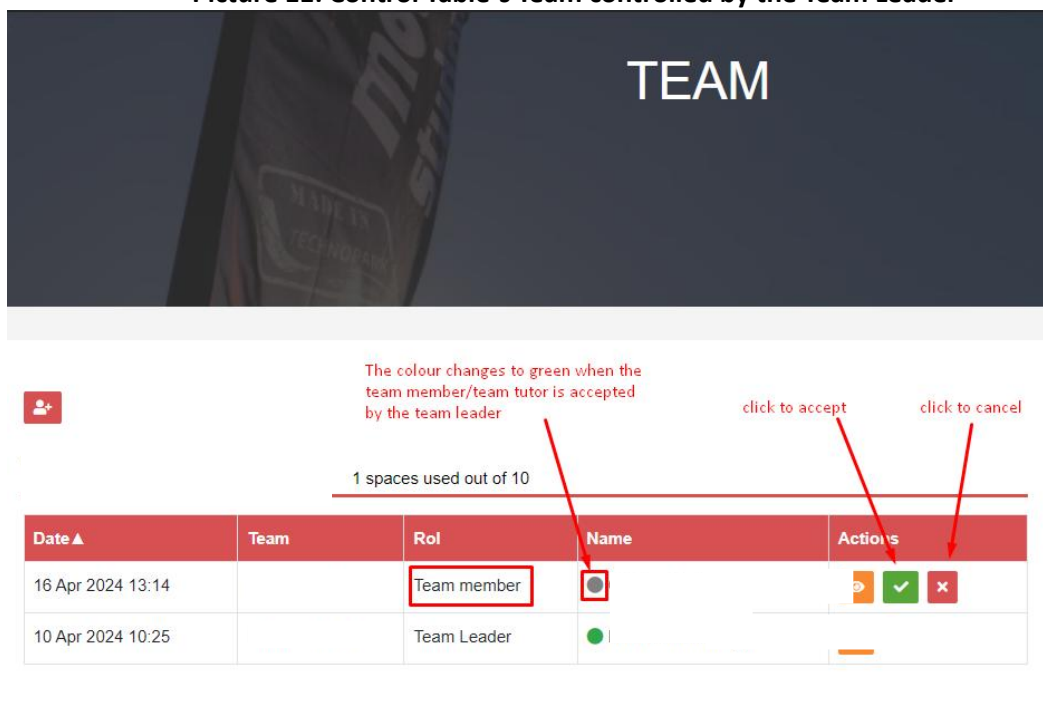


- The team leader has to enter in the MotoStudent Platform, click on Team Course → My Team and accept the team member/team tutor who is requesting to be registered.

**Picture 10: My team Space (depending on the team Electric or eFuel)**



**Picture 11: Control Table's Team controlled by the Team Leader**



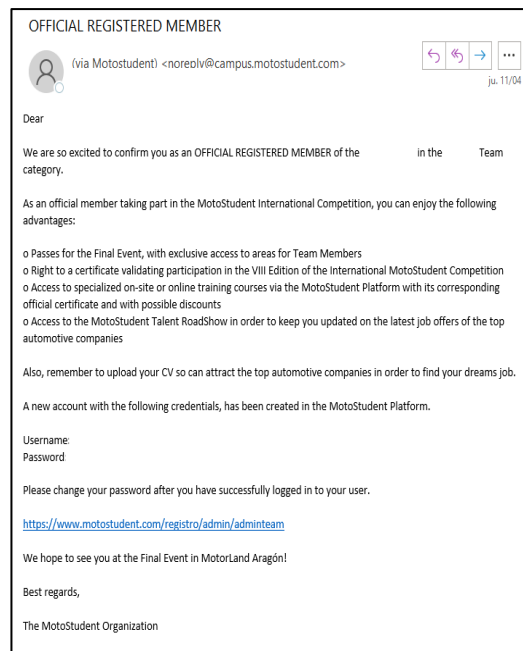
TEAM

1 spaces used out of 10

| Date ▲            | Team | RoI         | Name | Actions                                                                               |
|-------------------|------|-------------|------|---------------------------------------------------------------------------------------|
| 16 Apr 2024 13:14 |      | Team member |      | <input type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> |
| 10 Apr 2024 10:25 |      | Team Leader |      | <input type="checkbox"/>                                                              |

Once the Team Leader accepts, the team member/team tutor will receive an email confirming the official registration as a team member/ team tutor and will enjoy all the advantages of being part of the MotoStudent Community!

**Picture 12: Confirmation email received by the team member/team tutor when has been accepted by both administration department and the team leader (a username and password are sent and need to be changed)**



After these steps explained below, the Team Leader/Team Members/Team Tutors will be registered and can enjoy all the advantages of being part of the MotoStudent Community!

## **ANNEX 6: ENGINE MANAGEMENT PROCEDURE**

The Organization of the MotoStudent International Competition hereby establishes the following mandatory procedure for engine management, applicable to all registered teams.

In both categories (Electric and eFuel), the engines are the property of the MotoStudent Organization and are temporarily assigned to participating teams solely for the purpose of developing their prototypes and enabling fair participation in the Competition. These engines must be returned at the end of the respective edition in proper operating condition, except for normal wear resulting from their use during the Competition. Their use is strictly limited to the MotoStudent International Competition.

This measure aims to ensure the technical stability of the Competition over multiple editions, thereby facilitating the evolution of the prototypes without requiring constant adjustments due to engine changes.

### **SCOPE OF APPLICATION**

This procedure shall apply from the moment the Participation Kit is received by the Team until the conclusion of the Final Event of the corresponding edition, and the same procedure shall remain in force for subsequent editions.

### **MANDATORY RETURN OF THE ENGINE AFTER THE FINAL EVENT OF THE EDITION**

All teams are required to return the engine to the Organization at the end of the Final Event of the Edition, regardless of its conditions:

Teams present at the Final Event and competing in the race: on-site return at the assigned Parc Fermé.

Teams present at the Final Event but officially withdrawing from the competition: return the engine at the Technical Box.

Teams not attending the Final Event must ship the engine to the Competition Organization's facilities no later than thirty (30) days before the first day of the Final Event.

Shipping address:

Edificio Dr. Joaquín Repollés

Parque Tecnológico TechnoPark MotorLand

Alcañiz 4600 (Teruel), Spain

All returned engines will be inspected and serviced in order to be reused in future editions, in which teams will be re-assigned the same engine. Under no circumstances do teams acquire ownership of the engine. The ownership, acquisition cost and inspection of the engines are the sole responsibility of the MotoStudent Organization, provided that the engine is returned. The engine must not be opened, disassembled or modified by the Team under any circumstances.

## PROCEDURES RELATED TO THE ENGINE

In case of problems or failures related to the engine, the following situations may arise:

### Engine failure before the Final Event:

If an engine failure is detected at least 45 days prior to the Final Event, the team must:

Contact the Organization through the ticketing system, providing all relevant technical information.

Send the engine to the Organization's facilities for technical evaluation.

The Technical Direction will determine the origin of the failure:

**Failure due to normal use:** repair or replacement at no cost to the team.

**Electric:** All damage caused by manufacturing defects or flaws in the production process.

**eFuel:** Any damage resulting from manufacturing defects under normal operating conditions, as well as normal wear and tear.

**Failure due to misuse or negligence:**

**Repairable:** An invoice will be issued for the repair cost, to be paid by the Team.

**Electric:** Any damage to components such as the encoder, bearings, encoder connector and wiring, phase wiring, or the plastic seal of the motor that the Organization considers replaceable. If the repair cost exceeds the established engine replacement value, the engine will be considered irreparable and must be replaced.

**eFuel:** If the repair cost exceeds the established engine replacement value, the engine will be considered irreparable and must be replaced.

**Irreparable:** Full replacement of the engine, charged to the team based on the replacement fee. The new engine will also remain the property of the Organization and must be returned at the end of the edition.

**Electric:** Any damage to the stator, rotor, shaft, temperature sensor, or motor housing will be considered irreparable.

**eFuel:** Any damage not covered in the repairable section above indicated, such as

damage to crankcases, gearbox, etc.

#### **Engine failure during the Final Event or within 44 days prior to it:**

In case of engine failure during the Final Event or within the 44 days prior, teams must:

The team must notify the Organization immediately (at the technical box).

The engine will be retained for later inspection. Repairs will not be allowed until after the Final Event, due to the time required for proper assessment.

The team may request a courtesy engine on a rental basis (one-time per edition), under the following conditions:

Payment of the rental fee.

A security deposit, to be adjusted based on the technical diagnosis of the original engine.

The courtesy engine will be delivered only during the Final Event, upon request and after signing a rental agreement.

**Important:** If the courtesy engine suffers damage due to misuse, the team will be responsible for the full replacement cost. Requesting a second courtesy engine will not be permitted.

#### **FEES ASSOCIATED WITH THESE PROCEDURES**

The following fees must be considered once the procedures described above have been applied:

| Item                                               | Electric               | eFuel                  |
|----------------------------------------------------|------------------------|------------------------|
| Repair deposit                                     | € 1,500                | € 1,900                |
| Courtesy engine rental fee<br>(during Final Event) | € 500 (VAT included)   | € 500 (VAT included)   |
| Engine replacement cost<br>(irreparable failure)   | € 2,800 (VAT included) | € 3,900 (VAT included) |

Even if the engine failure is between the 45 days prior and the Final Event, you must contact the Organization at the Technical Box.

This procedure is intended to ensure the traceability and proper condition of the engines throughout the different editions of the MotoStudent Competition.

Participation in the competition implies the explicit acceptance of these conditions by all registered teams.

## ANNEX 7: MOTOR/ENGINE TEMPORARY LOAN TERMS AND CONDITIONS

By completing the registration of a team in the MotoStudent Competition and accepting its participation conditions, each team expressly acknowledges and agrees to be bound by the following terms and conditions:

**Every team registered in the MotoStudent Competition and recipient of this document available within the MotoStudent Rules & Regulations, shall hereinafter be referred to as the 'Borrower'.**

The Rules & Regulations of the present edition that the Borrower has accepted by registering one or more student teams provide for the delivery of a MotoStudent Kit to all Teams registered for the MotoStudent Competition, the components of which must be installed in the motorbike prototypes. Only the motors / engines included in the MotoStudent Kit must be returned to the Organization at the end of the Edition.

This annex sets forth the terms and conditions governing a temporary free loan of a motor/engine (electric or combustion), owned by MEF and described at the end of this document, to the Borrower, as a participant in MotoStudent.

MEF grants to the Borrower, and the Borrower accepts, a temporary, limited, free of charge and non-transferable loan to use the motor/engine described at the end of this document for the sole purpose of installing it on the prototype motorbike submitted to MotoStudent, in strict compliance with the Rules and the terms and conditions contained herein.

MEF expressly disclaims any and all liability arising out of the Borrower's use of the motor/engine, including civil liability for personal injury or property damage, whether or not suffered by Team Members or third parties.

MEF is the owner of the motor/engine. MEF shall have the right at any time to take any action consistent with ownership of the motor/engine. In case of bankruptcy, seizure, embargo or any other administrative or judicial act that disturbs the possession of the motor/engine, the Borrower will have to declare its condition of simple user.

All import duties and taxes and the costs of returning the motor/engine to MEF, including reimportation costs, if any, shall be borne by the Borrower in accordance with the Rules. Any other use, loan or transfer of the motor/engine to a third party is strictly prohibited.

The Borrower assumes all risks of damage, loss or theft of the motor/engine from the moment of delivery until its return to MEF, except for normal wear resulting from its proper use within the Competition. The Borrower agrees to cover the repairs or replacement costs if the motor/engine is damaged, lost, or not returned in proper working condition. In the event of any loss of, or damage to the motor/engine, the Borrower shall immediately notify MEF.

However, for the Borrower's own financial security and peace of mind, it is **recommended**, but **not mandatory**, to choose one of the following options:

- a) **Option 1 - Insurance**: Borrower may take out and maintain until the date of return of the motor/engine to MEF, an insurance policy covering all of the following: (i) civil liability for personal injury or property damage caused by the motor/engine to Borrower or any third party; (ii) any damage to, or loss of the motor/engine due to theft, fire, flood or any other circumstance causing total or partial loss or damage. If taken out, it is recommended that such insurance: (i) should be with a first-class insurance company; (ii) should name MEF as the direct beneficiary; and (iii) a copy of the insurance policy should be forwarded to MEF. (See the data and price at the end of this document)
  
- b) **Option 2 – Guarantee**: If the Borrower wishes to provide a guarantee, the price of which is set forth at the end of this document, such guarantee may be provided through a deposit into the bank account designated by MEF. MEF will return the deposit within fifteen (15) days following the receipt and inspection of the Motor, provided that it is in good working condition. If it is not in good condition, the deposit will not be returned and, in addition, the borrower will be required to pay the rest of the motor/engine price.

SWIFT-BIC CODE: BCOEESMM080

IBAN CODE: ES26 3080 0008 1121 0543 9422

The existence of such insurance or security deposit must in no way limit the Borrower's liability to MEF under this Agreement.

The Borrower is obligated to return the motor/engine to MEF at its own costs, either in person, after the Final Event of the MotoStudent, or as soon as the Borrower decides not to submit a prototype motorbike to the MotoStudent. Failure to return the motor/engine on time, or returning a damaged motor/engine, shall oblige the Borrower to pay MEF the price of the motor/engine as set out at the end of this document, or the repair costs invoiced by MEF, if any.

The loan of the motor/engine, as well as all actions related thereto, shall be governed by the common laws of Spain.

Any dispute or difference arising out of this Annex or the breach thereof, which cannot be amicably resolved by the Parties within one month, shall be submitted to the Parties' respective governing bodies for a further period of thirty days to attempt to resolve it amicably before either Party is entitled to bring a claim. This does not apply to claims with a shorter limitation period.

If no such amicable solution is reached in accordance with the preceding paragraph, the Parties, waiving any other jurisdiction to which they may be entitled, agree to submit to the following jurisdictions:

- a) If the BORROWER is domiciled in Spain or in a country of the European Economic Area (EEA), to the Courts and Tribunals of the City of Alcañiz, Teruel, Spain.
- b) If the BORROWER is domiciled outside the EEA and such country is a member of the 1958 New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards, to the arbitration of the Spanish Court of Arbitration -CEA-, which shall appoint a sole arbitrator and conduct the arbitration proceedings in accordance with its rules. The seat of the arbitration shall be Madrid, Spain and the language of the arbitration shall be English. The Parties expressly agree to execute the arbitration award in good faith and without further recourse.
- c) If the BORROWER's country is not a party to the 1958 New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards, the dispute shall be



submitted to any competent court of MEF's choice.

The Parties accept that any communication under this Annex may be made by electronic means through the MotoStudent Platform, through the ticketing system.

The registration of every team in the Competition through the MotoStudent Platform, acknowledges and accepts the terms for transferring of the engine/motor. Registration of a team in the Competition through the MotoStudent Platform constitutes full acceptance of the present Terms and Conditions by all team members.

### **MOTOR/ENGINE DESCRIPTION AND SHIPPING DETAILS**

#### **ELECTRIC CATEGORY**

**Powered by: CIRCE**

**Property of MEF:** Fundación Moto Engineering Foundation (VAT Number: ESG44223477)  
P.T. TechnoPark MotorLand - Edif. 2WTC, 44600 Alcañiz – Teruel (Spain)

**Motor/Engine Replacement Price (in case of insurance): € 2,800**

**In case of provide a guarantee: € 1,500**

#### **eFUEL CATEGORY**

**Powered by: KTM**

**Property of MEF:** Fundación Moto Engineering Foundation (VAT Number: ESG44223477)  
P.T. TechnoPark MotorLand - Edif. 2WTC, 44600 Alcañiz – Teruel (Spain)

**Motor/Engine Replacement Price (in case of insurance): € 3,900**

**In case of provide a guarantee: € 1,900**

## ANNEX 8: MODEL DECLARATION OF RESPONSIBLE FOR TIRES FOR THE TEAMS

|                                                                                      |             |   |
|--------------------------------------------------------------------------------------|-------------|---|
| Reference number: _____                                                              | Team: _____ | — |
| Name and Surname, ID number and role in the Team (Tutor or Team Leader):<br><br><br> |             |   |

The participating Team in the current Edition of the International MotoStudent Competition (eFuel / Electric Category) hereby submits this declaration to inform the Competition Organization of the identification codes of each tire (front and rear) corresponding to the set of tires that may be used during the Final Event.

By means of this declaration, the Team confirms that the tires listed below are the only ones that will be presented for technical verification and, consequently, the only ones that may be used during the tests and sessions of the Final Event.

By signing this document, the Team also declares that it has read and understood Article B.9.2 – Tires of the Competition Regulations and undertakes to comply fully with its provisions.

| FRONT TIRE CODES |  | REAR TIRE CODES |  |
|------------------|--|-----------------|--|
|                  |  |                 |  |

Signature:

|                                  |
|----------------------------------|
| <b>ORGANIZATION OBSERVATIONS</b> |
|                                  |